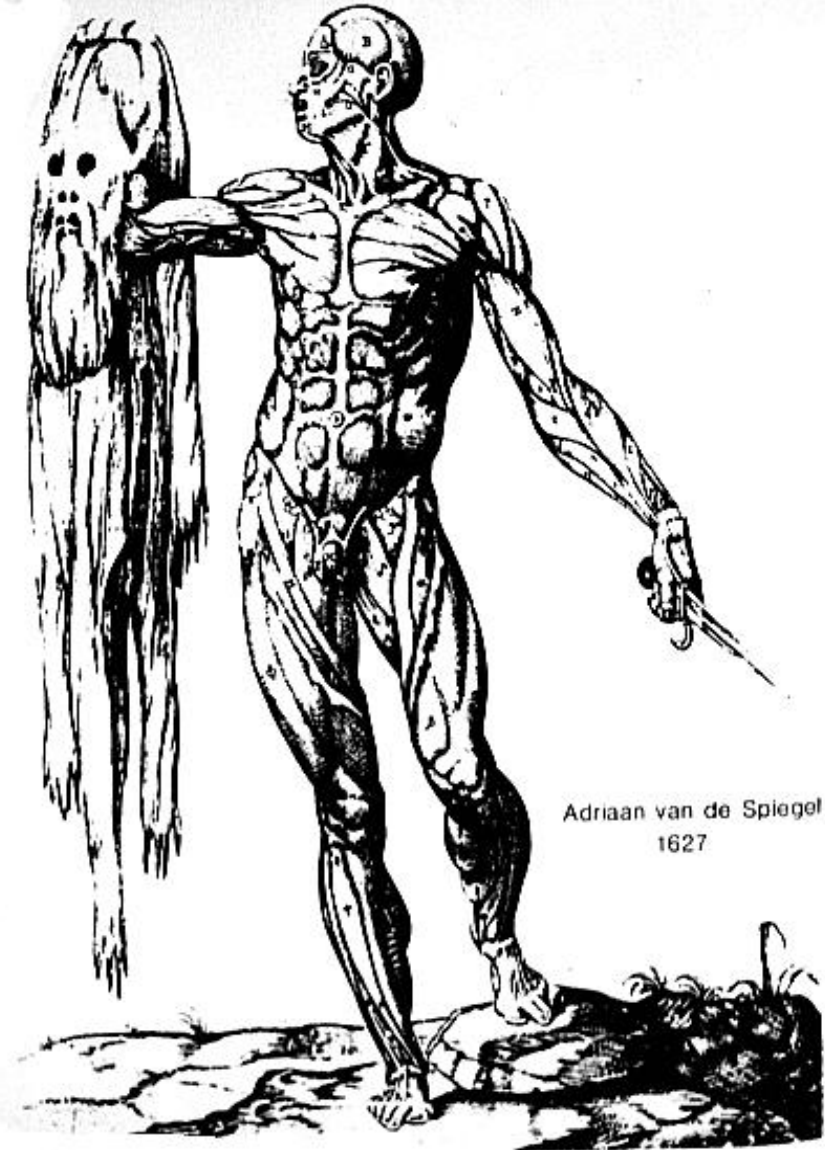


The Integumentary System

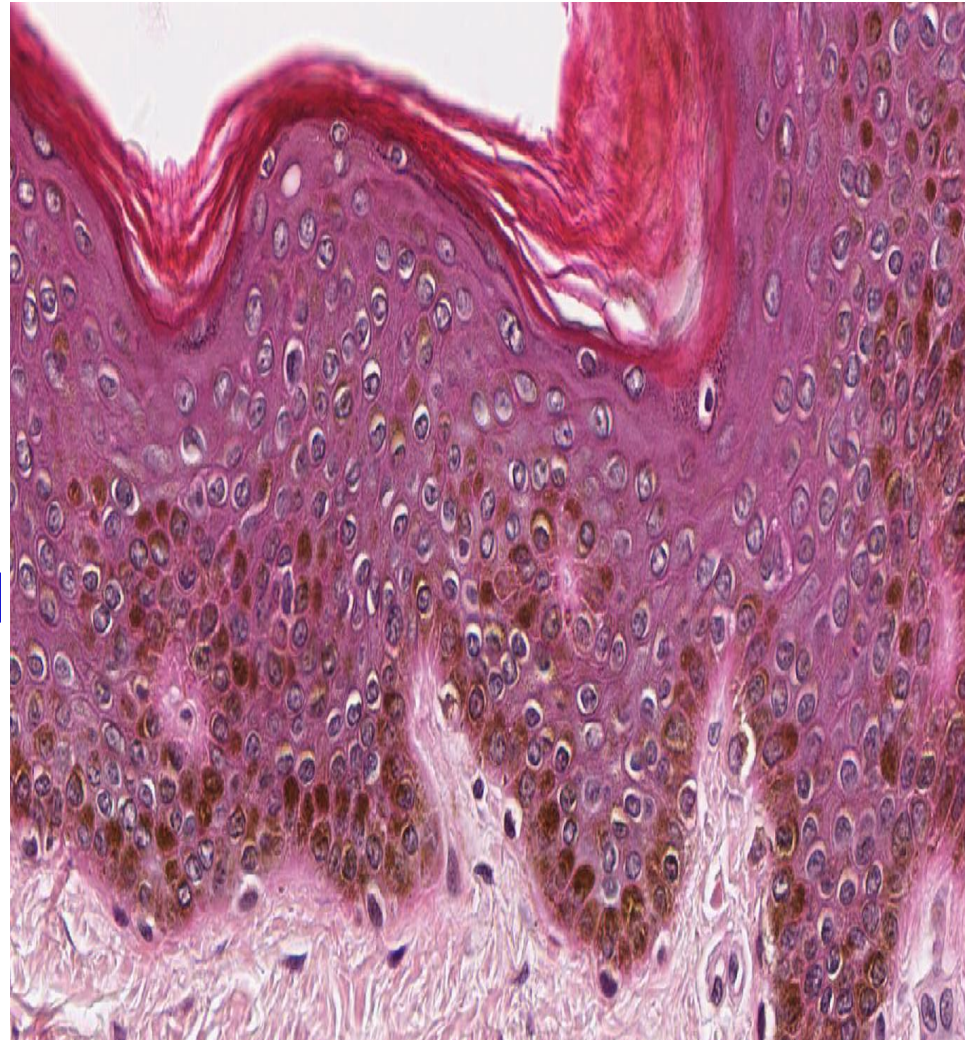
The Integumentary system

- Is one of the 11 organ systems; it is the most superficial system.
- Consists of the **skin** (i.e., the cutaneous membrane) plus all the skin appendages (or skin accessory organs) which include
 - **Hairs**
 - **Nails**
 - **Skin glands:**
 - **Eccrine/merocrine sweat glands,**
 - **Apocrine sweat glands,**
 - **Sebaceous/Oil glands**
 - **Mammary glands**
 - **Sensory receptors**



General Functions of the Integumentary System

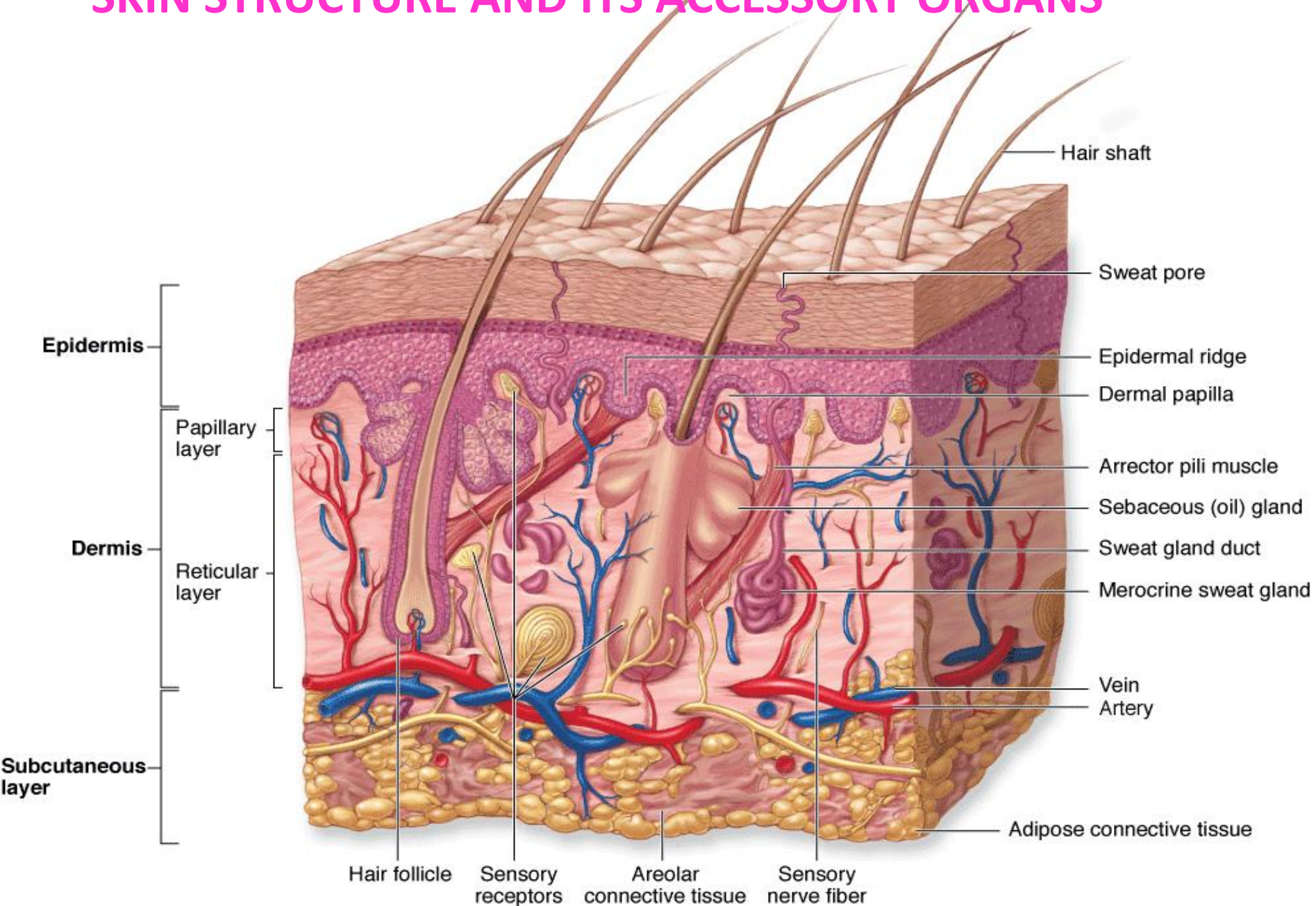
- **Protection** from mechanical injury
- **Physical protection** of pathogen entry
- **Chemical prevention** of pathogen growth
- Provide immunological information
- **Sensory reception** (general sensation)
- **Thermoregulation**
- Metabolic functions
- **Beauty** and **sexual signaling**



SKIN: Introduction

- also known as the **integument** (L. *integumentum*, covering) or **cutaneous membrane**.
- Skin is the largest single organ of the body, typically accounting for 10–15% (roughly 1/6th) of total body weight; covers 1.5–2 m² of surface area in the adult.
- Histologically , skin is composed of the **epidermis**, an outer epithelial layer of ectodermal origin, and the **dermis**, an inner, connective tissue layer of mesodermal origin.
- The junction of dermis and epidermis is irregular, projections or evaginations of the dermis called **dermal papillae** interdigitate with invaginations of the epidermis known as **epidermal ridges**.
- Skin appendages are epidermal derivatives and include hairs, nails, and skin glands.

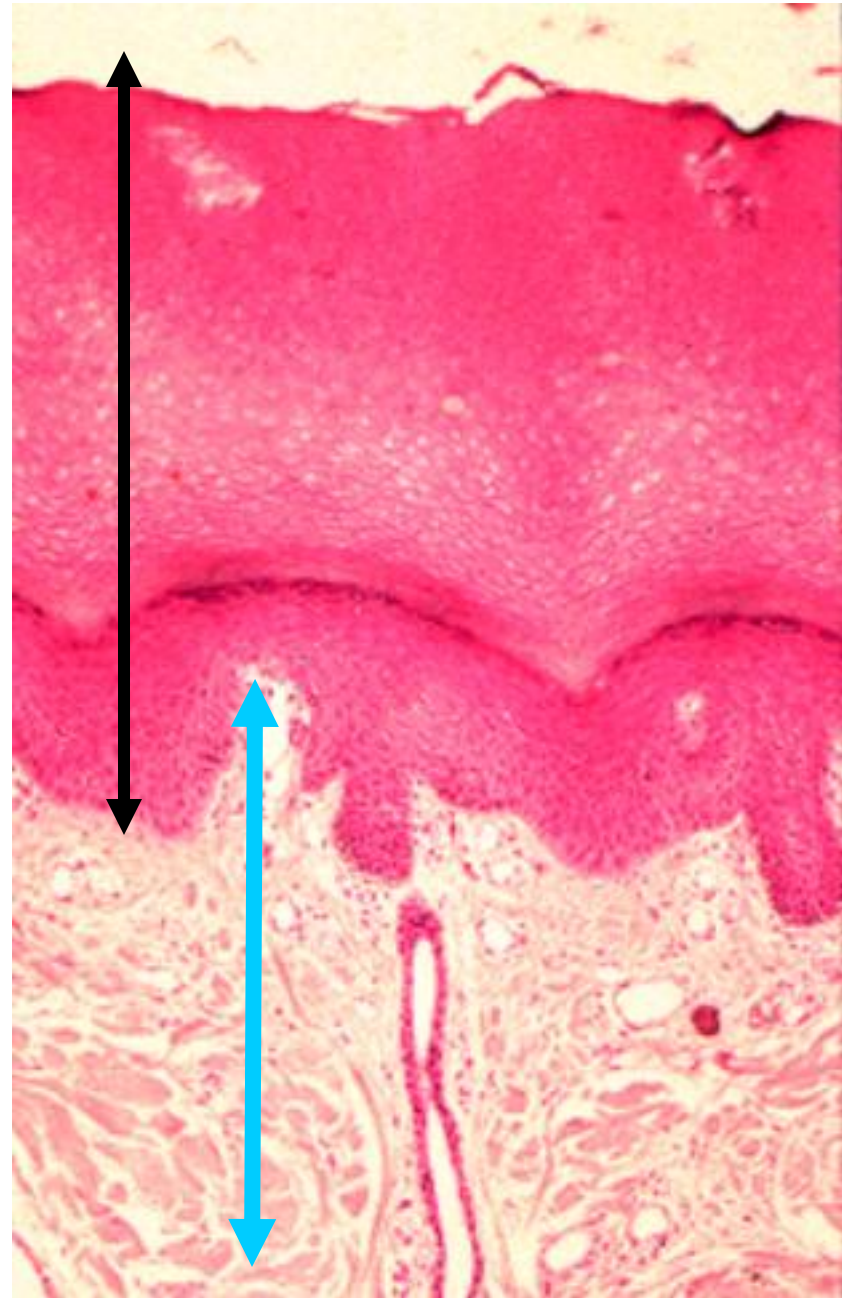
SKIN STRUCTURE AND ITS ACCESSORY ORGANS



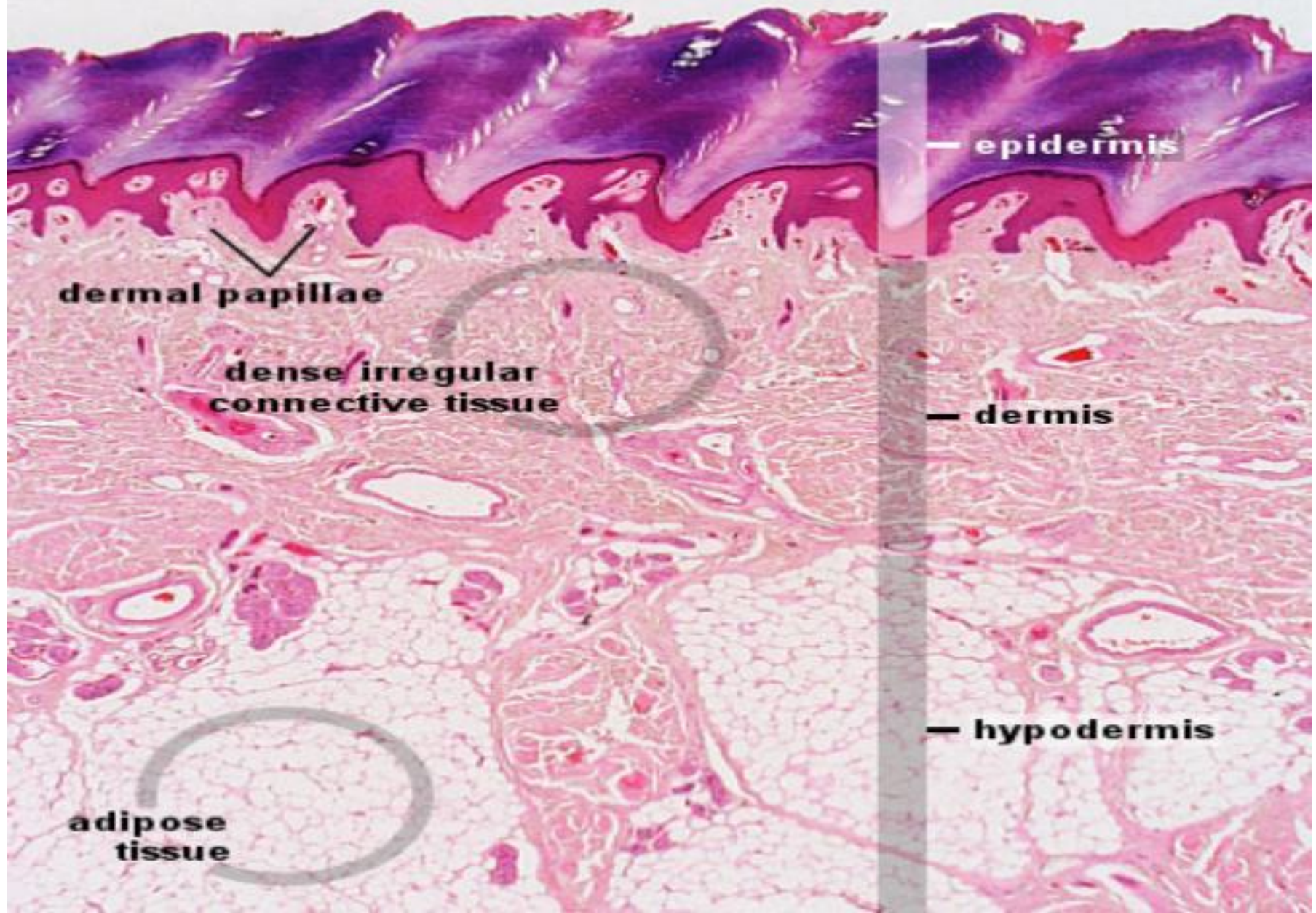
Basic Skin Structure

The skin has 2 main layers:

1. **The superficial, avascular epidermis** *epi* means “above” and *dermis* means “skin”) consisting of 4-5 epithelial strata resting upon basement membrane.
2. **The deep, vascular dermis** consisting of fibro-elastic CT.
 - Contains multiple blood vessels, and the accessory skin appendages.



Skin, thick H&E

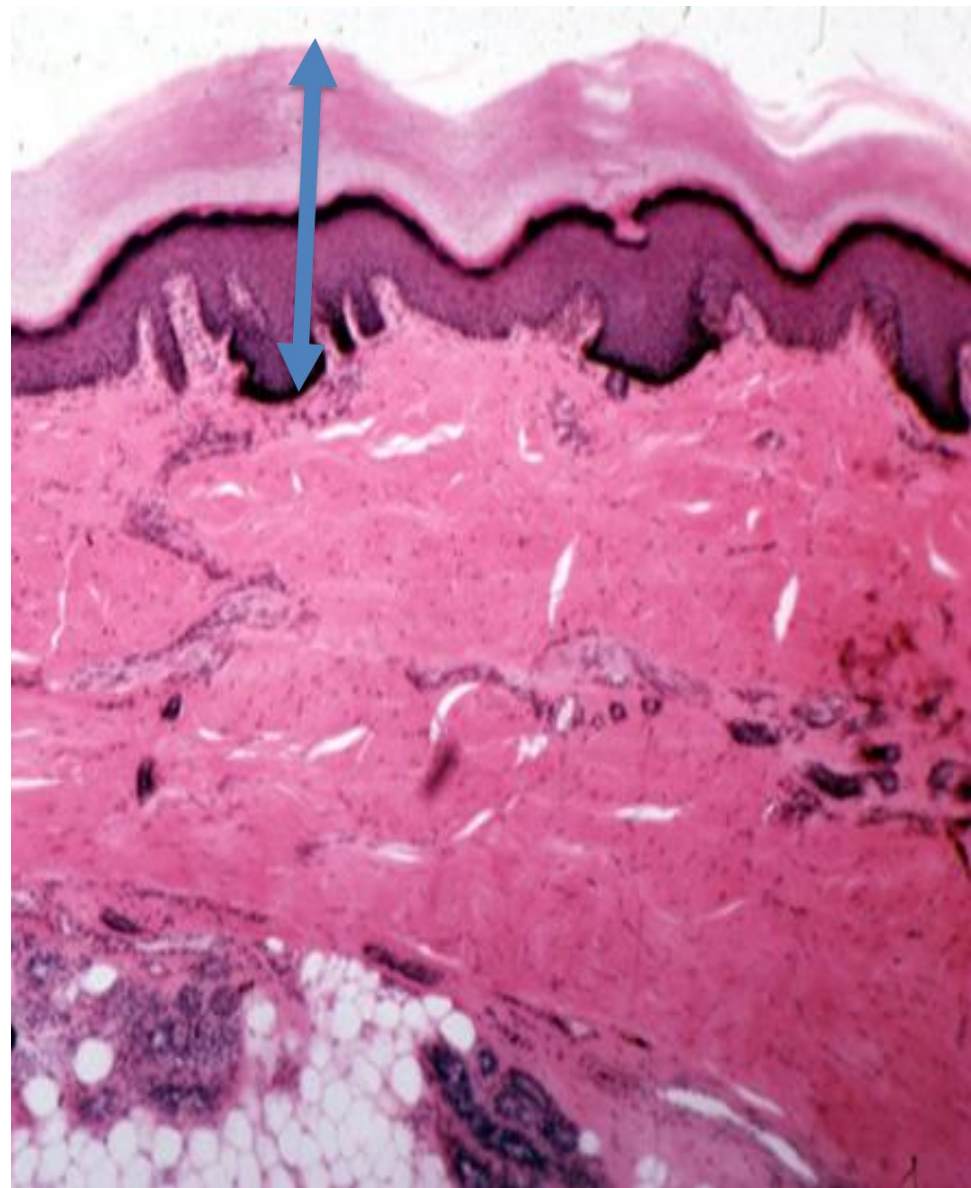


The Epidermis

- Keratinized stratified squamous epithelium
- Avascular
- Consists of 4 distinct cell types arranged in 4 or 5 distinct strata.

Epidermal cell types:

1. Keratinocytes (*structure*)
2. Melanocytes (*pigment*)
3. Merkel cells (*sensation*)
4. Langerhan's cells
(*phagocytes - immune cells*)

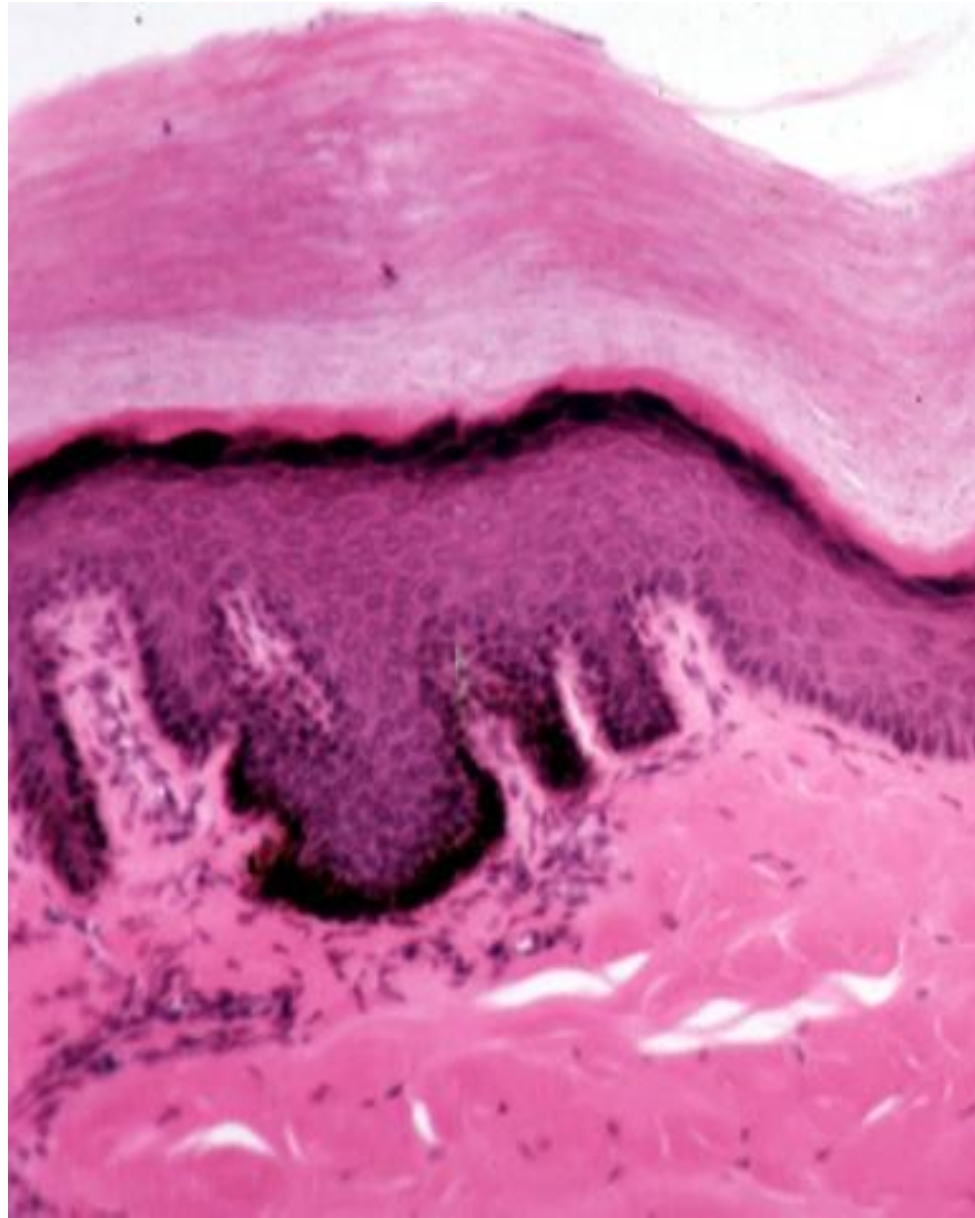


The arrow indicates the epidermis of thick skin

Structure of Epidermis

- consists of keratinized stratified squamous epithelium containing four types of cells: keratinocytes, melanocytes, Langerhans cells and Merkel's cells.
- Epidermis consists of four or five strata; from inward to outward:
 - 1. Stratum Basale/germinativum(Basal layer): deepest**
 - a single layer of basophilic columnar or high cuboidal cells resting on the basal lamina at dermal-epidermal junction.
 - 2. Stratum spinosum:**
 - consists of 5-10 layers of polygonal cells with central nucleus and cytoplasmic processes filled with bundles of keratin filaments (tonofibrils).
 - 3. Stratum granulosum:**
 - composed of 1–3 layers of flattened ovoid cells containing a centrally located nucleus, and a cytoplasm filled with two types of coarse basophilic granules: free kerato-hyaline granules and lamellar membrane-bounded granules.
 - 4. Stratum lucidum: found in thick skin only**
 - thin layer of translucent flattened eosinophilic cells, cell organelles and nuclei are no longer evident.
 - 5. Stratum corneum: horny/cornified cell layer- outermost**
 - 10-20 layers of flattened non-nucleated keratinized cells and their cytoplasm is filled with a birefringent filamentous scleroprotein, Keratin.
 - The cells are nothing more than scale-like bags of keratin

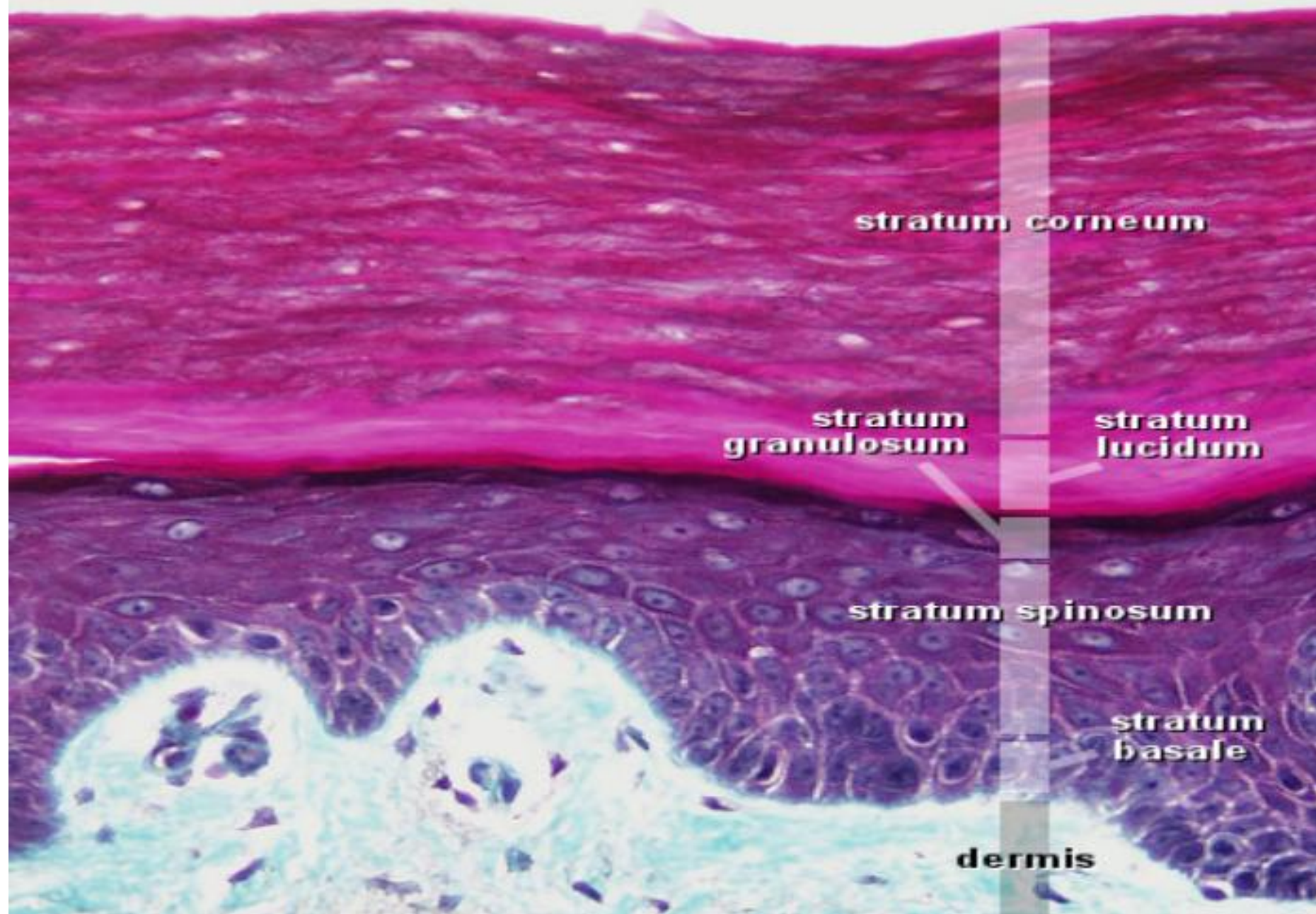
Skin Types



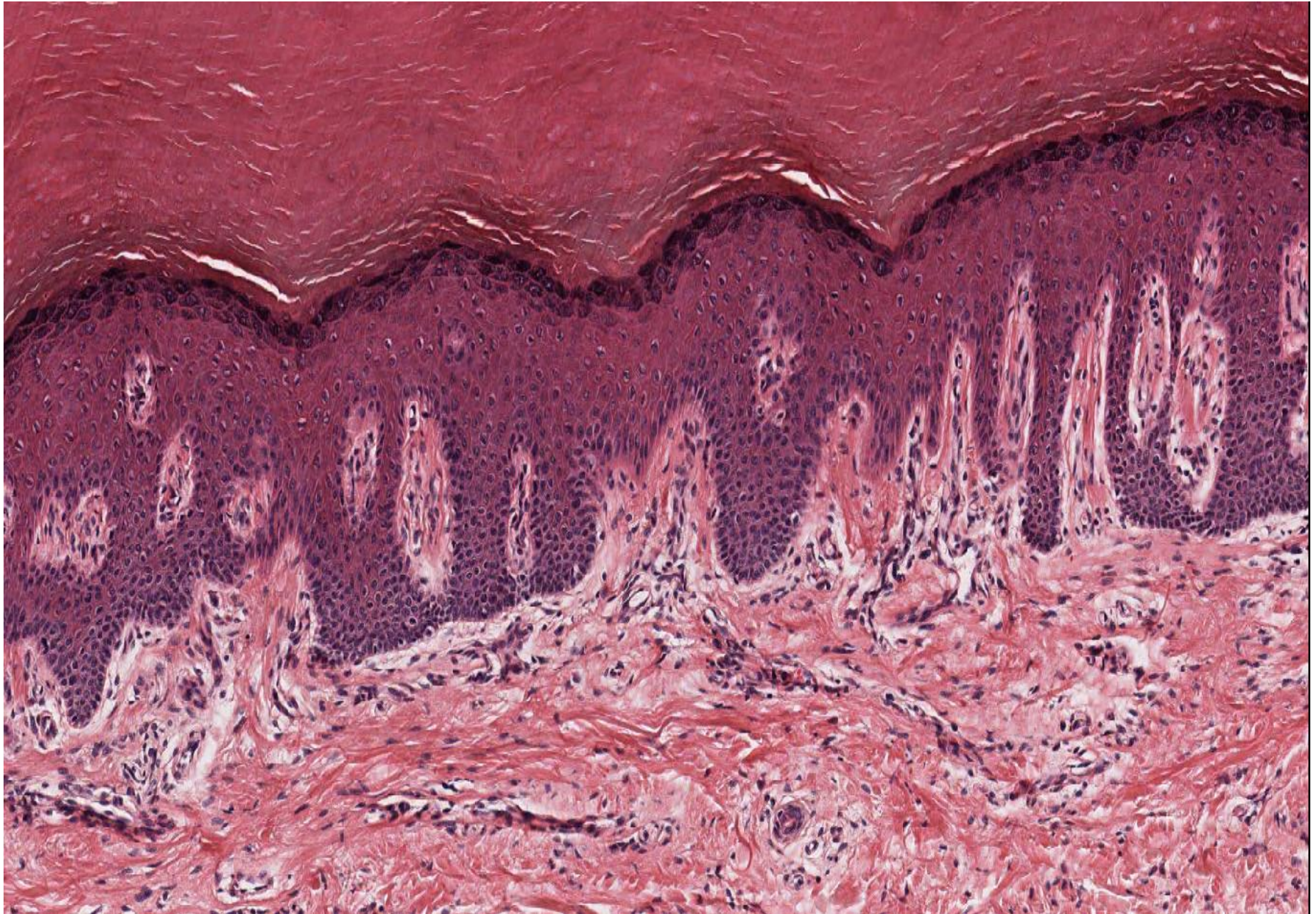
1. Thick Skin (Glabrous skin)

- Found on soles of feet and palms of hands and corresponding parts of fingers and toes.
- Lacks hair follicles, arrector pili muscles & sebaceous glands; as a result has smooth surface
- Epidermis consists of five layers from inward to outward as follows:
 1. Stratum basale
 2. Stratum spinosum
 3. Stratum granulosum
 4. Stratum lucidum
 5. Stratum corneum

Skin, thick trichrome

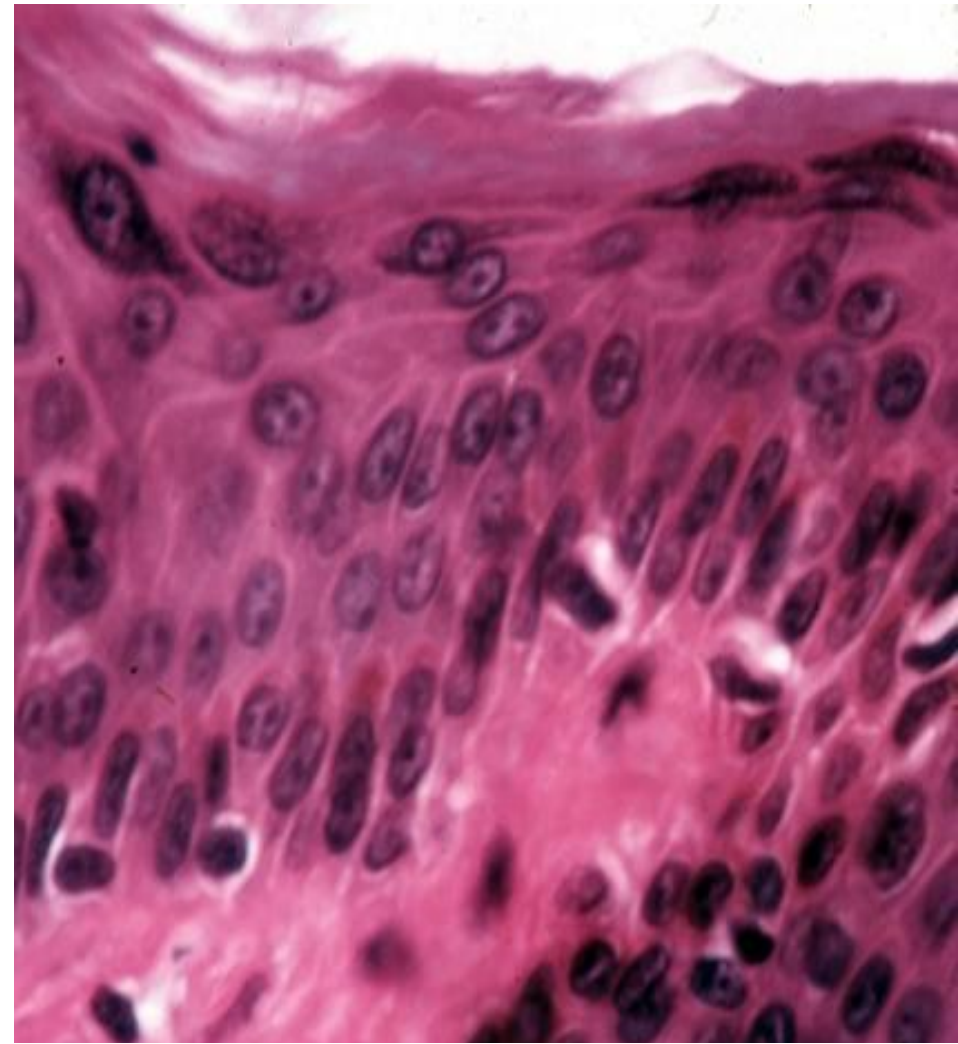


Thick skin

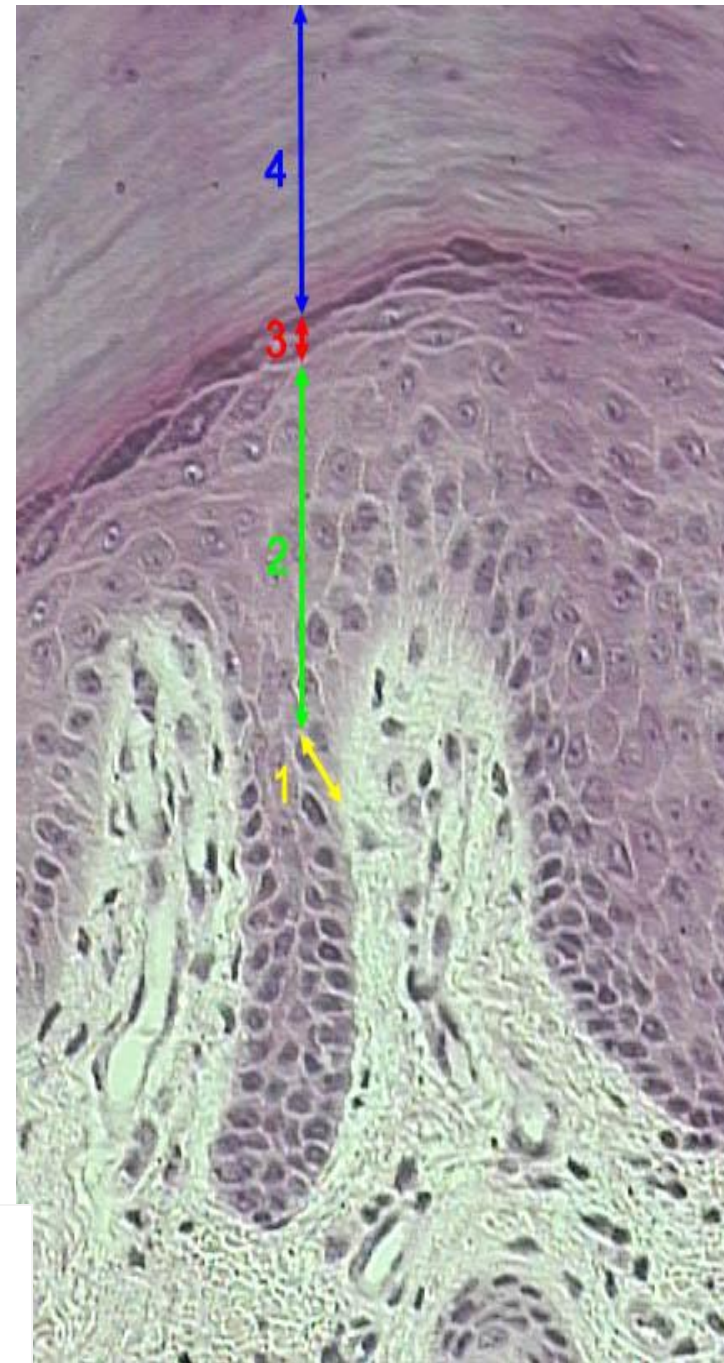
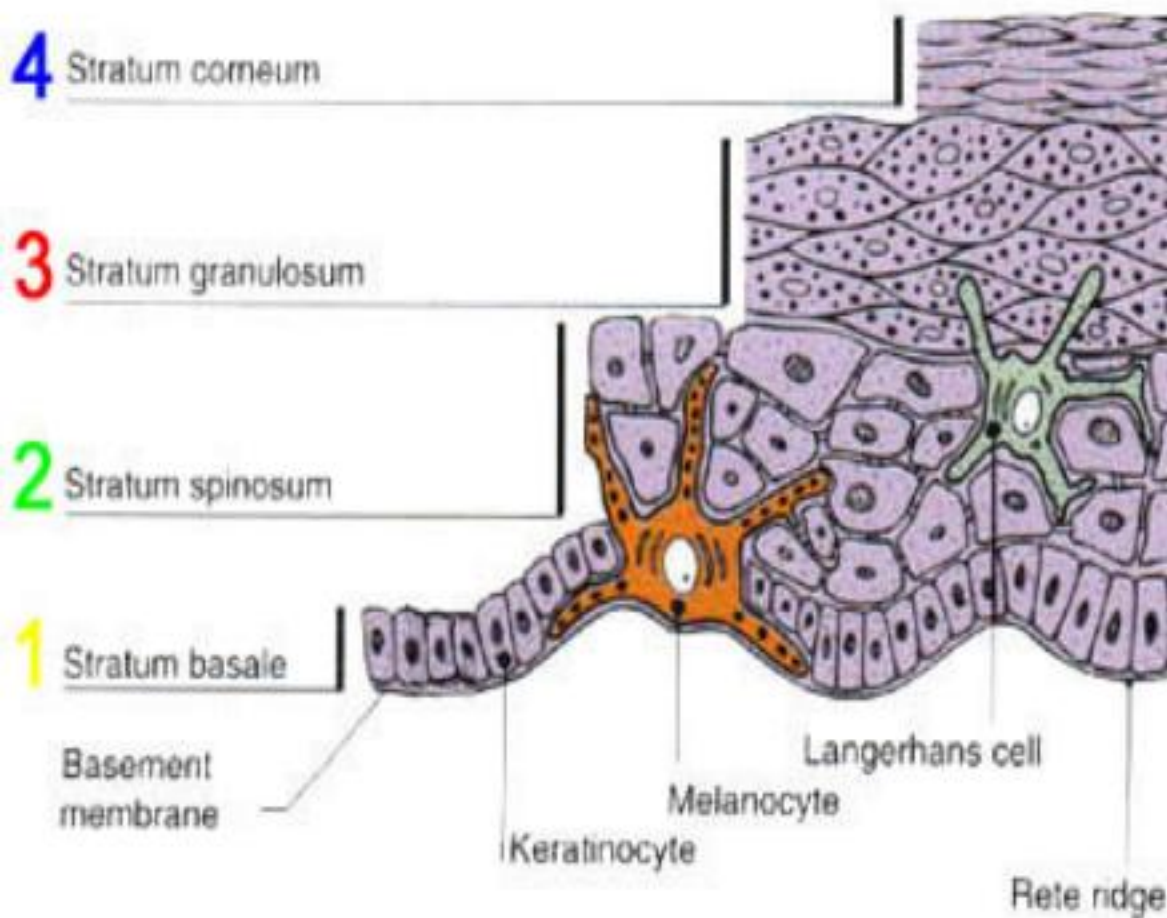


2. Thin skin

- Found everywhere else on the body.
- Contains only 4 epidermal strata, lacks stratum lucidum.
- The remaining 4 layers are thinner than those of thick skin.
- Why is thick skin found on the palms and soles? What is the advantage of that?



Note: this slide is at a higher mag. than the thick skin slide on the previous page

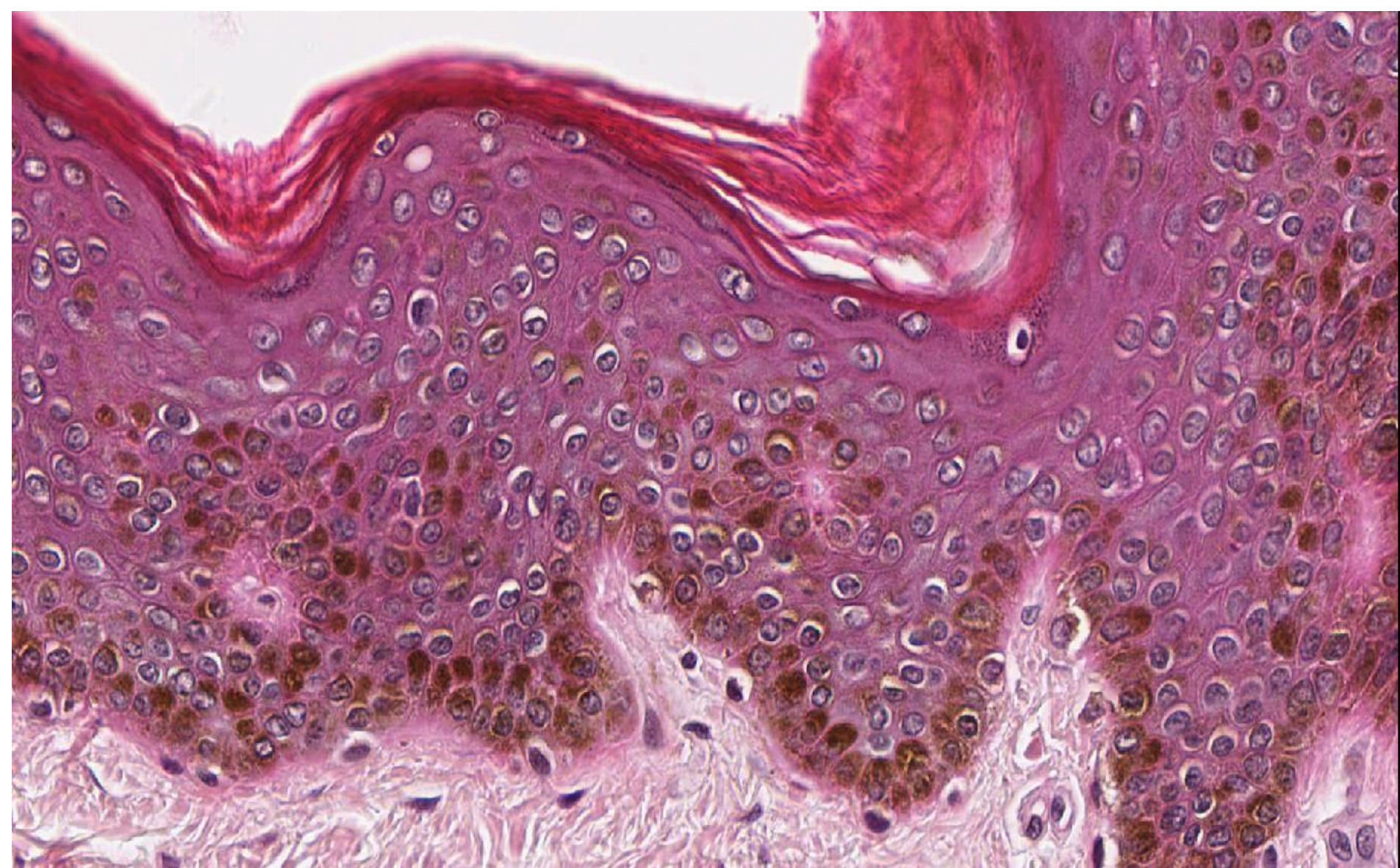


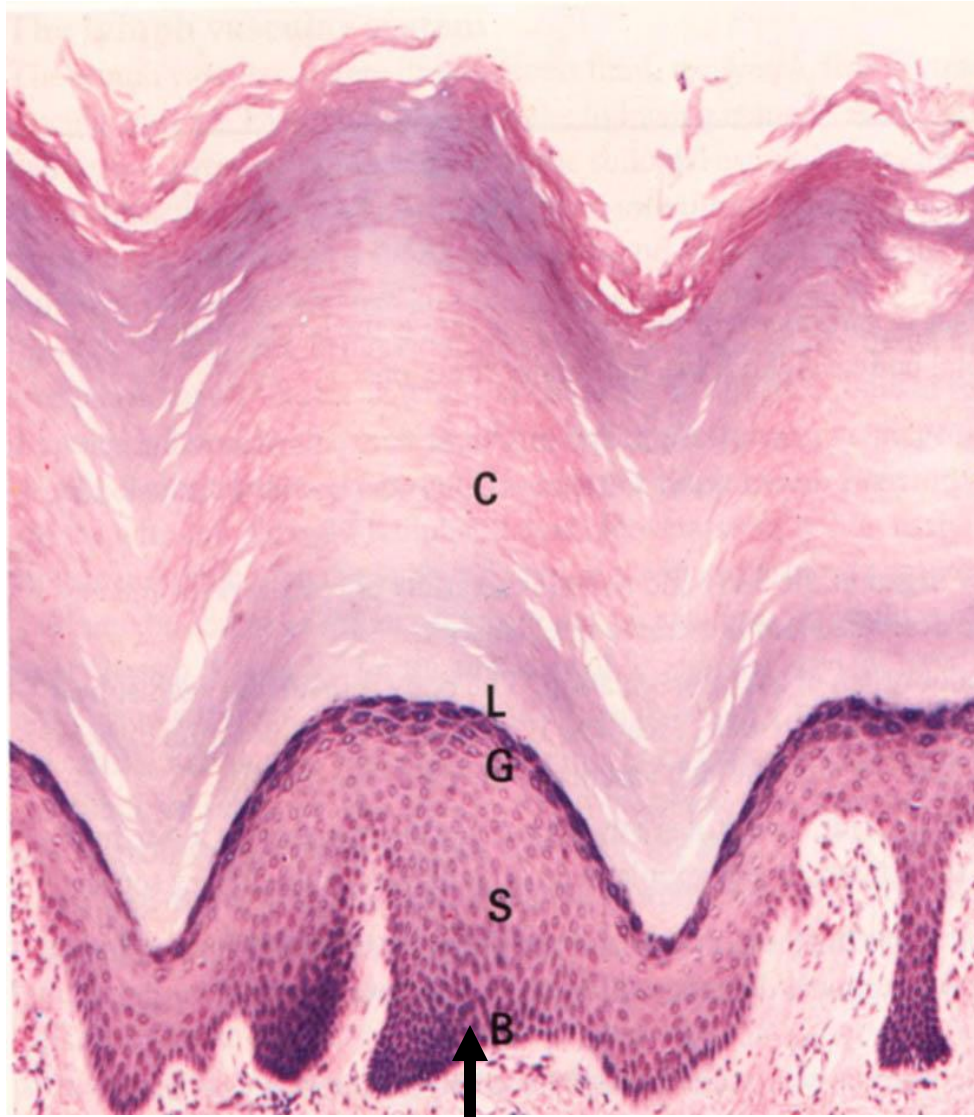
Notice the 4 layers of thin skin in both the cartoon and the photomicrograph.

Skin, thin H&E



Thin heavily pigmented skin





Thick skin

Thin skin



- The designations "thick" and "thin" refer to the thickness of the epidermal layer, which varies from
 - **75 to 150 μm in thin skin and**
 - **400 to 1400 μm (1.4 mm) in thick skin.**
- Total skin thickness (epidermis plus dermis) also varies according to site.
 - For example, skin of the back between the scapulae is the thickest skin in the body about 4 mm thick, whereas that of the scalp is about 1.5 mm thick.

Thick skin

- Palms, soles & skin of the palmar aspect of fingers and plantar aspect of toes
- Epidermis: 400–1400 μm thick

Thick epidermis, formed of the classic layers as follows:

- (1) thick Malpighian layer
- (2) thick granular layer
- (3) a thin clear layer
- (4) Very thick horny layer.

No hair follicles

No sebaceous glands

Has prominent ridges & folds

No hair muscle (Arrector pili)

Thin skin

- All other parts of the body
- Epidermal thickness: 75—150 μm

Thin epidermis formed of: (1) thin Malpighian layer

- (2) Very thin granular layer
- (3) No clear layer
- (4) Very thin horny layer.

Presence of hair follicles

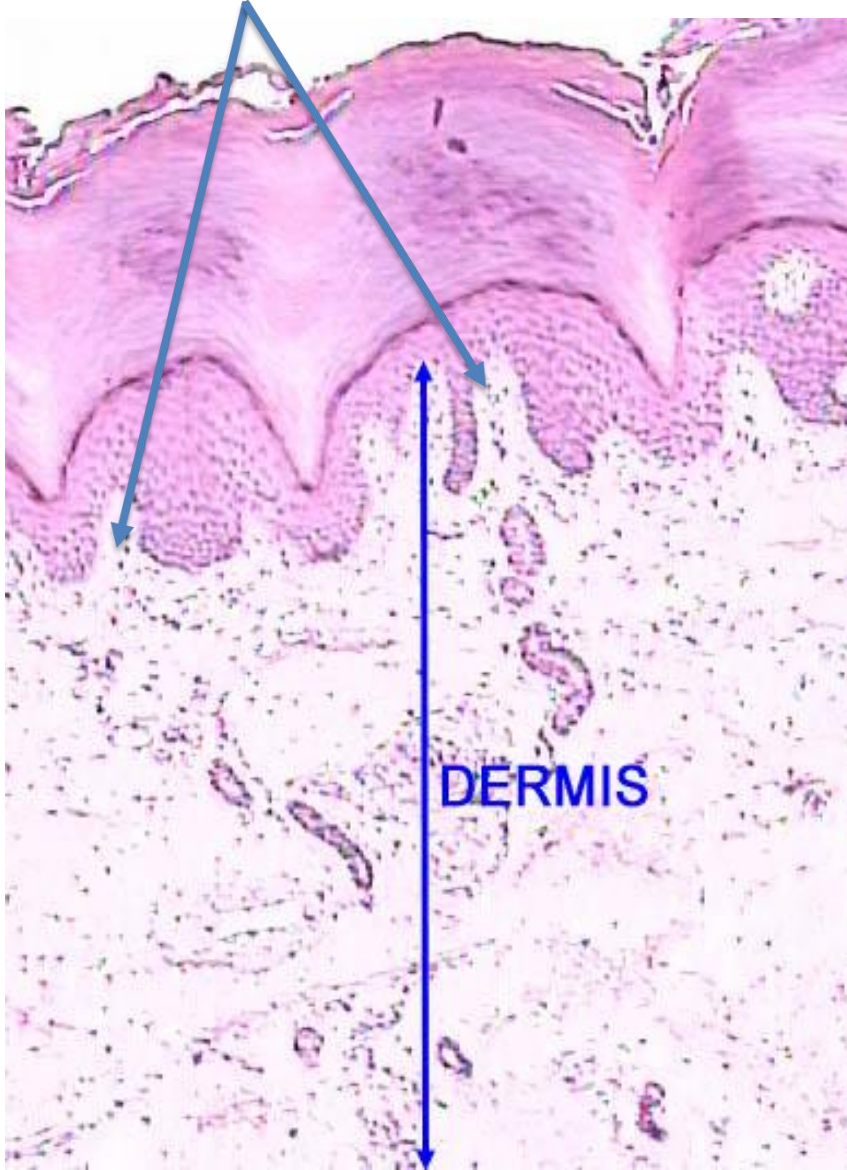
Present

Less developed ridges & folds

Present

DERMIS (CORIUM)

dermal papillae



- Strong, flexible fibro-elastic connective tissue.
- Divided into papillary dermis and reticular dermis.

Papillary dermis is the upper $\frac{1}{5}$ th of the dermis and consists of loose (areolar) CT.

- Provides an arena for immune cells to fight invaders.
- Projects upward as dermal papillae in thick skin to interdigitate with the epidermis & form a strong connection.
- Heavily invested with blood vessels – they constrict in cold weather and dilate in warm weather. Why?
- Also contains multiple sensory receptors.

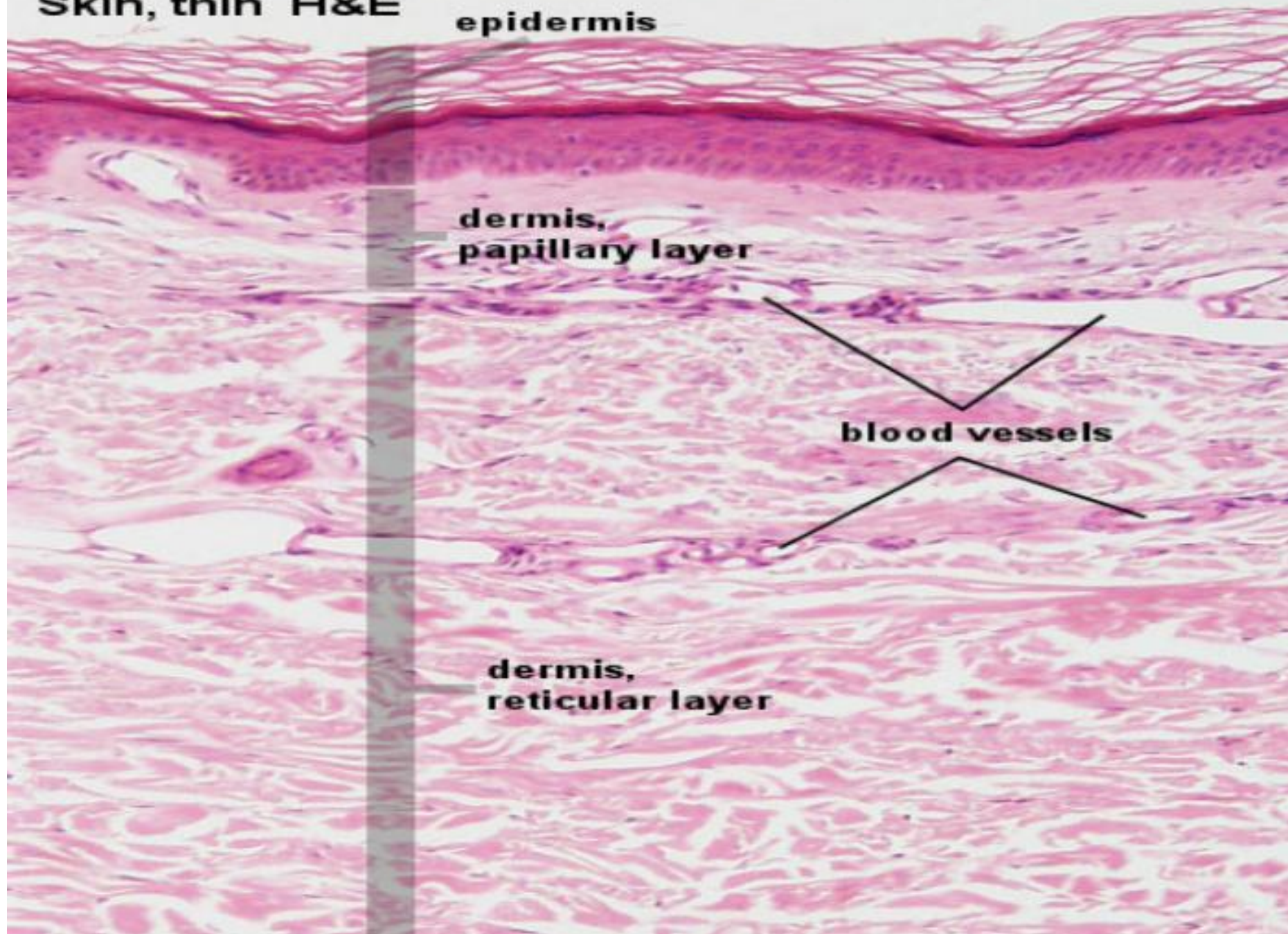
Skin, thin H&E

epidermis

**dermis,
papillary layer**

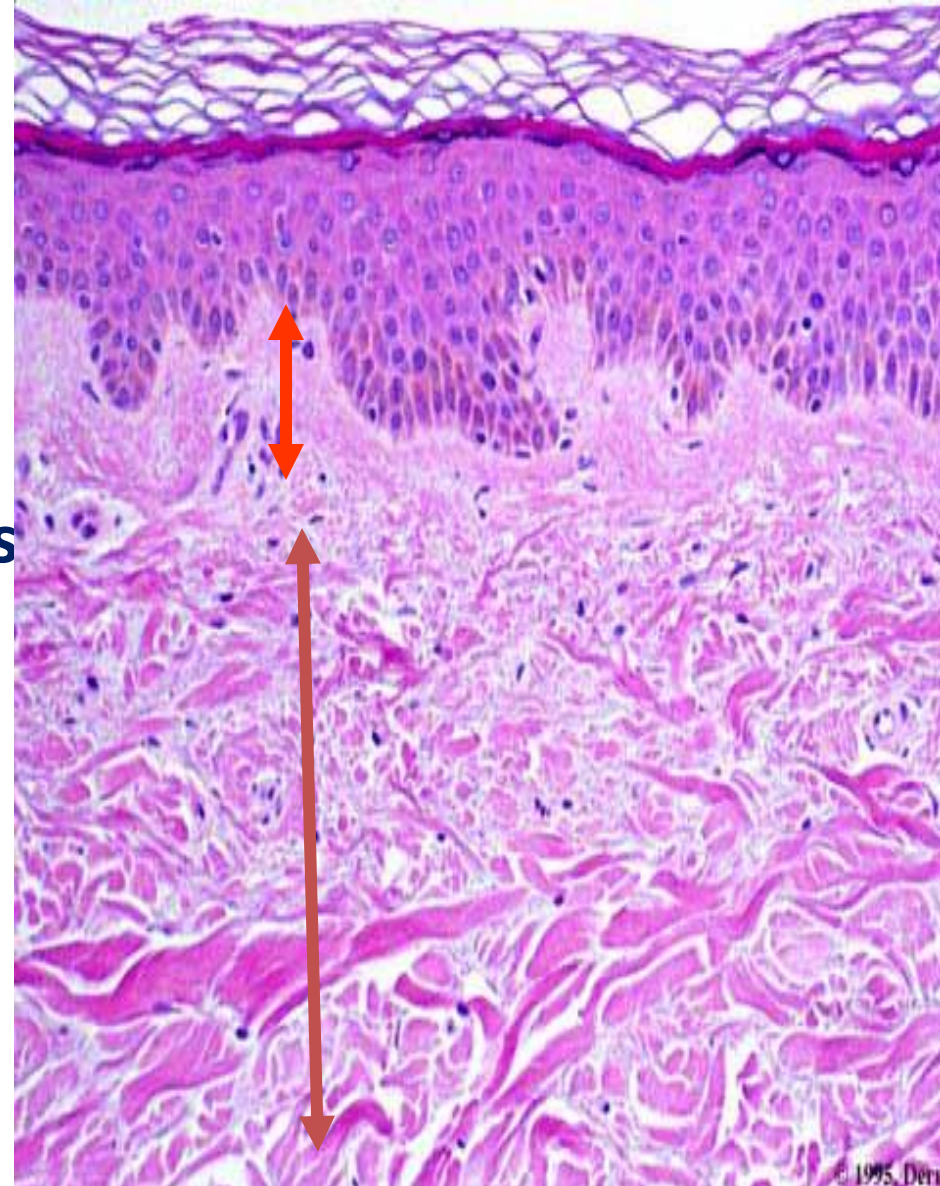
blood vessels

**dermis,
reticular layer**



Reticular Dermis

- So named because of interlacing/interwoven disposition of its collagen and elastic bundles
- is the lower 4/5th & consists of dense irregular CT.
- The fibers provide the skin its strength and resiliency.
- Elastic fibers give the skin the ability to stretch & recoil.
- Despite their epidermal origin, the majority of skin appendage are housed within the dermis & hypodermis.



The short arrow indicates the papillary dermis and the long arrow indicates the reticular dermis

Skin, thick van Gieson & elastin

epidermis

stratum
spinosum

stratum
basale

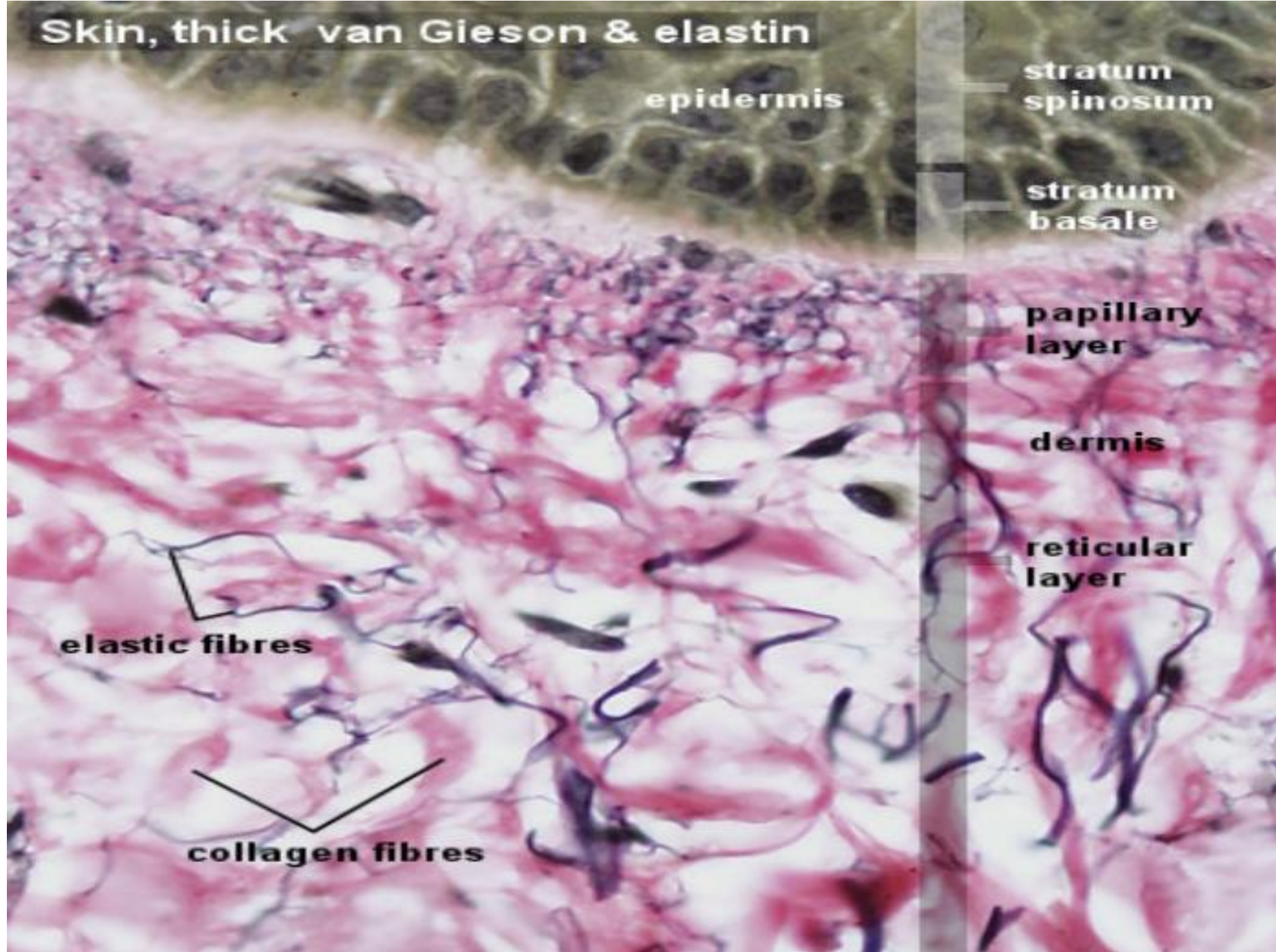
papillary
layer

dermis

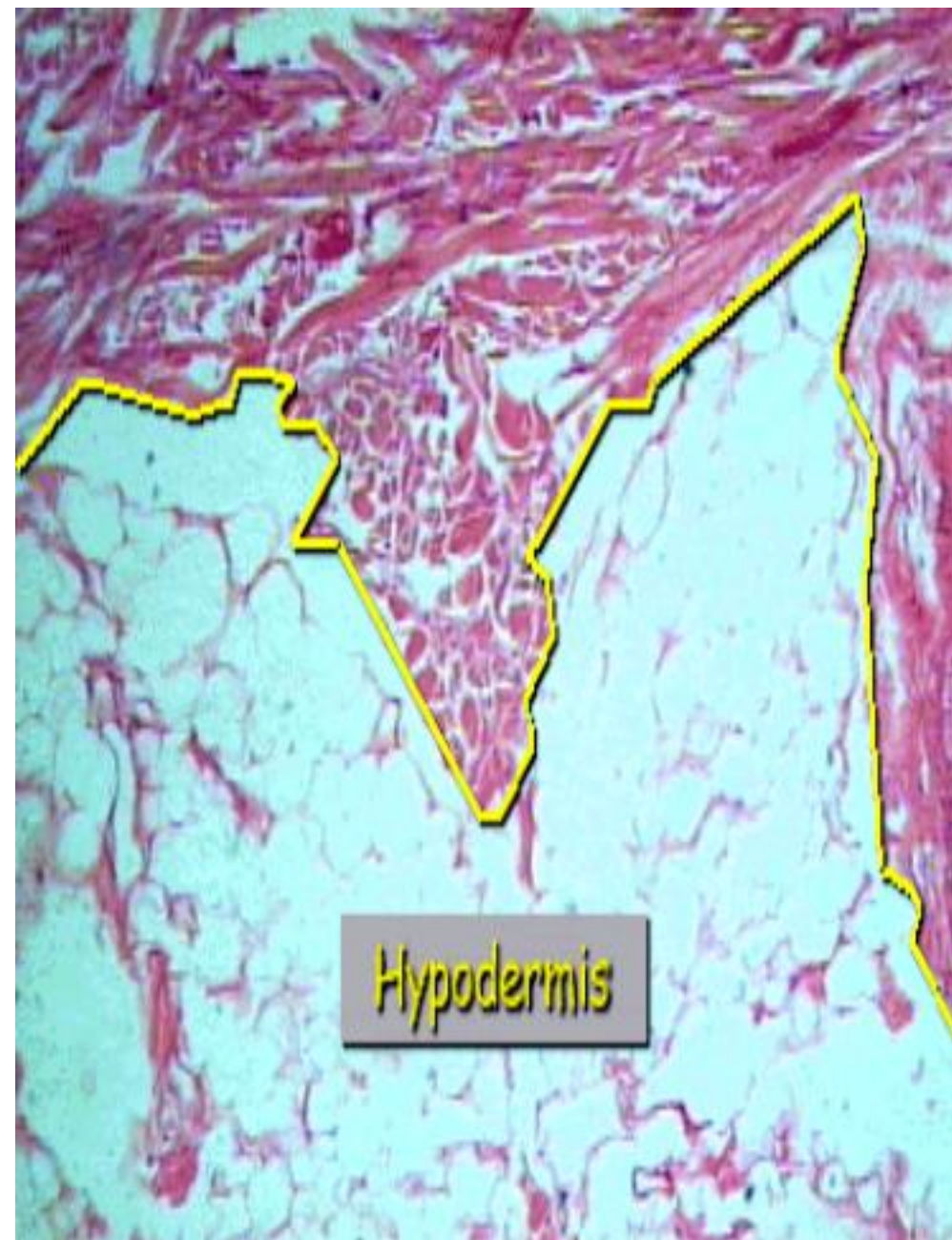
reticular
layer

elastic fibres

collagen fibres

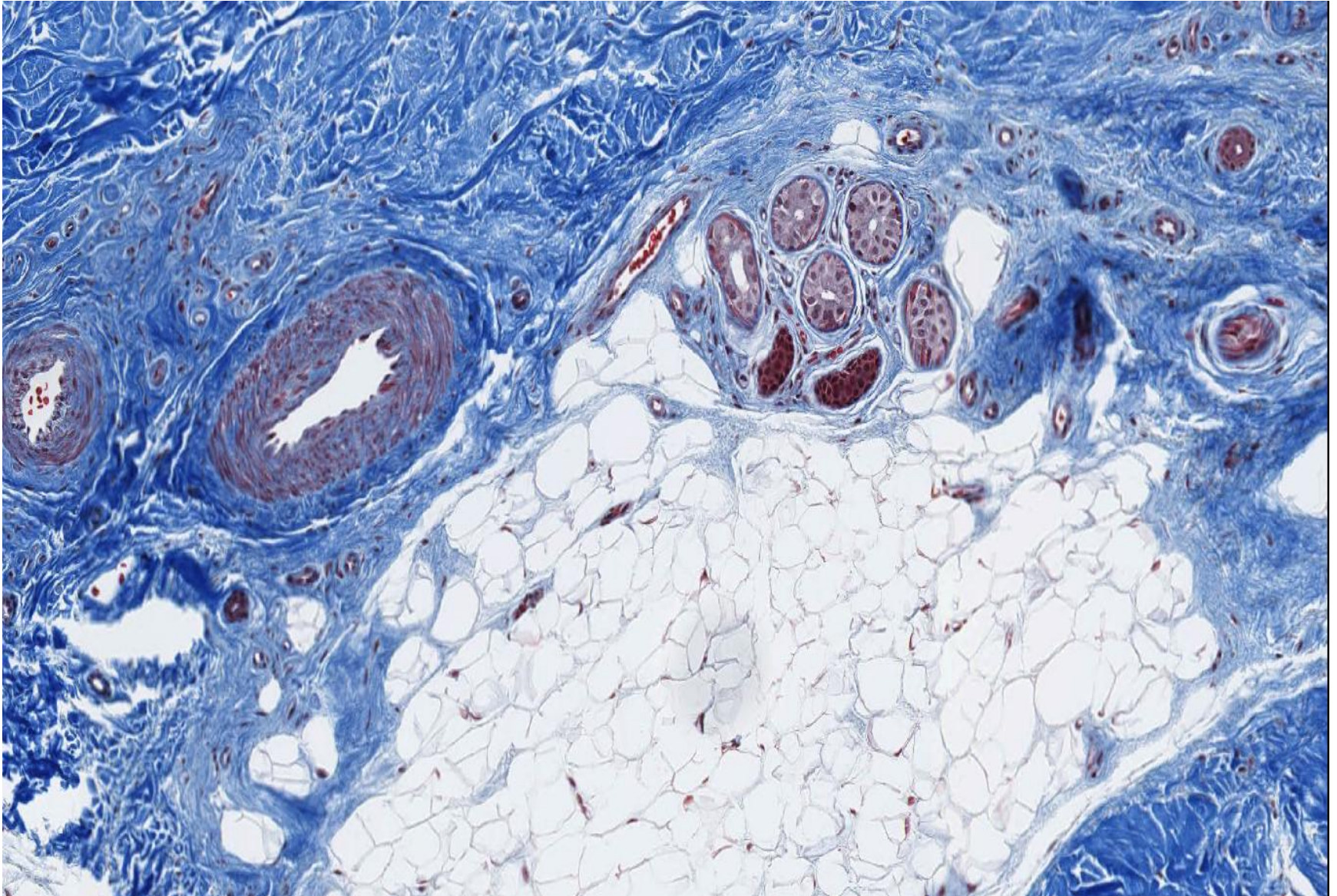


Subcutaneous Tissue

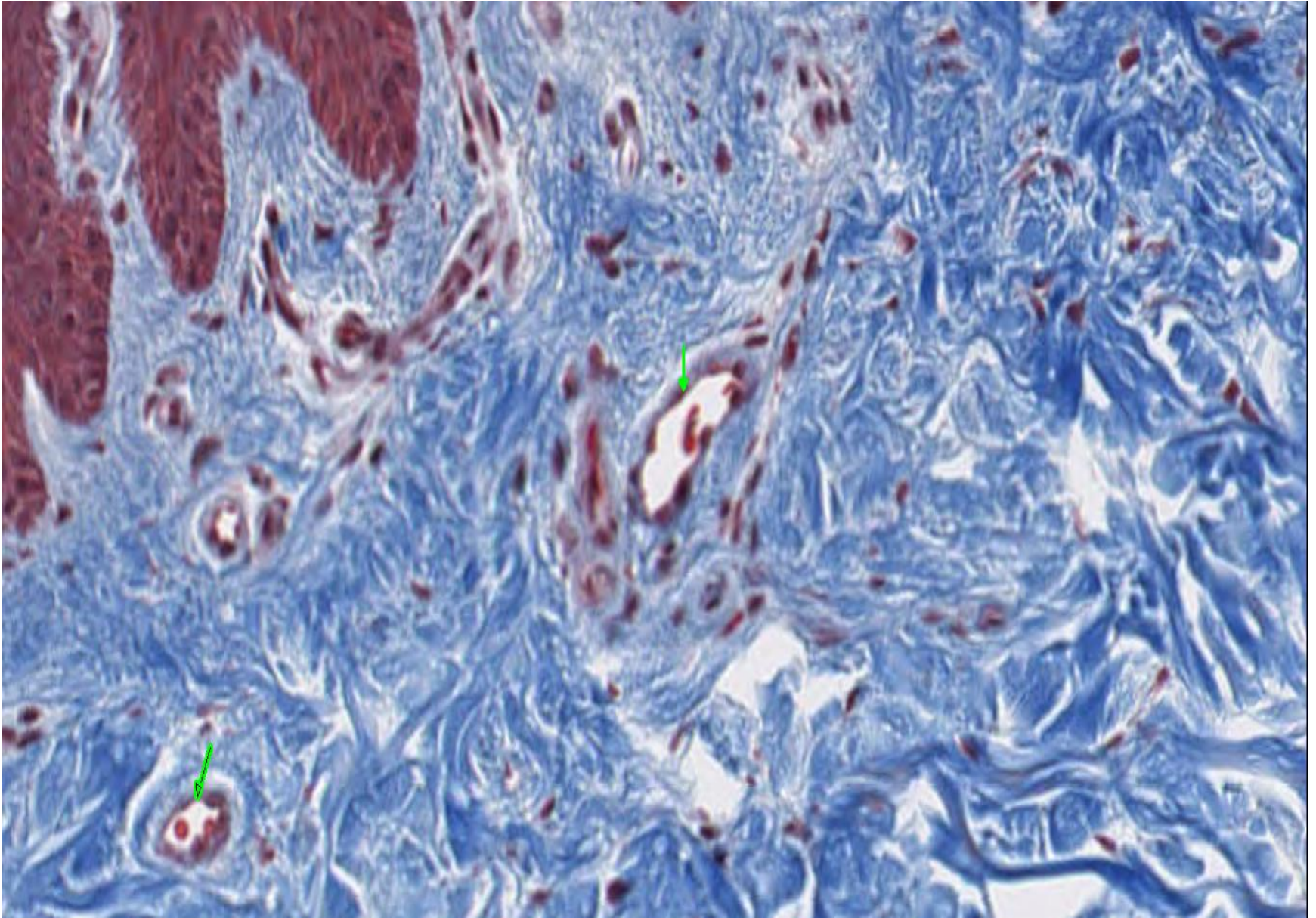


- A.k.a. the **hypodermis** or superficial fascia.
- Deep to the dermis.
- Consists of primarily adipose tissue plus some loose areolar CT.
- Stores energy, provides insulation and padding.
- Anchors the skin to underlying structures, especially muscles.
- Different distribution between the sexes.

Cutaneous vascular plexuses of Skin



Subpapillary vascular plexuses



Skin Color

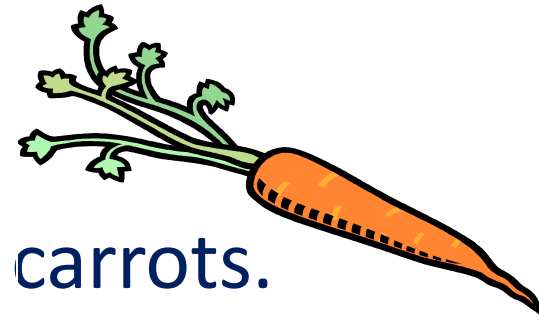
- Determined by **different** factors (pigments):
 1. Melanin
 2. Carotenoids (carotene)
 3. Hemoglobin & its oxygenation
- Of these, only melanin is made in the skin.



1. Melanin:

- Polymer of amino acid tyrosine. Its synthesis is catalyzed by an enzyme called tyrosinase.
- **Albinos** lack this enzyme, probably due to gene mutation.
- Ranges in color from yellow to reddish brown to black.
- All people have the same # of melanocytes, individual variations in skin color are due to how much and what type of melanin is made.
- **Freckles** and **moles** are local accumulations of melanin.
- **Vitilligo** are local loss of melanin from the skin

2. Carotenoids (Carotenes)



- Yellow to orange pigment found in vegetable plant products such as carrots.
- When large amounts are eaten, it can be deposited in the subcutaneous adipose tissue and/or stratum corneum of thick skin.

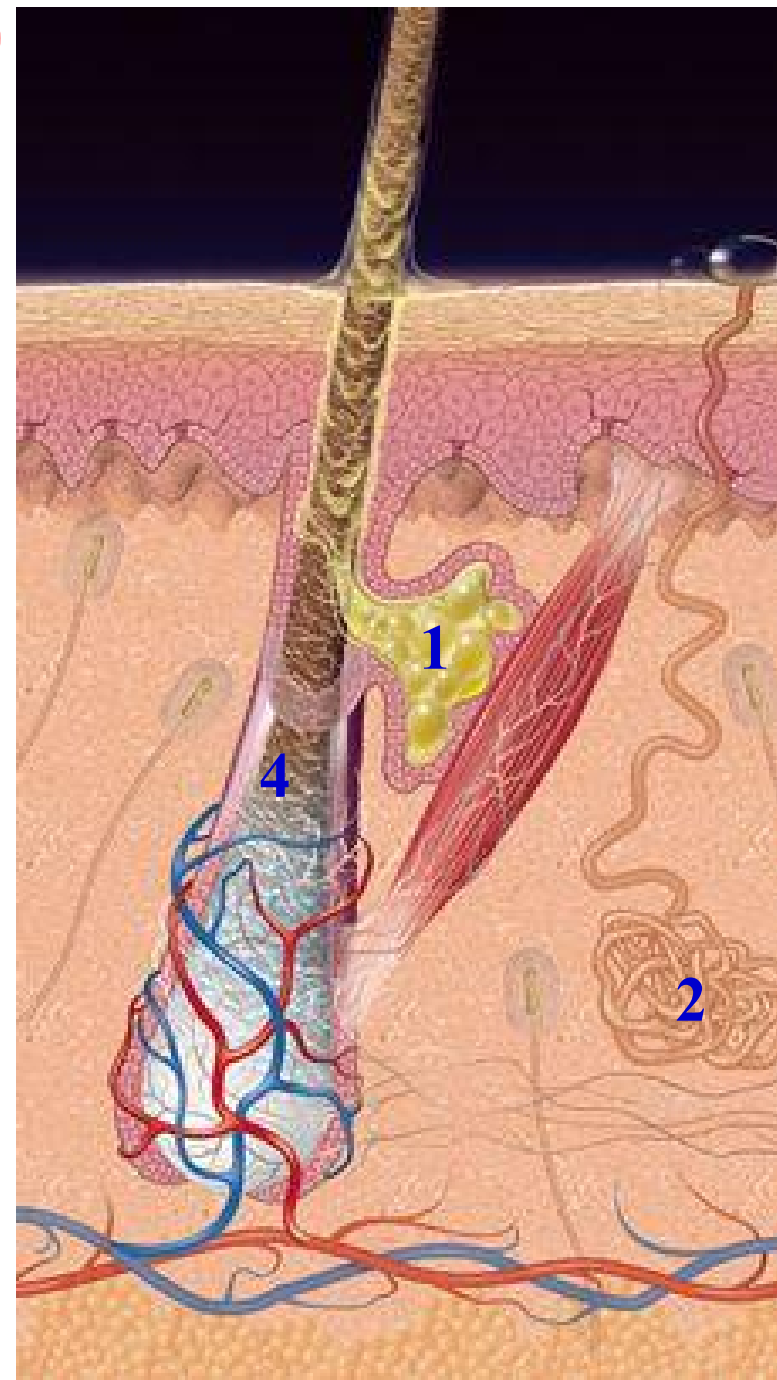
3. Hemoglobin

- Pigmented protein that transports oxygen within the blood.
- In Caucasians, the fair skin allows the crimson color of oxygenated blood to make the skin have a somewhat **pinkish hue**.
- The amount of capillaries and the quality (degree of oxygenation) and quantity of the hemoglobin they contain affects skin color.

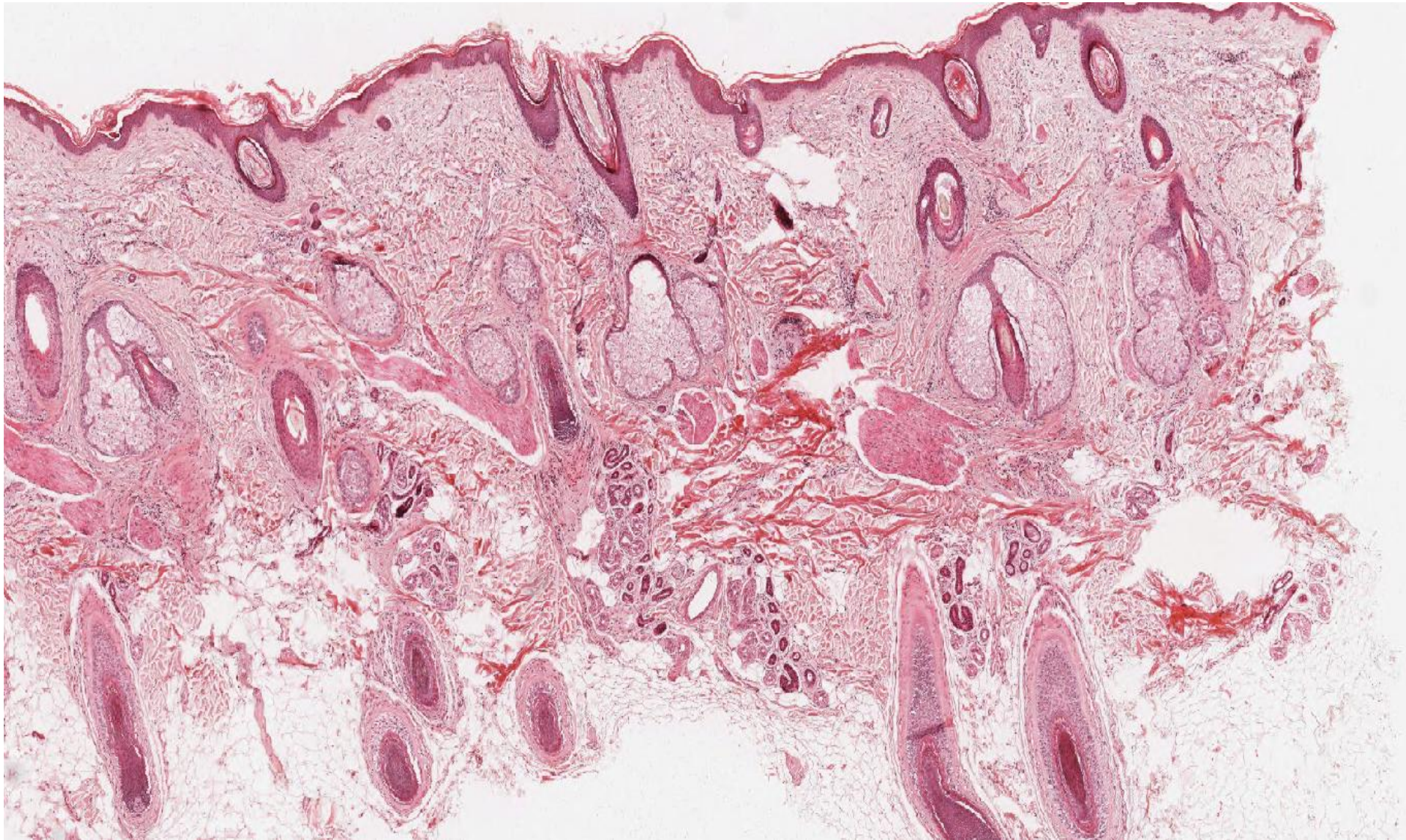
Skin Appendages (Adenexa)

- Are found in the dermis and hypodermis
- Includes

1. Hairs
2. Nails
3. Skin glands
 - a. Sebaceous glands
 - b. Sudoriferous/eccrine sweat glands
 - c. Apocrine sweat glands (e.g. Ceruminous glands)
 - d. Mammary glands



Skin appendages

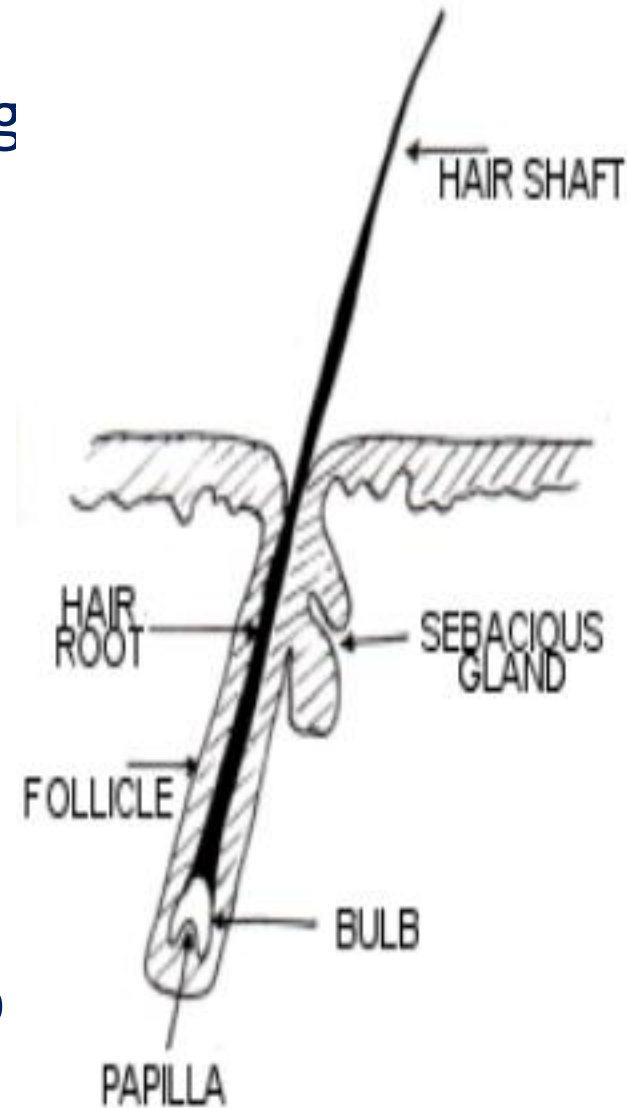


Hairs (pili) and Hair Follicles

- Hair is a flexible strand made of highly keratinized dead cells.
- The keratin in both hair and nails is hard keratin, a stiffer, more compact version of the soft keratin that dominates the epidermis.
- It is tougher and its cells do not individually flake off.
- The hair is made by the living hair follicle, in the region known as hair matrix.
- The color, size, shape and texture of hairs vary according to age, genetic background, and region of the body.
- All skin has at least minimal hair except that of the palms, soles, lips, glans penis, clitoris, and labia minora. The face has about 600 hairs/cm² and the remainder of the body has about 60/cm².

Hair and Hair Follicles

- **Hair** consists of a **shaft** which protrudes from the skin and the **root** which is within the skin.
- At the base, the root gets wider forming the **hair bulb**.
- Hair consists of 3 concentric regions:
 1. **Medulla** → the innermost region consisting of large, loosely arranged keratinized but non-pigmented cuboidal cells separated by air spaces.
 2. **Cortex** → surrounds the medulla and consists of flattened keratinized and pigmented cells
 3. **Cuticle** → outer layer of thin flattened scale-like keratinized cells that overlap like shingles with their free edges pointing upward.



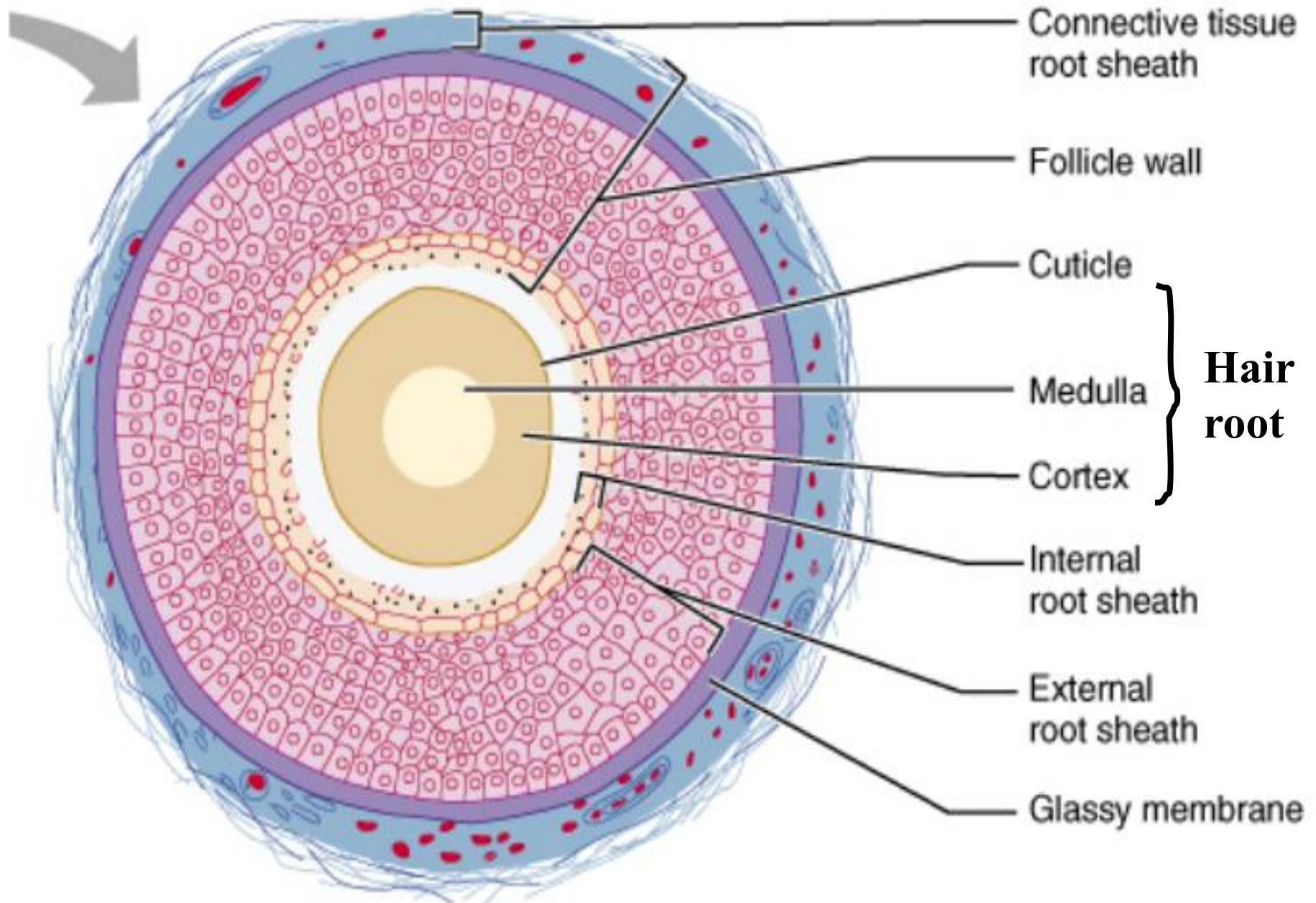
Root of hair

- is the embedded part of the hair surrounded by inner and outer root sheaths of the hair follicles and dermal CT sheath.

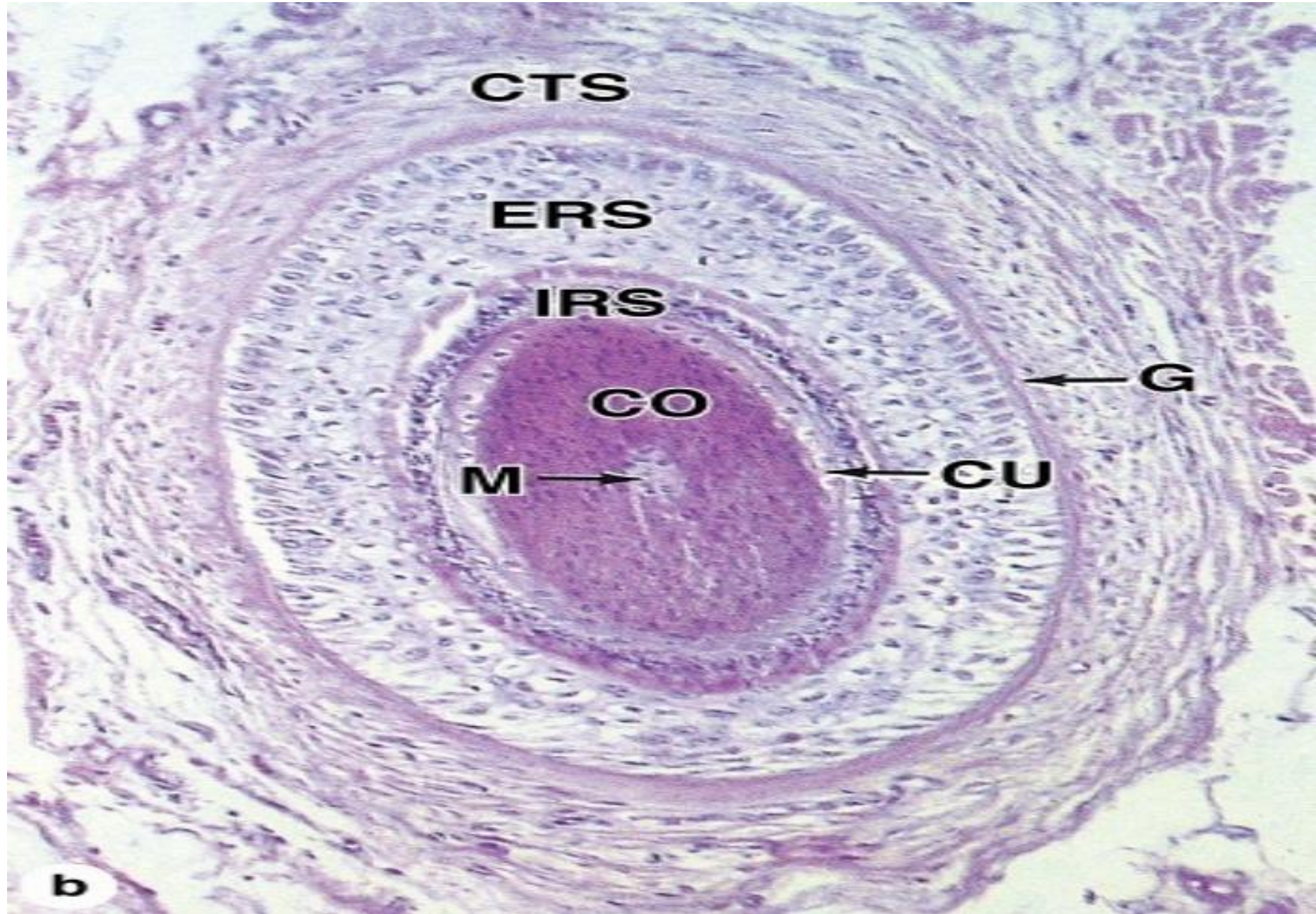
Hair follicle

- is formed of outer **collagenous root sheath** and **inner epithelial** (internal & external) **root sheaths**.
- **Internal (inner) epithelial root sheath:** is formed of two sub layers: inner Huxley's layer and outer Henle's layer.
 - **Huxley's layer:** the inner layer of cells of the root sheath of a hair.
 - **Henle's layer:** the outer layer of cells of the internal epithelial root sheath of a hair;
- **Outer (External) epithelial root sheath (EERS)** is downward continuation of the living layers of epidermis.
 - The outer most cells of the **EERS** rest on a thickened basal lamina known as **Glassy membrane**.

Notice the 3 layers of the hair shaft itself

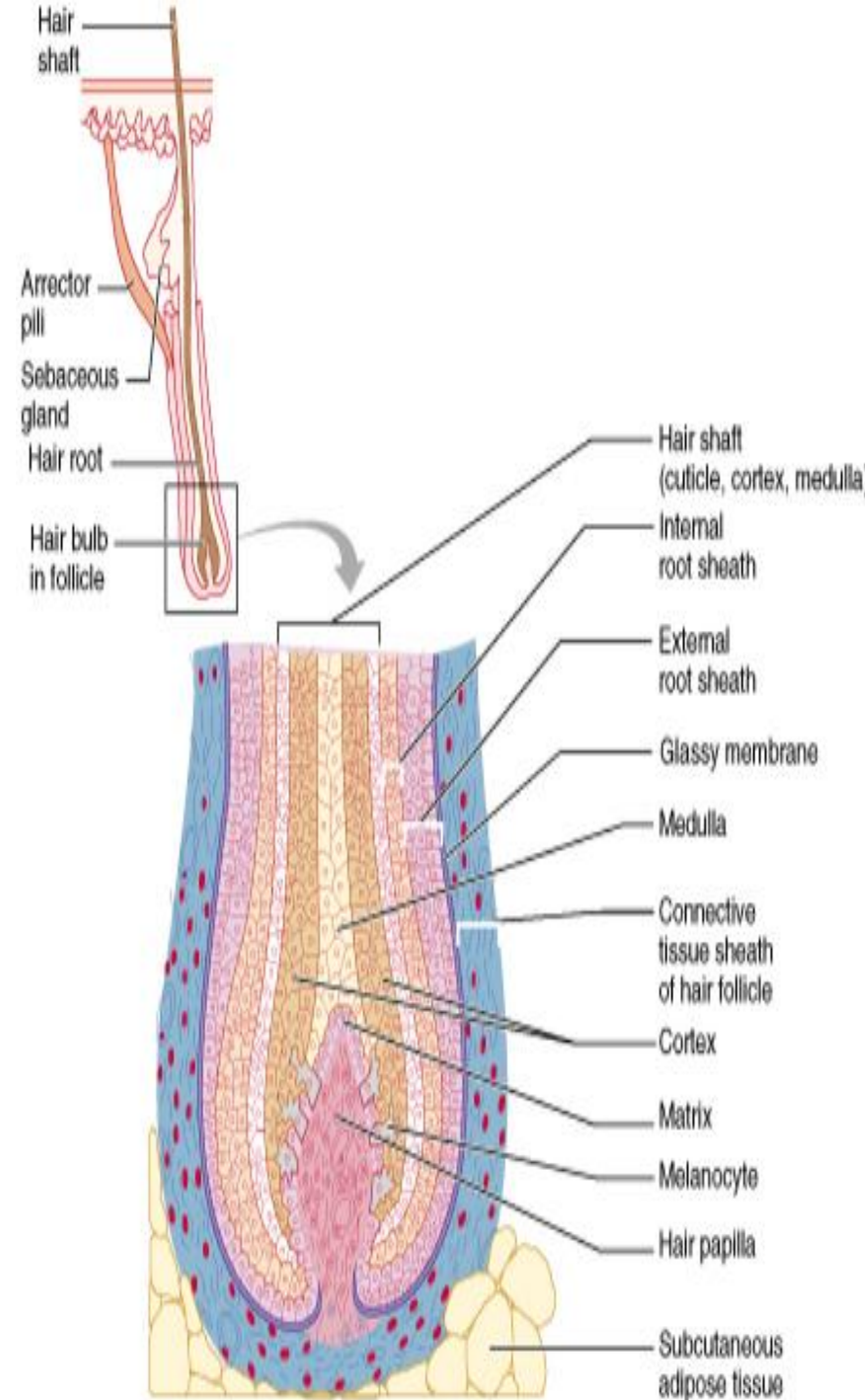


Oblique Section through Hair Root and its Follicle



Hair and Hair Follicles

- The hair follicle surrounds much of the hair root.
- It contains an outer CT sheath & inner epithelial root sheath.
- At the base of the hair follicle is a single layer of mitotic cells derived from the stratum basale. This is the hair matrix.
- All the cells of the hair are derived from the **hair matrix**.
- Just beneath the hair matrix is an obvious dermal papilla called the **hair papilla**. It contains the blood vessels that nourish the matrix and the cells of the hair follicle.



Skin, hairy H&E

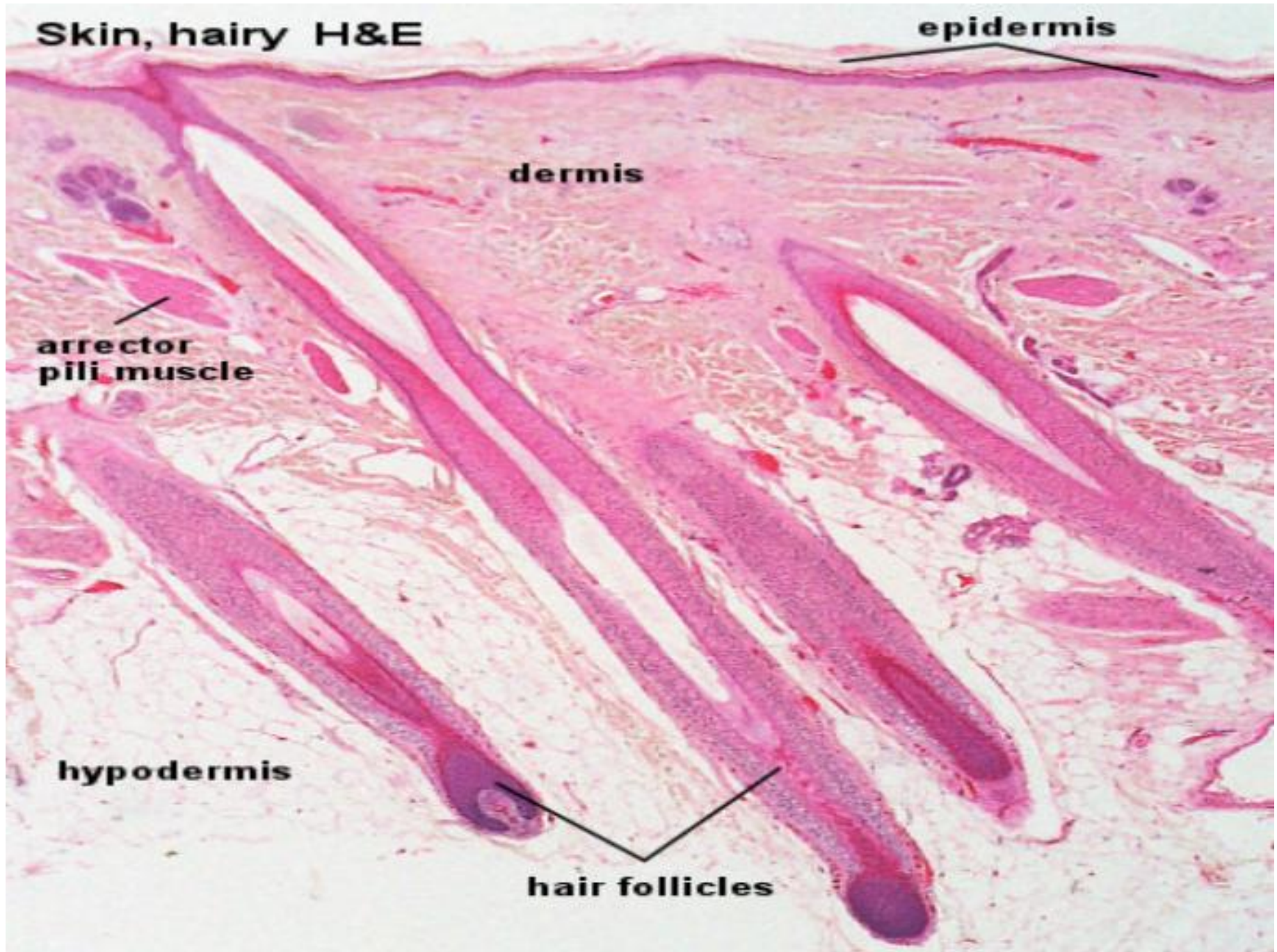
epidermis

dermis

**arrector
pili muscle**

hypodermis

hair follicles



Skin, hairy H&E

hair

dermal
sheath

inner and outer
root sheath

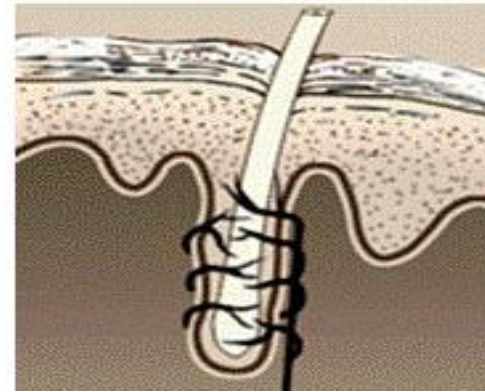
bulb

dermal papilla

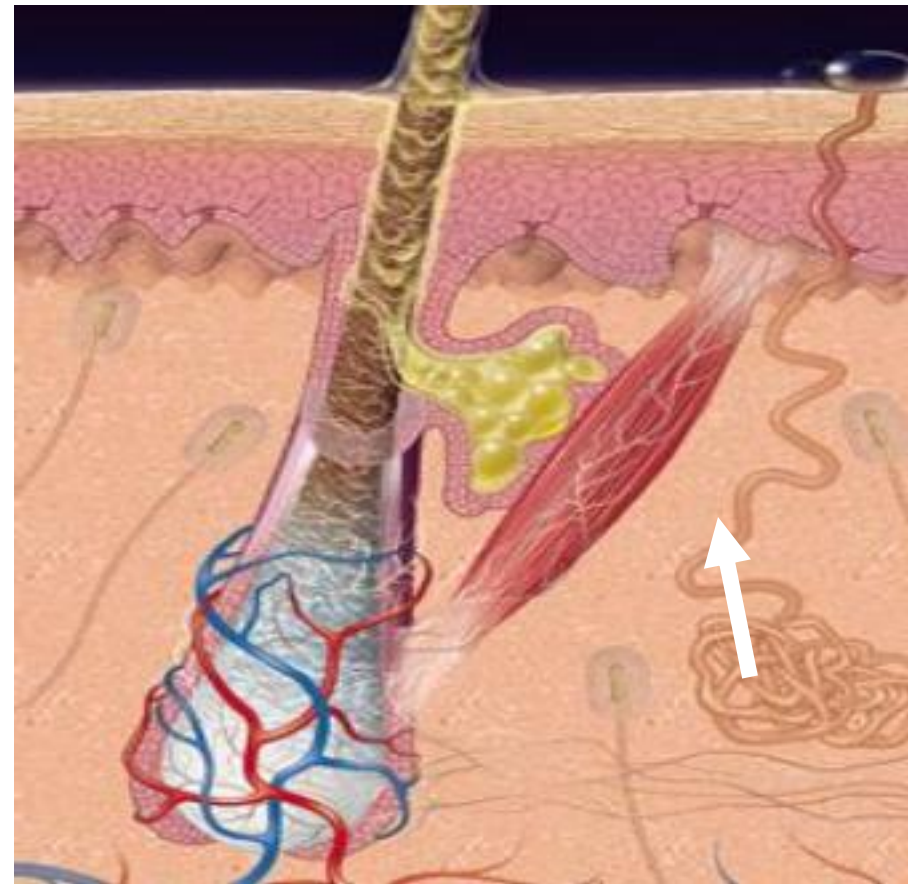


Hair and Hair Follicles

- Wrapped around the bulb of the follicle is a network of sensory nerve endings known as the **hair root plexus**. Allow the hairs to serve a sensory function.
- Attached to each hair is a small bundle of smooth muscle known as an **arrector pili muscle**.
- In times of fright or cold, these muscles contract and cause the hair to stand on end – and produces goose bumps.



*Hair
Receptors*



The arrow indicates an arrector pili muscle. In this picture, you should also try to identify the shaft, root, follicle, hair papilla, and sebaceous gland.



Types of hair

- Not all hair is alike, even on one person. Over the course of our lives, we grow three kinds of hair; **lanugo**, **vellus** and **terminal hair**.
- 1. **Lanugo** is fine, downy, unpigmented hair that appears on the fetus in the last 3 months of intrauterine development.
- 2. **Vellus** by the time of birth, most of lanugo hair is replaced by similar fine, unpigmented hair called **vellus**.
 - Except for the **eyebrows**, **eyelashes**, and **hair of the scalp**, all of the hair of children, **two-thirds of the hair of women** and **one-tenth of the hair** of **men** is **vellus**.
- 3. **Terminal hair** is longer, coarser, and pigmented. It occurs on the scalp, eyebrow, and eyelashes; at puberty it replaces the vellus in the axillary and pubic regions, on the face of males (to form the beard), and to varying degrees on the trunk & limbs.

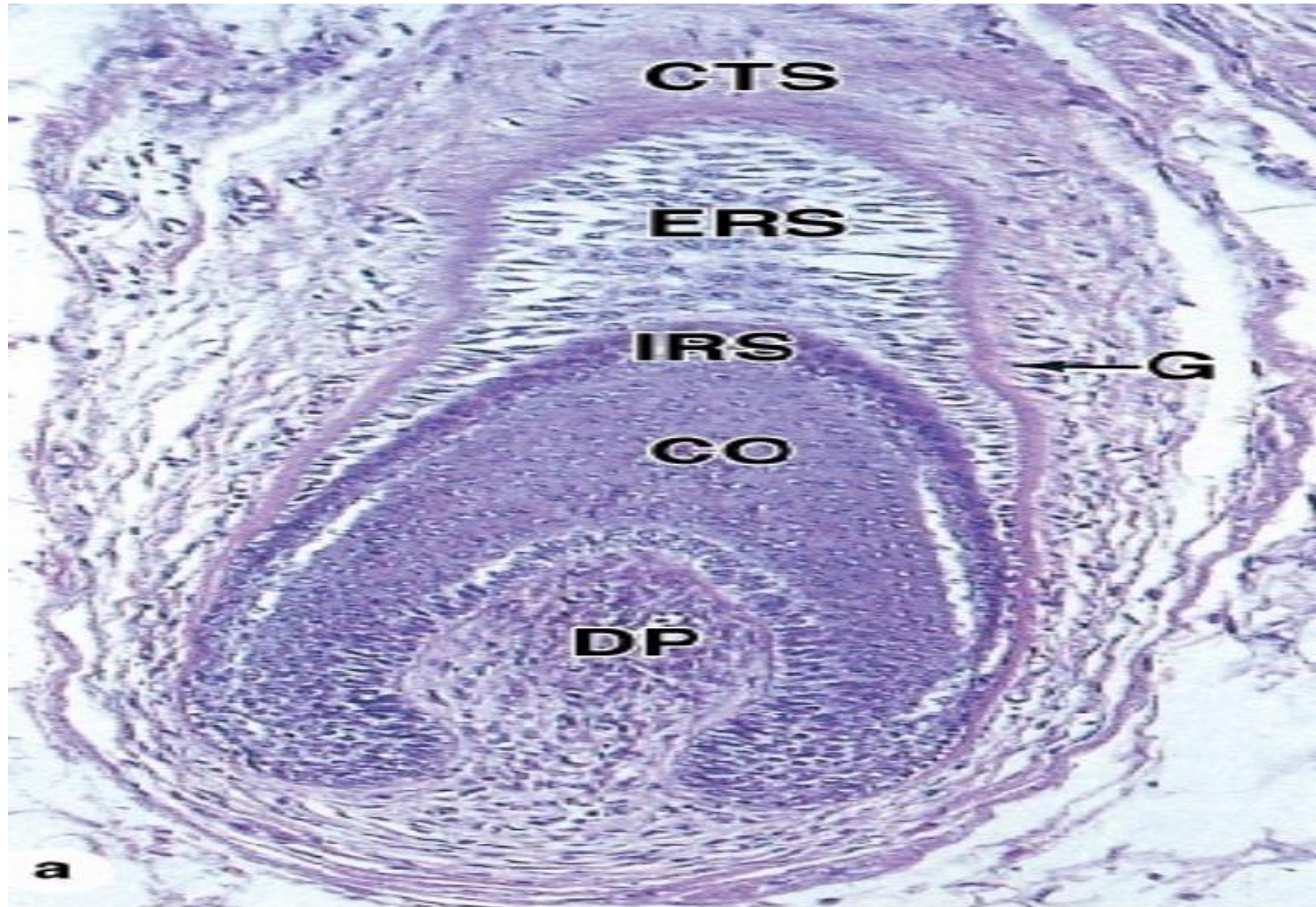
HAIR GROWTH CYCLE

- Hair growth is not continuous process.
- Each hair follicle lives in a cycle consisting of a long period of growth followed by a relatively short period of rest.
- There is a genetic predisposition of each hair follicle to follow a particular pattern of growth and rest.
- Hair follicles are programmed to stop producing hair & spend more time in the resting stage with the progression of age.
- The growth cycle of a hair consists of 3 stages:

1. ANAGEN (GROWING) STAGE

- is the growing period of a hair follicle.
- for the hair follicles in the scalp typically lasts about 3 to 5 years.
- Hair matrix cells at base of hair root produce new cell population which elongate the hair.

Growing Hair



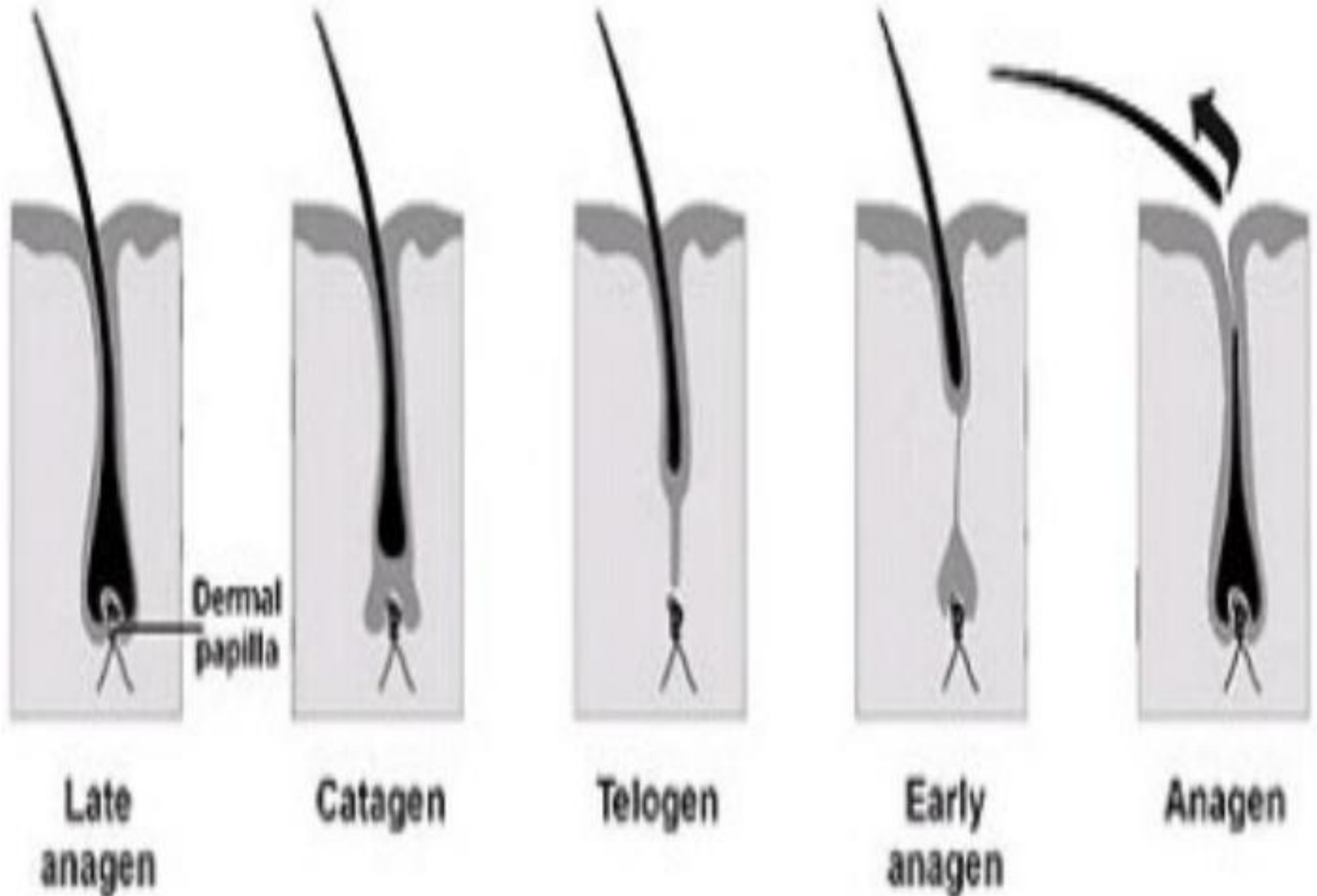
2. CATAGEN (INTERMEDIATE) STAGE

- At the end of the growth period, hair follicles prepare themselves for the resting phase.
- This transition period of a hair follicle from growth to rest is called the catagen stage.
- usually lasts about 1 to 2 weeks or so.
- During the catagen phase the deeper portions of the hair follicles start to collapse.

3. TELOGEN (RESTING or SHEDDING) STAGE

- This is the resting period of a hair follicle.
- During the resting period the hair is still attached to the hair follicle, but it is not growing.
- After the resting phase the hair is shed and a newer hair begins to grow, thus, starting a new hair growth cycle.
- It is usually 3 to 4 months and at the end of this period older hairs that have finished their life will fall out (shedding) and newer hairs will begin to grow.

Hair growth cycle



- The growing (anagen) phase constitutes about 90% (1000 days or more) of the growth cycle of a hair follicle,
- While intermediate (catagen phase; 10 days) and shedding (telogen phase; 100 days) phases constitute only 10% of it.
- That is, at a given time, about 10% of hair follicles are in the intermediate and shedding phases; thus, not growing.
- These hair follicles, however, are randomly distributed over the scalp, so that no bald spots are seen.

NAILS

- made up of tightly packed and highly compressed **nail plate** of dead squamous epithelial cells filled with hard keratin.
- cover and protect the dorsal surface of each distal phalanx
- Nail consists of **nail root** (proximally), **body** and **free edge** (most distally).

The Nail Root

- is embedded in the **proximal skin fold** which is thin and lacks both hair and glands.
 - The epidermal stratum corneum extending from the **proximal nail fold** forms the upper **cuticle**, **eponychium**.
 - **Cuticle** is a fold of stratum corneum on the proximal and under the distal ends of nail body.
- Internally, surrounded **dorsal and ventral nail matrix**.
- Nail growth occurs in the **nail matrix** and **lunula**

Structure of Nail



The free edge of a nail is the portion of the nail plate beyond hyponychium & appears grey.

Nail Body

- Part of a nail beyond the eponychium
- Made up of nail plate which is bound to a bed of epidermis called the **nail bed**, which contains only the basal and spinous layers.
- Pink visible portion due to rich capillaries (nail capillary) in the underlying dermis. ➡ cue in patients degree of

illness

Nail root

- buried under skin layers
- **lunula** is crescent whitish part of root due to thickened stratum basale.

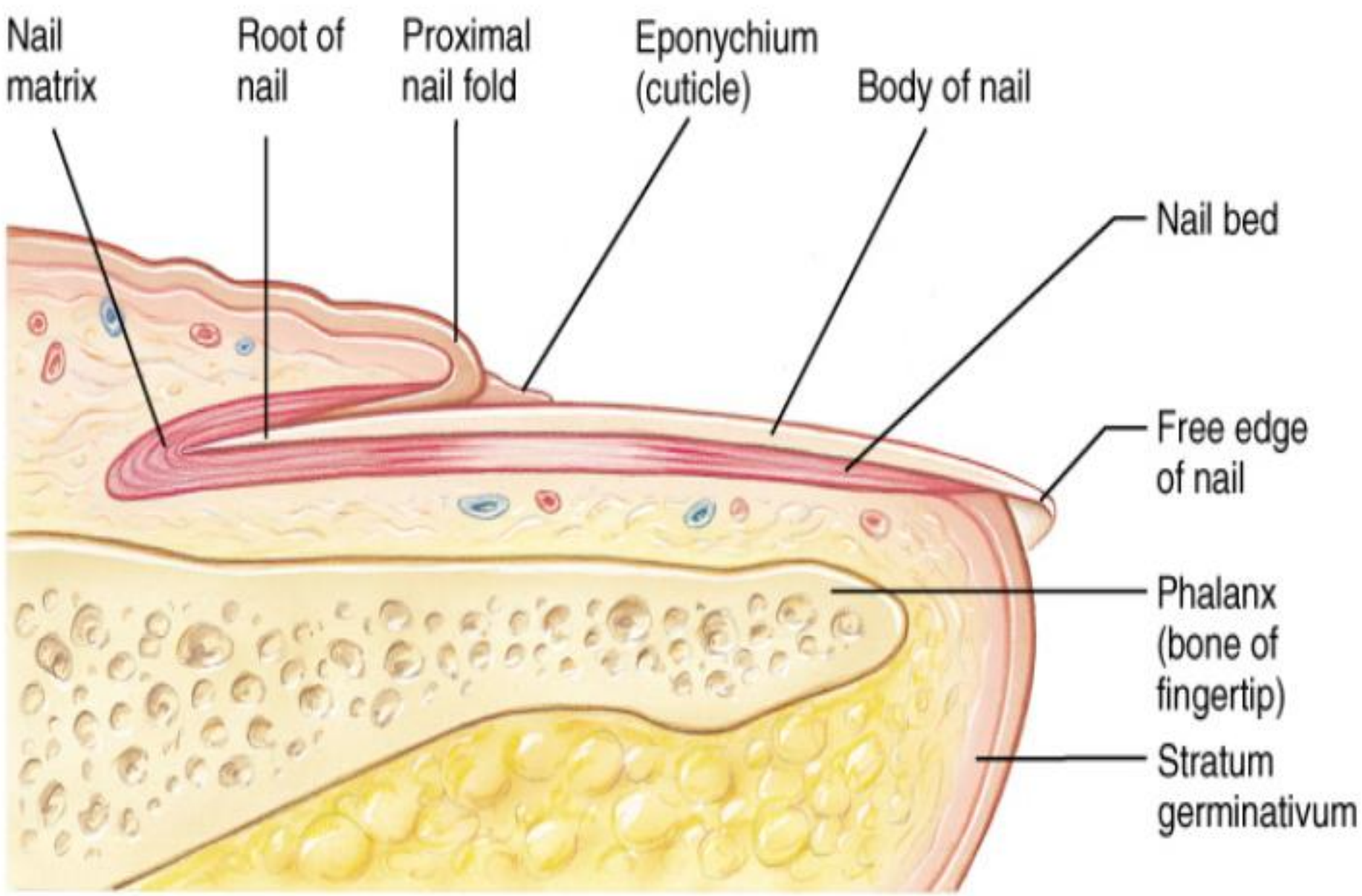
Eponychium: cuticular extension of PNF derived from stratum corneum.

Hyponychium– a thickened epidermal layer, that secures the undersurface of the nail plate

NAILS

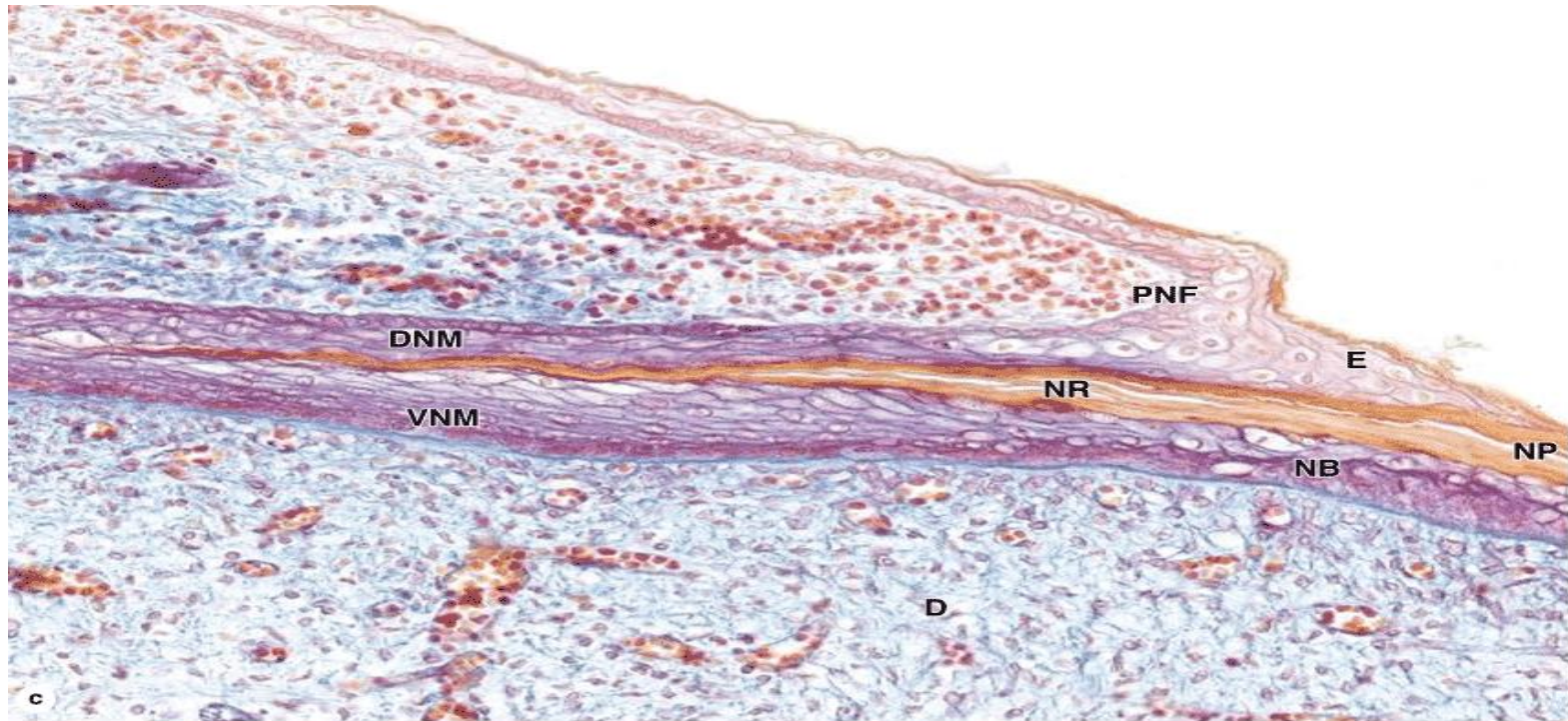
- The keratinized **nail plate** is bound to a bed of epidermis called the **nail bed**, which contains only the basal and spinous layers.
- The nail plate arises from the **nail matrix**, which extends from the nail root.
- Cells of the matrix divide, move distally, and become keratinized, forming the nail plate.
- This matures as the nail plate, which continuous growth in the matrix pushes forward over the nail bed (which makes no contribution to the plate) at about 3 mm/month for fingernails and 1 mm/month for toenails.
- The distal end of the nail plate becomes free of the nail bed at the epidermal fold called the **hyponychium** and is worn away or cut off.
- The nearly transparent nail plate and the thin epithelium of the nail bed provide a useful window on the amount of oxygen in the blood by showing the color of blood in the dermal vessels.

- **Nail bed:** skin overlaid by nail plate
- Nail growth originates from **nail root** in **nail matrix**; cells become keratinized
- **Cuticle:** part of proximal nail fold attached to the upper surface of the nail plate



(b)
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micrograph of a sagittal section of a fetal finger

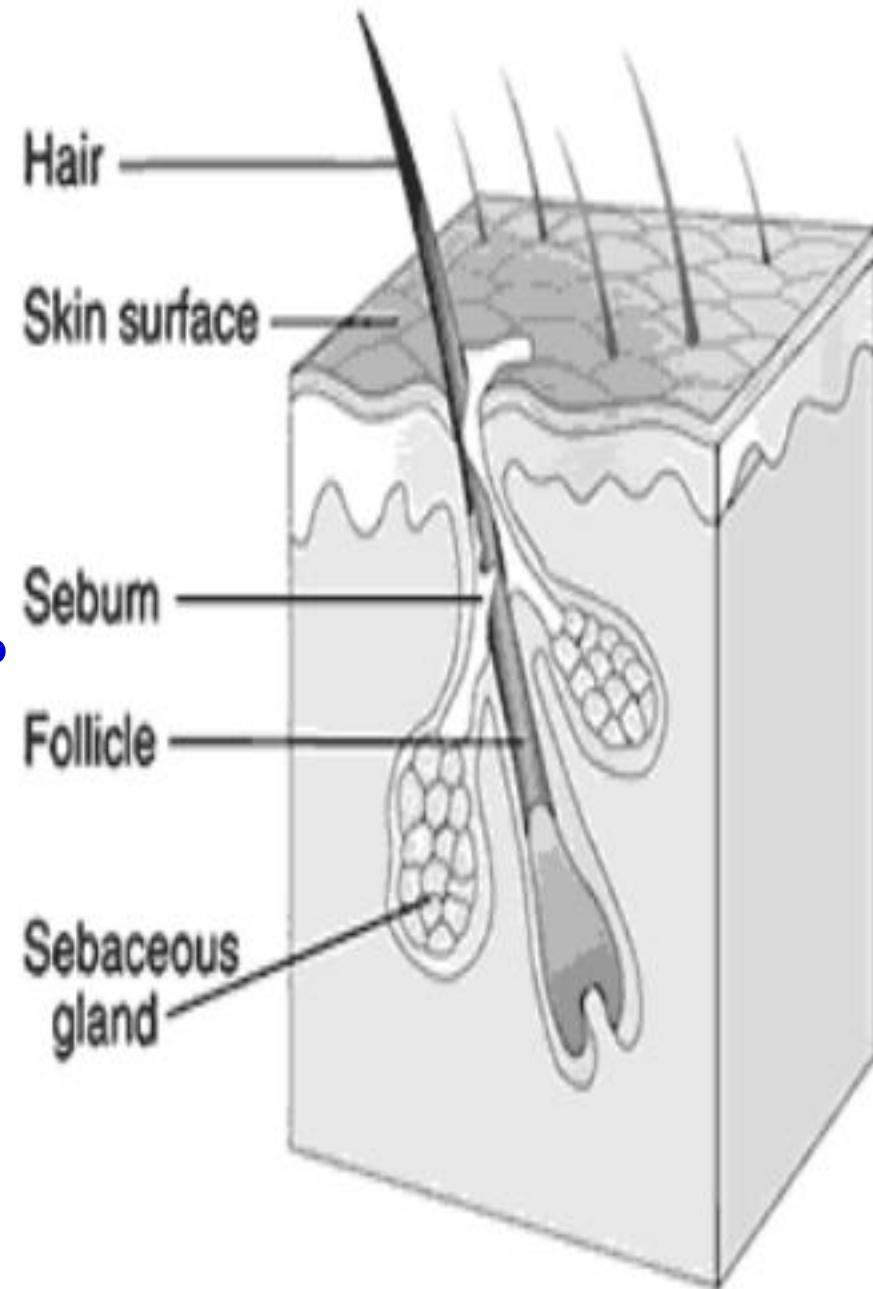


- Note the proximal nail fold (PNF) and its distal continuation, eponychium (E) or cuticle.
- The nail root (NR), the most proximal region of the nail plate (NP), is formed like the hair root by a matrix of proliferating, differentiating keratinocytes. These cells make up the dorsal nail matrix (DNM) and ventral nail matrix (VNM), which contribute keratinized cells to the nail root.
- The mature nail plate remains attached to the nail bed (NB), which consists of basal and spinous epidermal layers over dermis (D), but is pushed forward on this bed by continuous growth in the nail matrix.

Sebaceous Glands

- Simple branched alveolar glands found everywhere except palms of the hands and soles of the feet.
- On average of about 100 glands exist per square centimeter of skin, but the density increases to 400–900/cm² in the face and scalp.
- Do the ducts of these glands branch?
- Are these glands exo- or endocrine?
- Secrete an oily, lipid-rich secretion called **sebum**.
- Sebum is typically secreted by **sebocytes** which undergo autolysis into a hair follicle or occasionally onto the body surface.
- Sebum softens and lubricates the skin. It is also quite bactericidal.

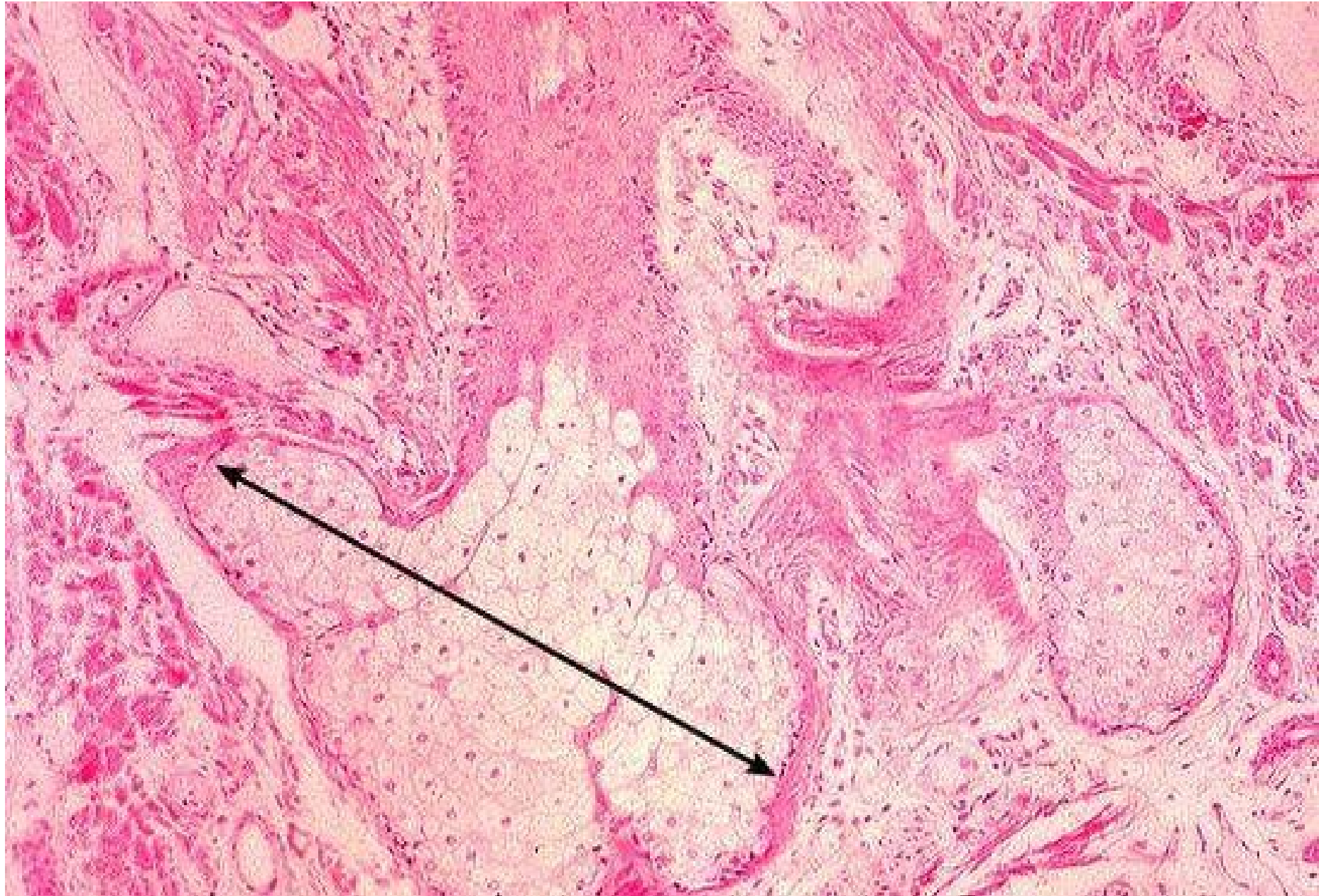
Skin glands



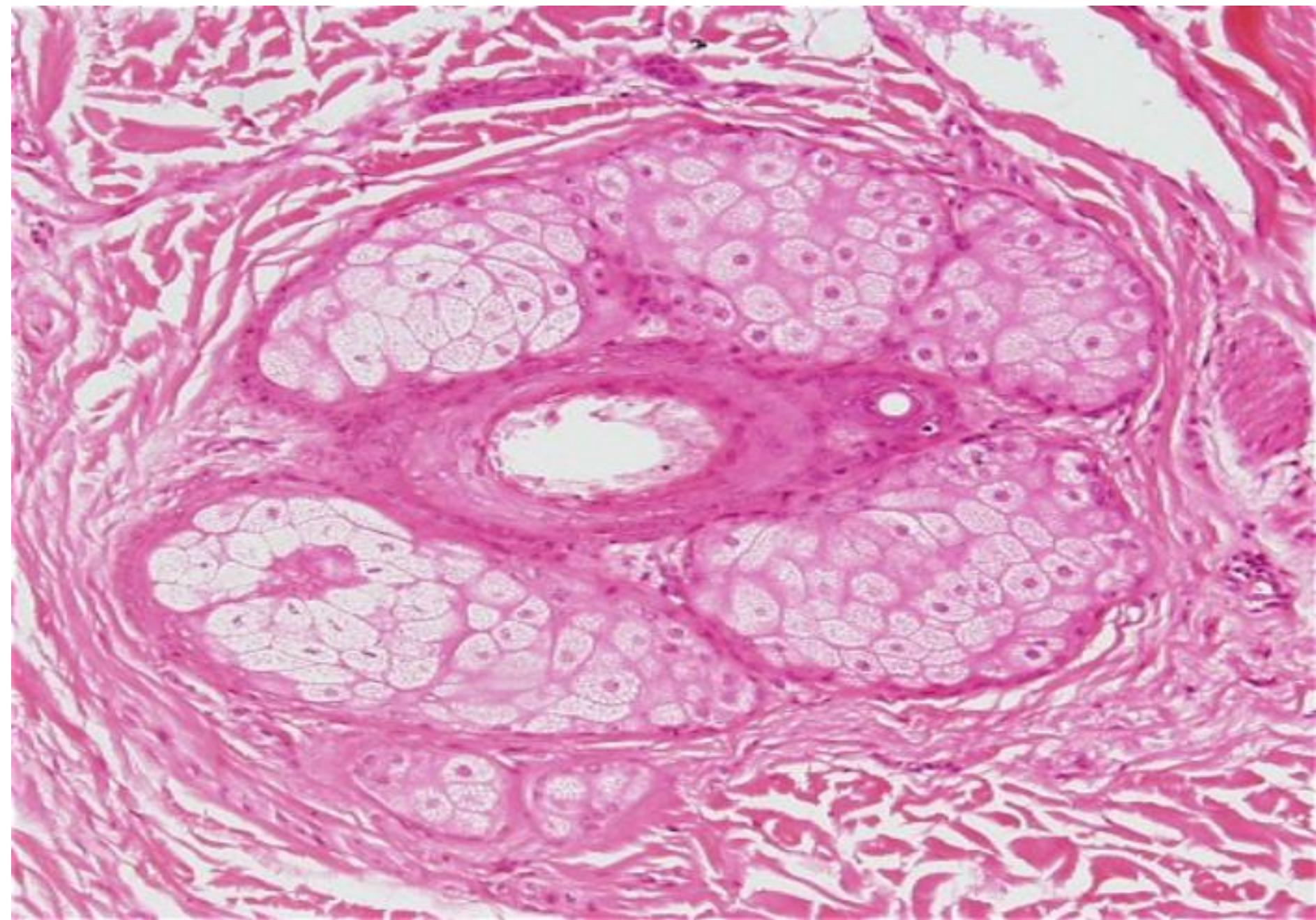
The sebaceous gland is indicated by the arrow. Note how its duct is unbranched and how it empties into a hair follicle.



Sebaceous gland



Pilosebaceous unit



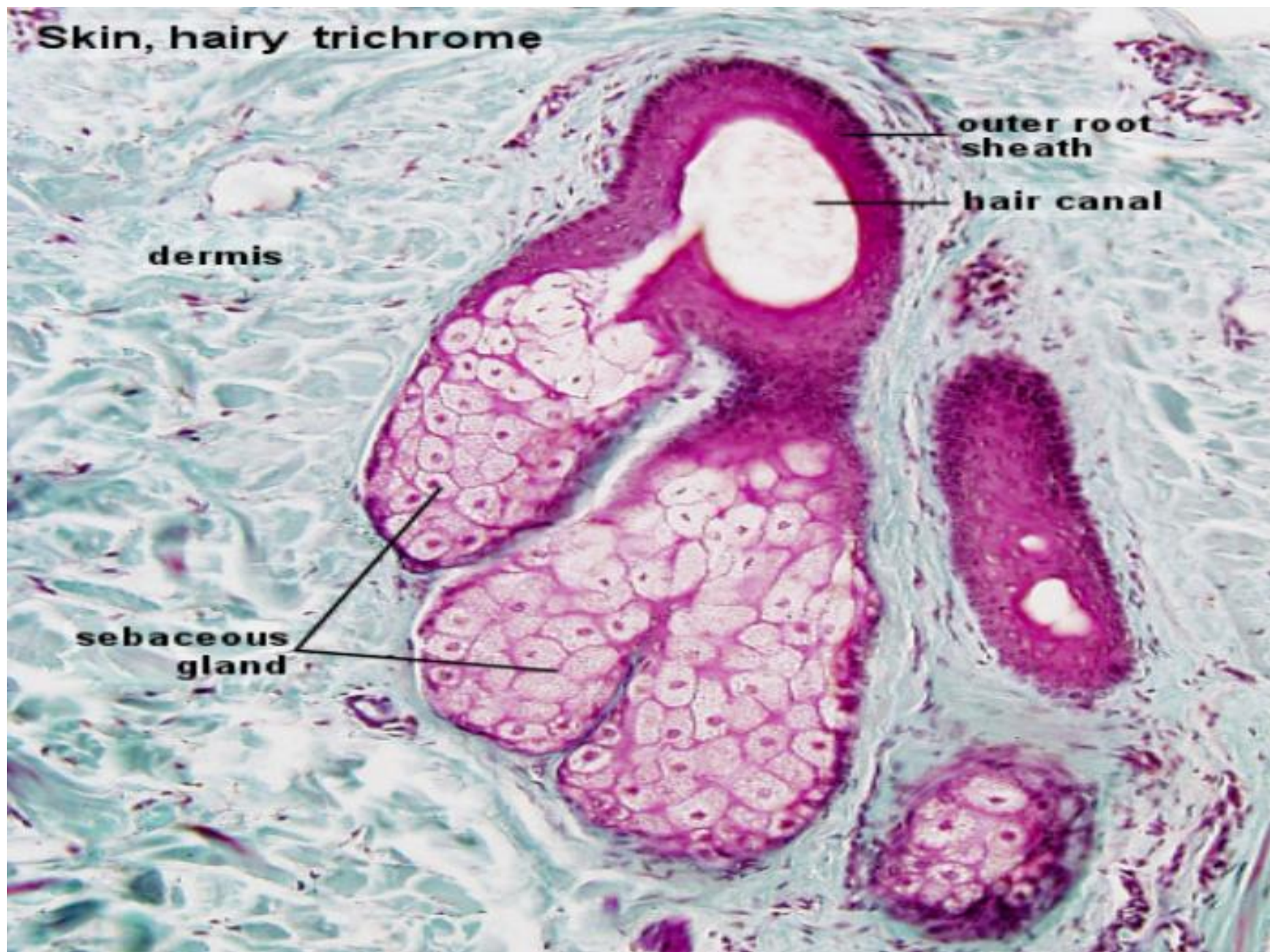
Skin, hairy trichrome

dermis

outer root sheath

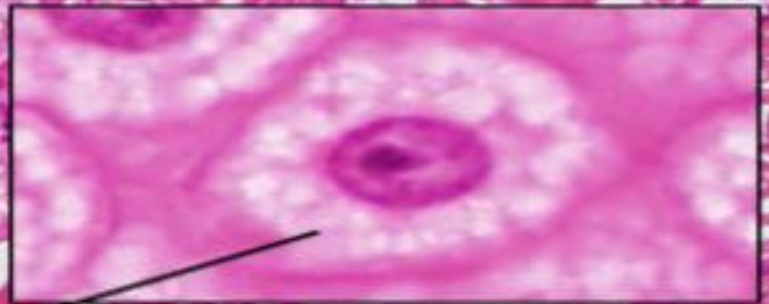
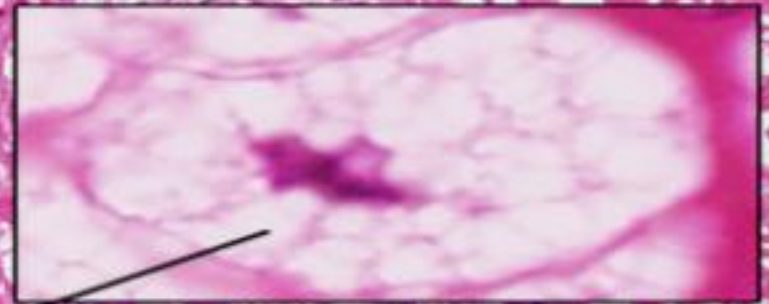
hair canal

sebaceous gland



Skin, hairy H&E

**disintegrating sebum-containing
secretory cell**



**intact sebum-containing
secretory cell**

MODIFIED SEBACEOUS GLANDS

- are sebaceous glands not associated with hair roots

1. Areolar glands (of Montgomery)

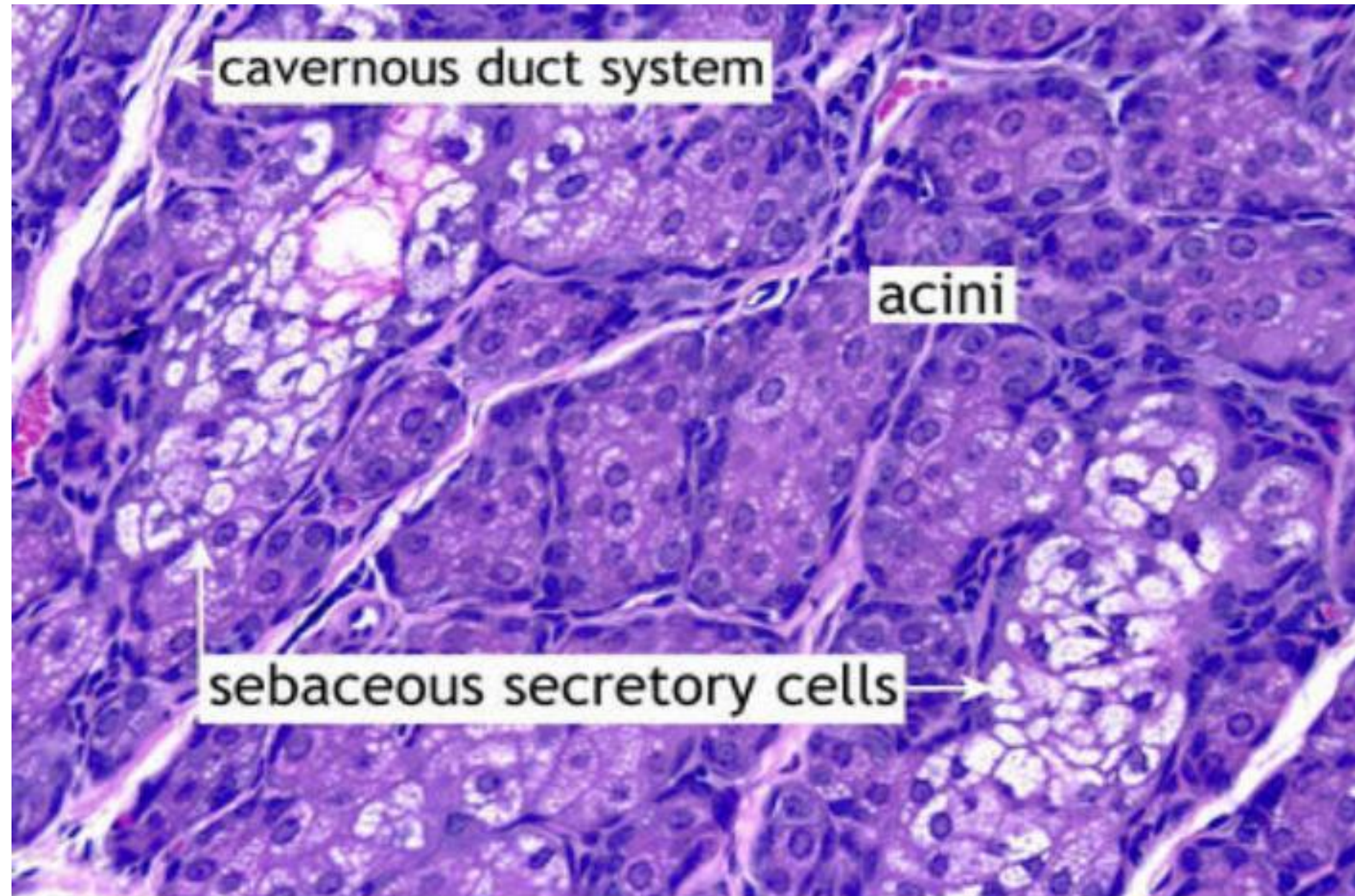
- are modified sebaceous glands in the skin covering the nipple (the areola)
- The oily secretion lubricates the nipple & assist the the infant during suckling.

2. Meibomian glands/ Tarsal glands of eyelid

3. Glands of Tyson

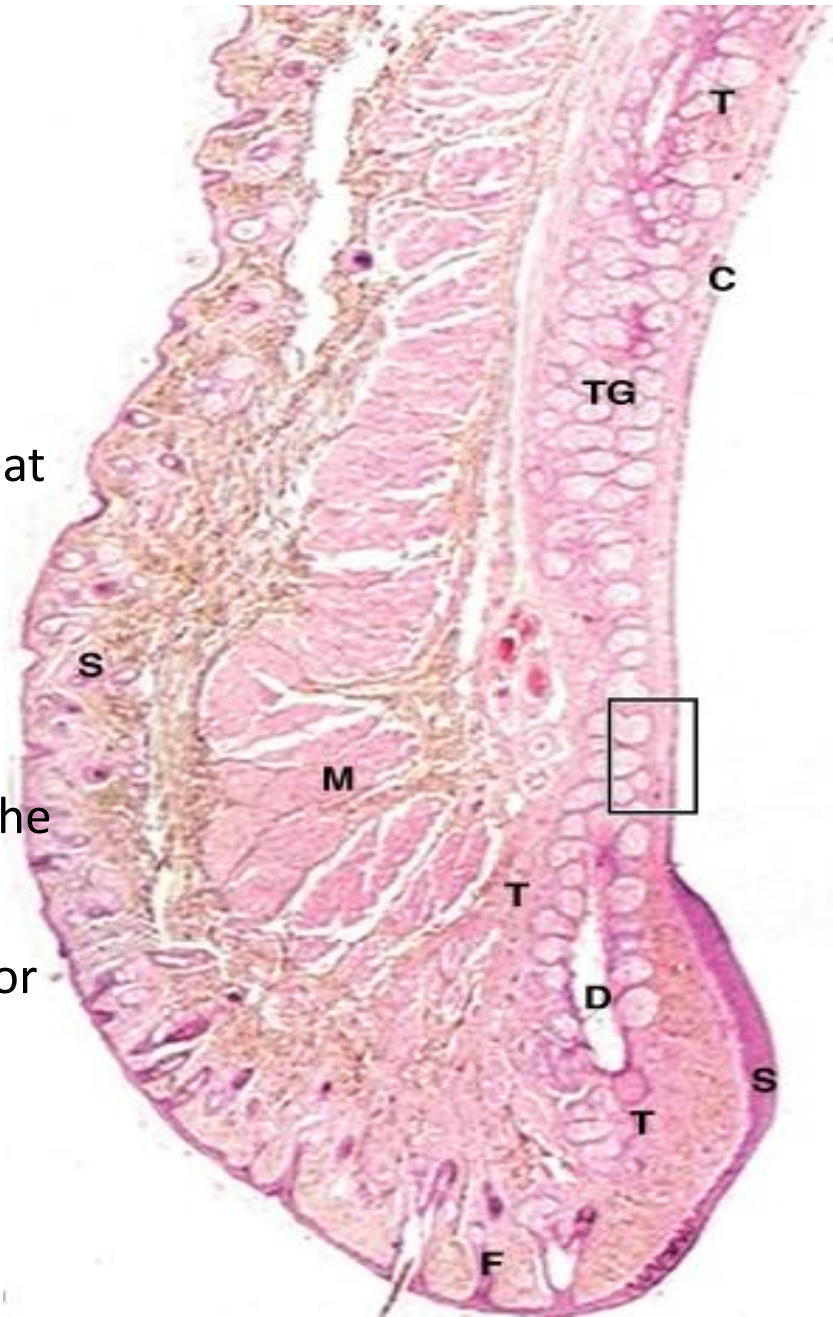
- Are preputial glands found in prepuce (a retractile fold of skin that contains connective tissue with smooth muscle in its interior).
- present in the internal fold and in the skin that covers the glans penis or clitoris; in clitoris, they are called **clitoral glands**.

Human clitoral gland

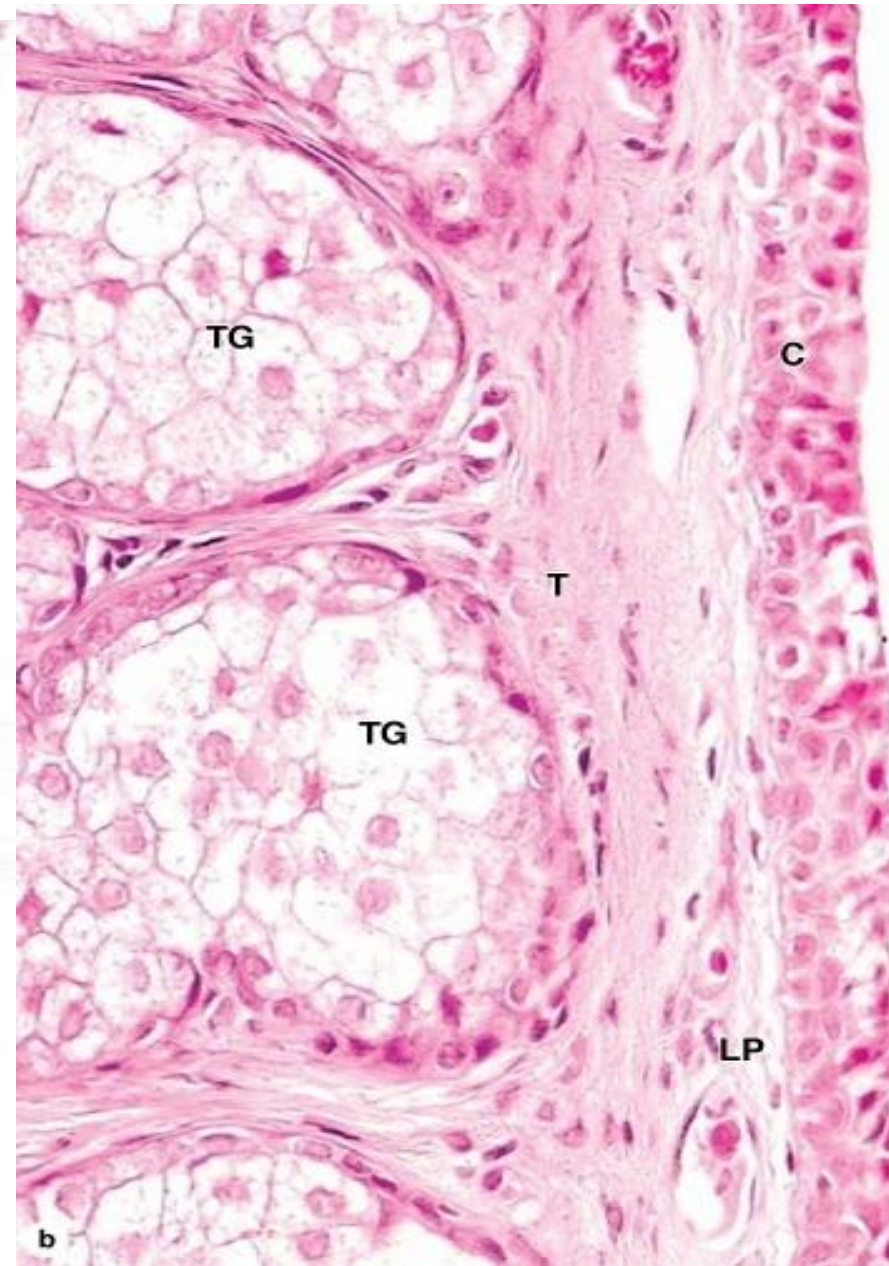
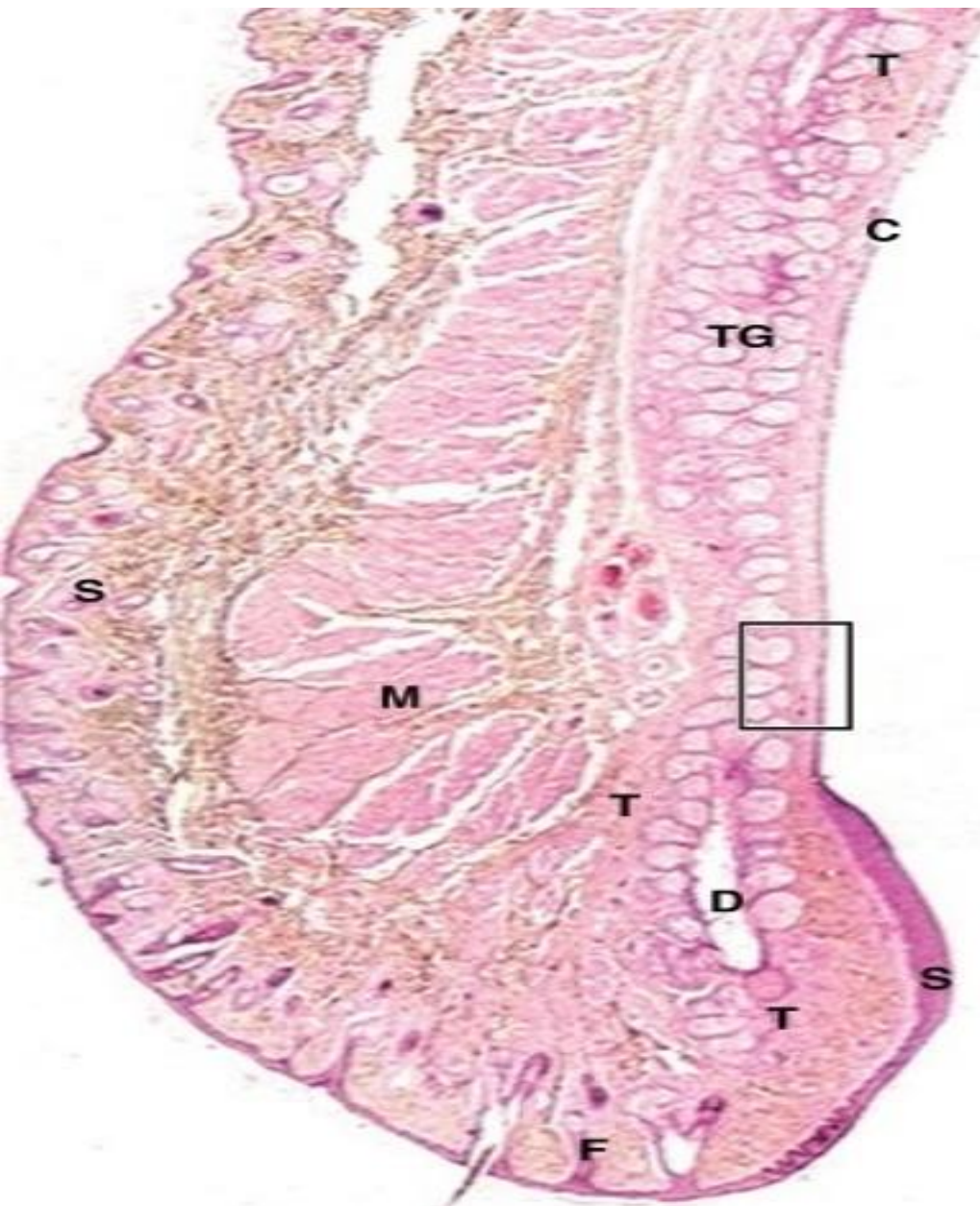


Sagittal section of upper eyelid

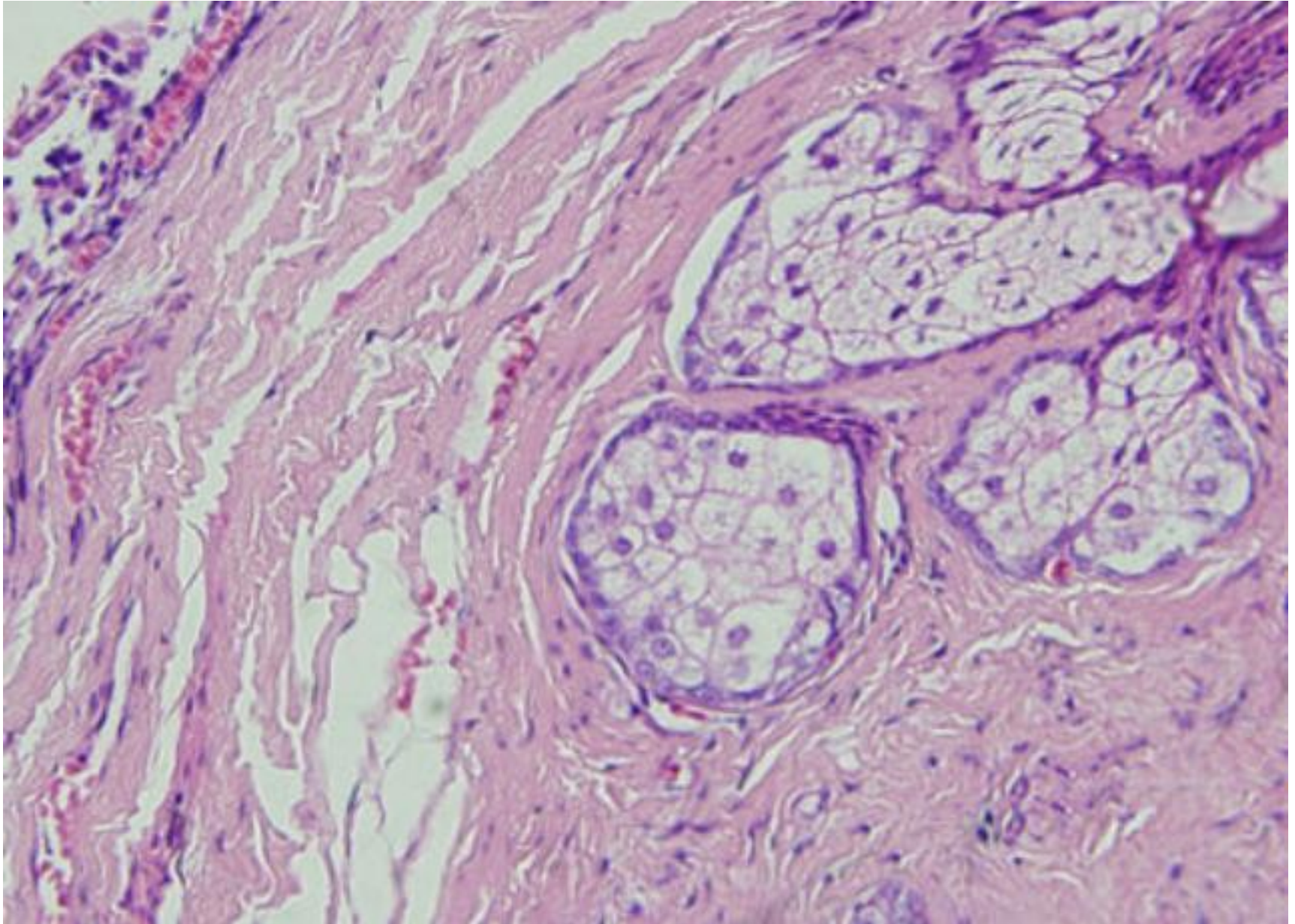
- The eyelid is a pliable tissue with thin skin (S) covering its external surface and smooth conjunctiva (C) lining its inner surface.
- At the outer rim of the eyelid are a series of large hair follicles (F) for the eyelashes.
- Associated with these hair follicles are small sebaceous glands and modified apocrine sweat glands.
- Internally eyelids contain fascicles of striated muscle (M) comprising the orbicularis oculi muscle and closer to the conjunctiva a thick plate of fibroelastic connective tissue called the tarsus (T).
- This tarsal plate provides structural support for the eyelid and surrounds a series of large sebaceous glands, the tarsal glands (TG) (aka Meibomian glands), with acini secreting into long central ducts (D) that empty at the free edge of the eyelids.



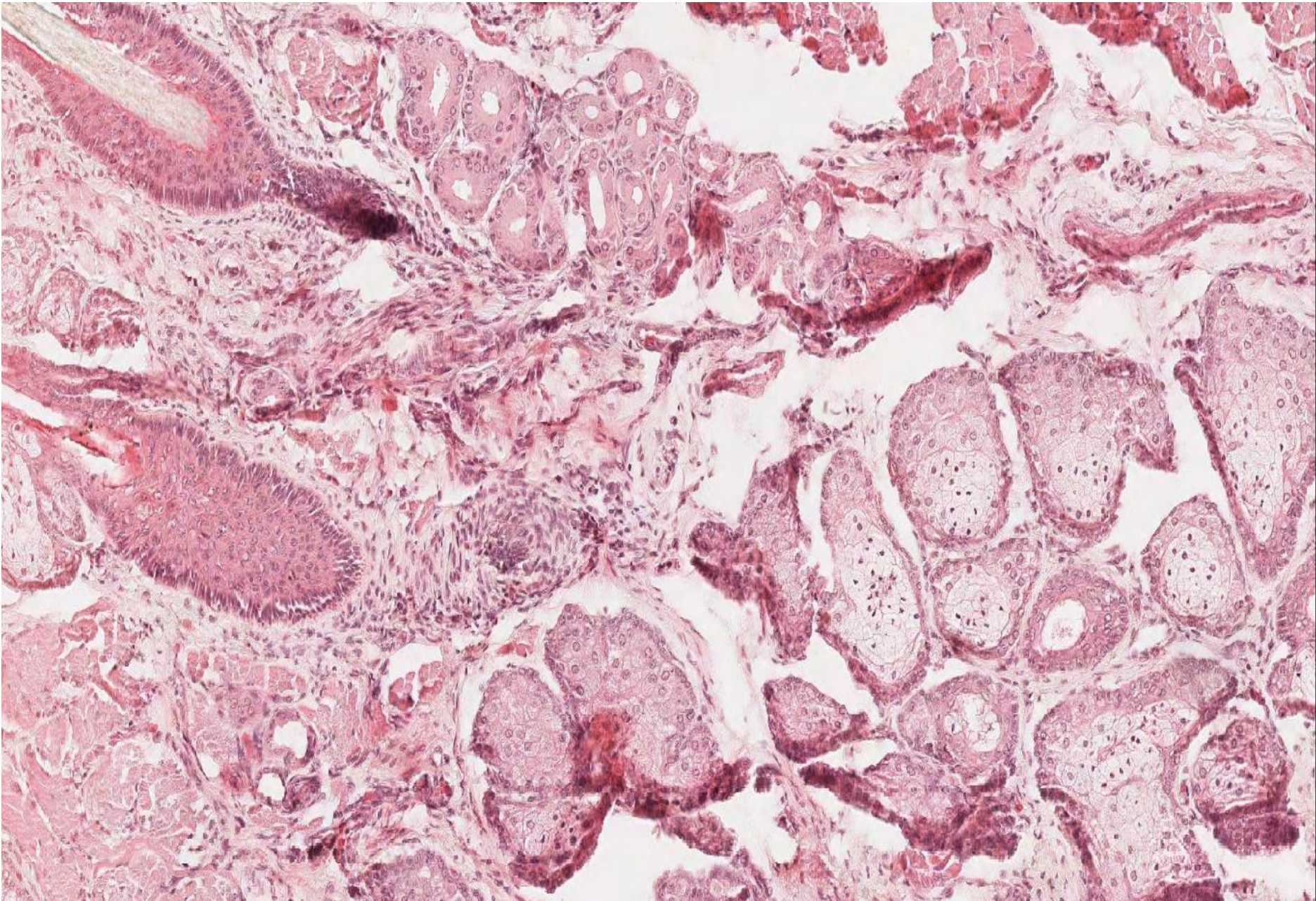
EYE LID and Tarsal glands in the tarsus



Human tarsal plate

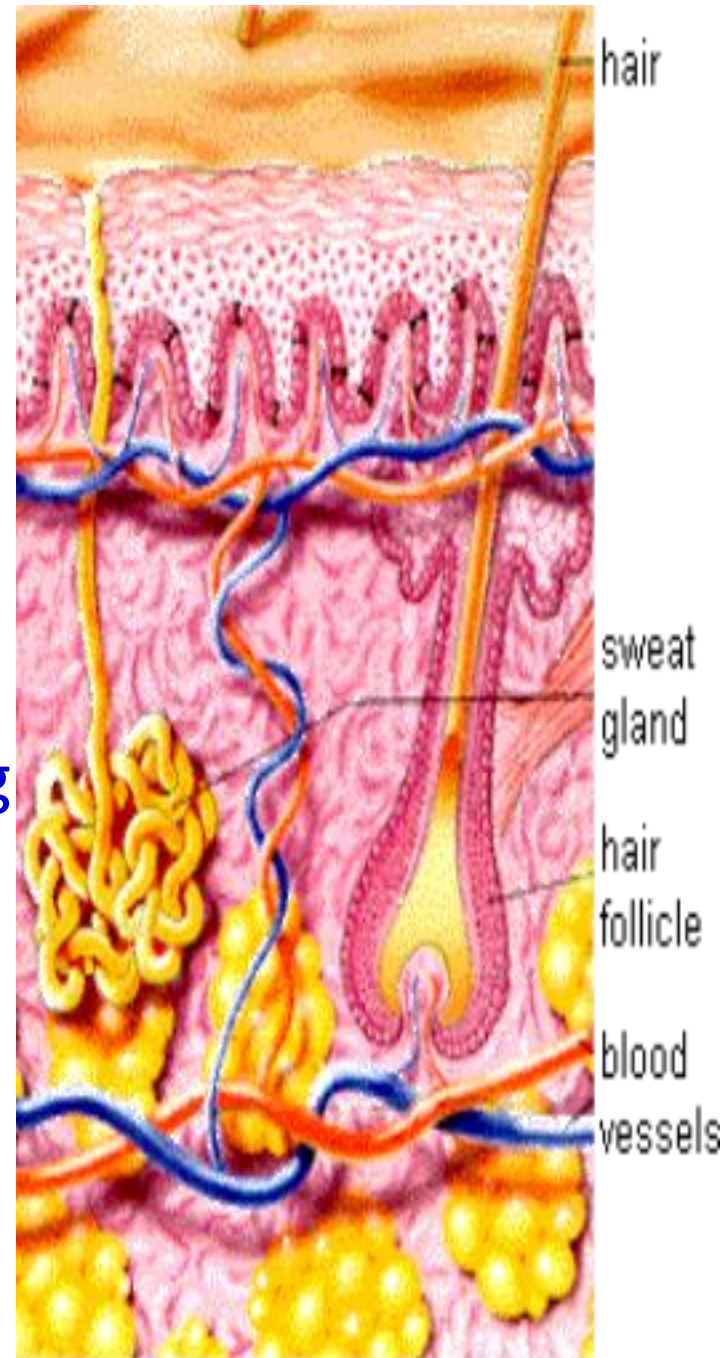


Eye lid: Tarsal (Meibomian) glands, Glands of Moll, & Glands of Zeiss



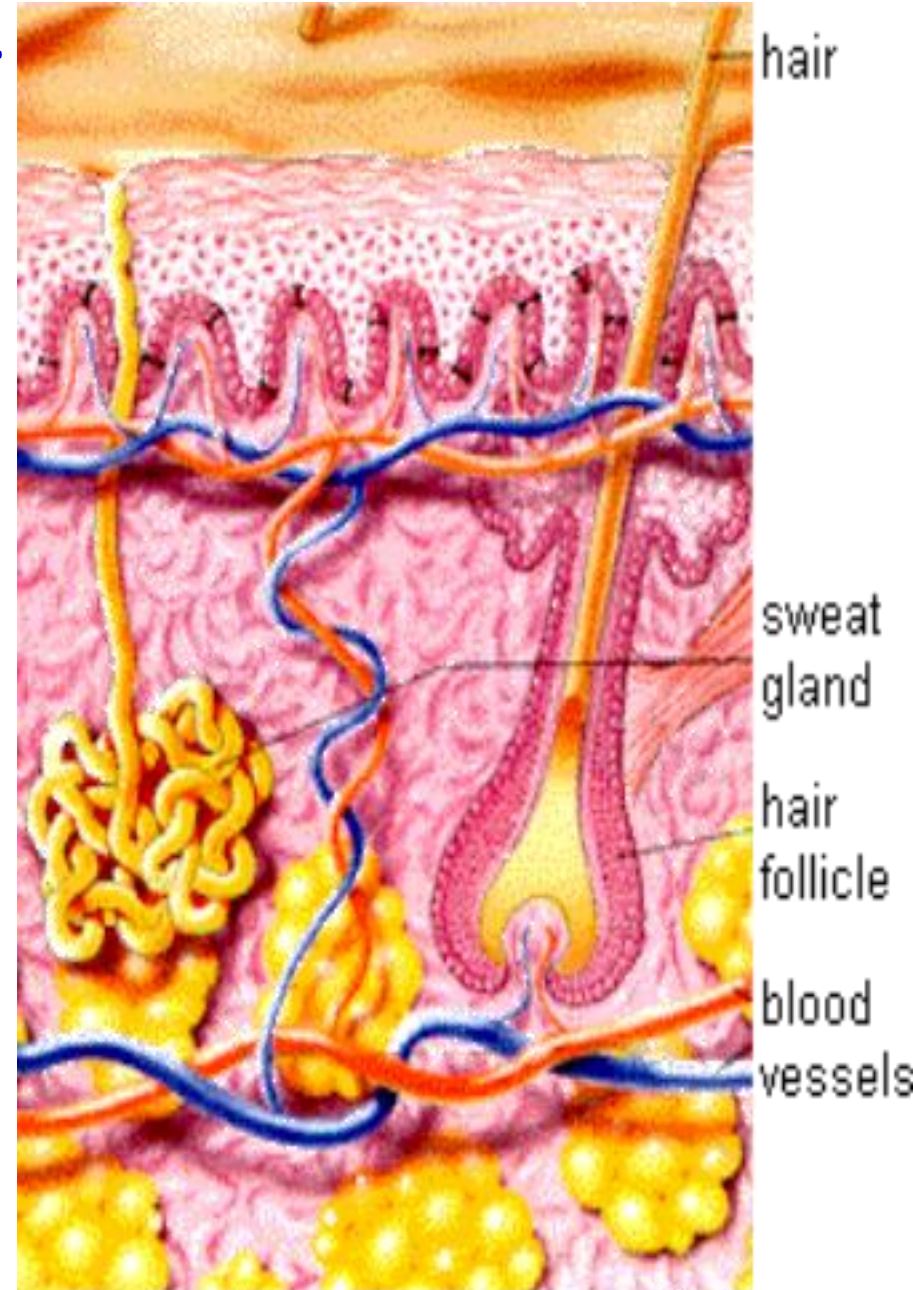
Sweat Glands

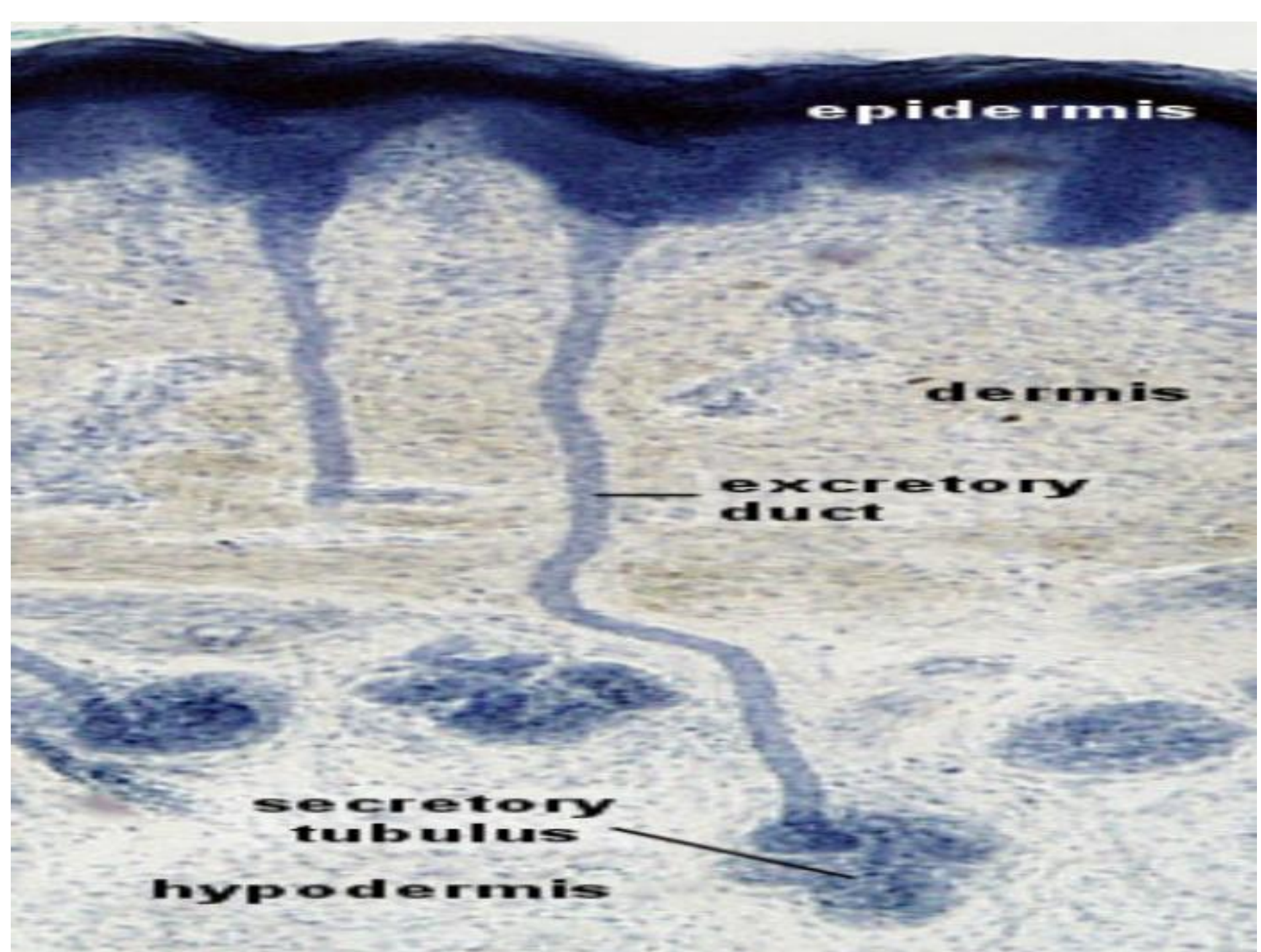
- Simple, coiled, tubular glands.
- Distributed over most of the body
- Over 2.5 million per person.
- Collectively the 2.5-3.0 million sweat glands of the average person roughly equal the mass of a kidney and can produce as much as 10 L/day, a secretory rate far exceeding that of other exocrine glands.
- Are of Two types with different distributions, functions, and structural details:
 1. Merocrine sweat glands
 2. Apocrine sweat glands



Sudoriferous/Eccrine/Merocrine Sweat Glands

- Simple, coiled, tubular glands.
- Distributed over the entire body except the nipples and portions of the external genitalia.
- are widely distributed in the skin & are more numerous than apocrine sweat glands.
- Especially prominent on the palms, soles, and forehead.
- and are most numerous on the soles of the feet (620/cm²).





Skin, thick H&E

dermis

vein

hypodermis

**merocrine
sweat gland**



Skin, thick H&E

excretory duct

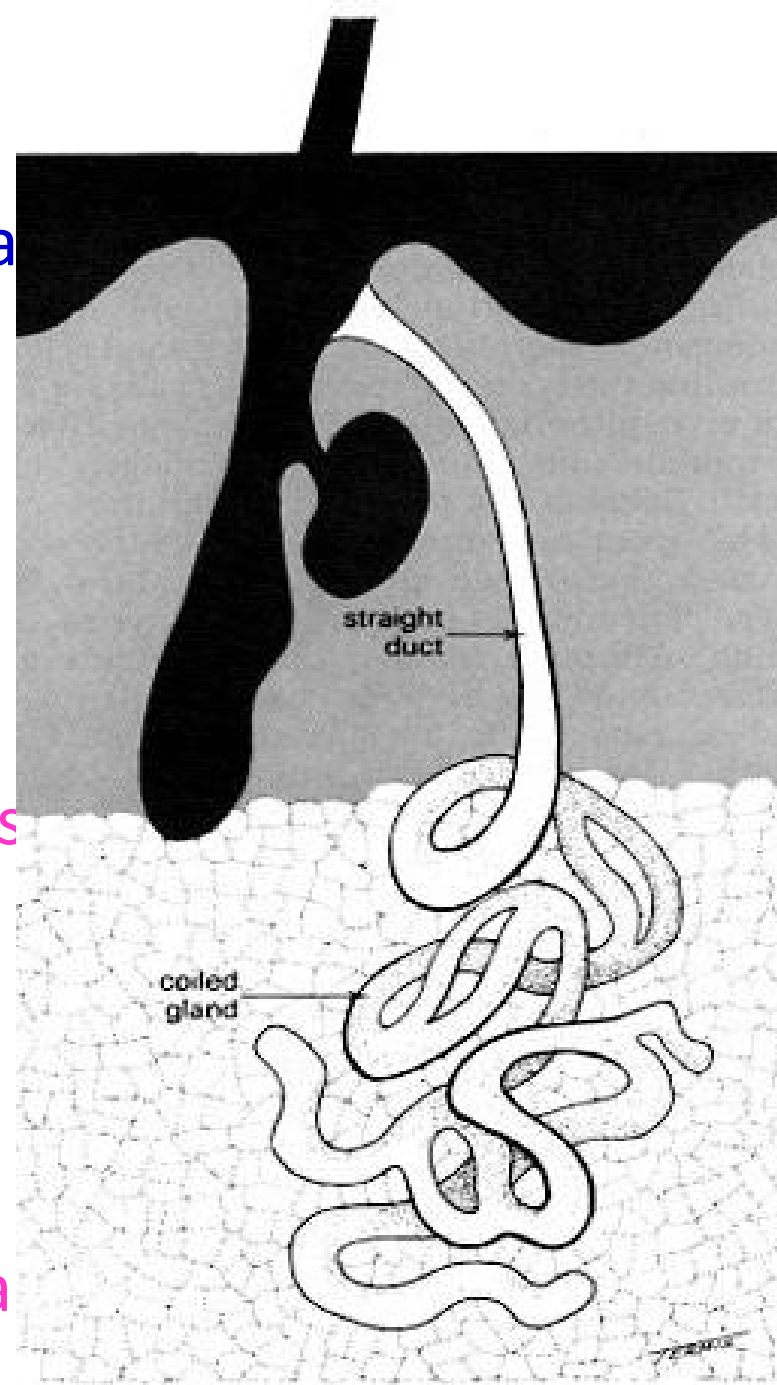
**secretory
tubulus**



- Both the secretory portions & ducts of merocrine sweat glands are coiled & have small lumens lined with stratified cuboidal epithelium.
- **The secretory part is generally more pale-staining than the ducts and has stratified cuboidal epithelium consisting of three cell types:**
 - 1. Pale pyramidal or columnar clear cells**
 - produce the sweat, having abundant mitochondria and microvilli to provide large surface areas.
 - Interstitial fluid from the capillary-rich dermis around the gland is transported through the clear cells, either directly into the lumen or into intercellular canaliculi that open to the lumen.
 - 2. As numerous as the clear cells are dark pyramidal cells**
 - line most of the luminal surface and do not touch the basal lamina
 - Dark cells are mucoid and filled with glycoprotein-containing granules whose functions are not well-understood but include components of innate immunity with bactericidal activity.
 - 3. Myoepithelial cells: rest** on the basal lamina and produce contractions that help discharge secretion into the excretory duct.

Apocrine Sweat Glands

- Found primarily in the nipple, axillary, pubic, and anal (circumana glands) regions of the body.
- Also found in the facial region in men only.
- Larger in size and less coiled than merocrine sweat glands.
- Ducts may empty into hair follicles
- Apocrine sweat is thicker than merocrine sweat since it contains more lipids and proteins.
- When it's degraded by epidermal bacteria, foul odors (malodour) can be produced.



Nipple H&E



Apocrine Sweat Glands

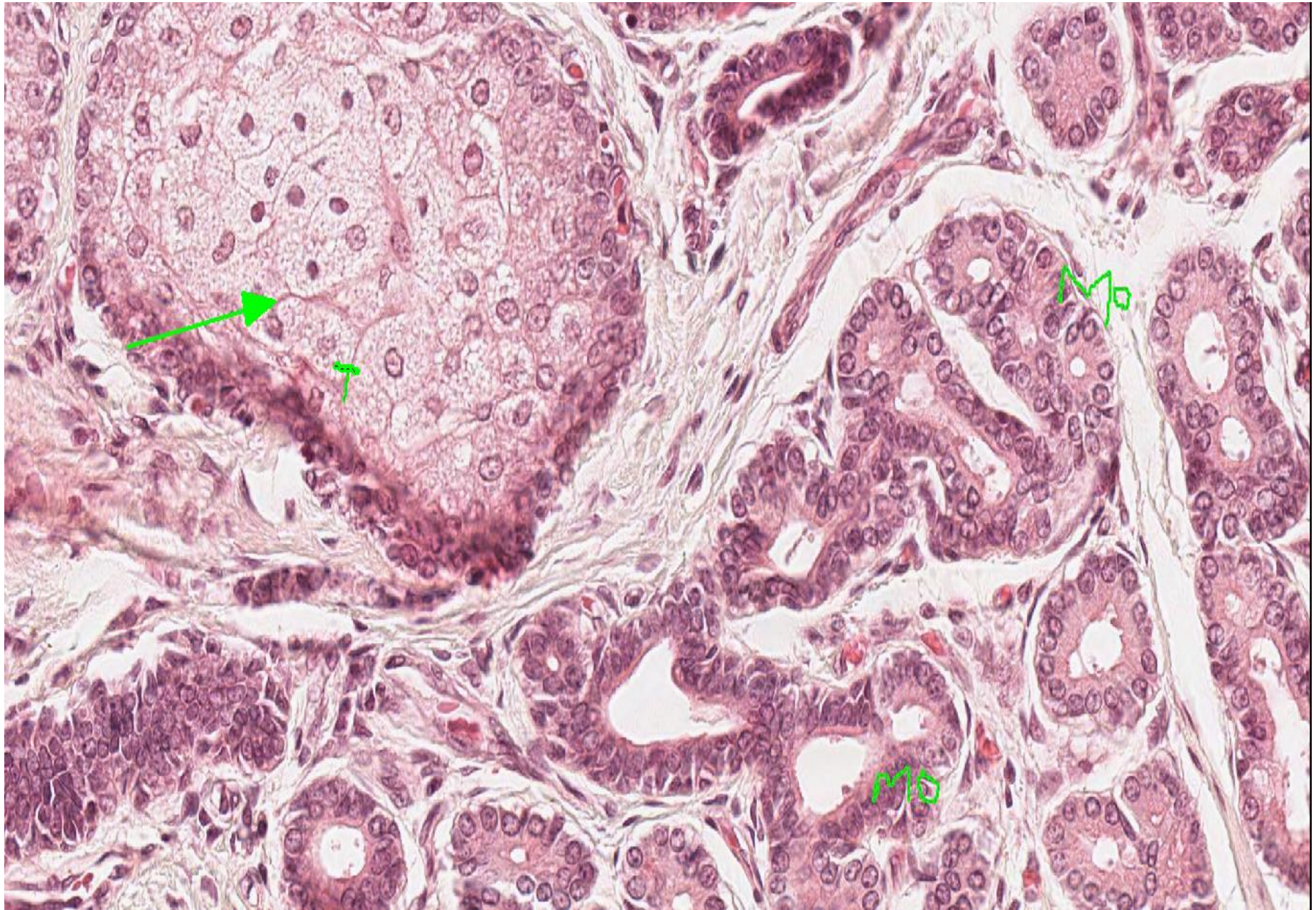
- Their development (but not functional activity) depends on sex hormones and their development is not complete until puberty.
- The most obvious histological differences between the eccrine and apocrine sweat glands are:
 1. the much larger lumen of apocrine glands
 2. less coiled secretory tubules of apocrine glands
 3. **simple secretory epithelium:** of the secretory portions of apocrine sweat glands which consist of eosinophilic cells with numerous apical secretory granules that undergo exocytosis.
- Thus the glands are misnamed: their cells show merocrine, not apocrine, secretion.
- Lumens of apocrine glands often show stored, protein rich products
- Myoepithelial cells help move into ducts opening into hair follicles
- The wall of the ducts is similar to that of the merocrine glands.
- Apocrine sweat glands are innervated by adrenergic nerve endings, whereas eccrine sweat glands receive cholinergic fibers.

Apocrine Sweat Glands

- Become fully formed at puberty.
- Secrete **protein-rich product** during times of pain, stress, and sexual activity.
- May (??) have a possible pheromone-like function.
- Play no role in thermoregulation.
- Are highly modified at two regions in the body:
 1. **External ear canal (Ceruminous glands)**
 2. **Ciliary border of the eye lid (Glands of Moll)**

Remember: *Glands are derivatives of epithelial tissue. So even though many reside in the dermis, they are not made from connective tissue.*

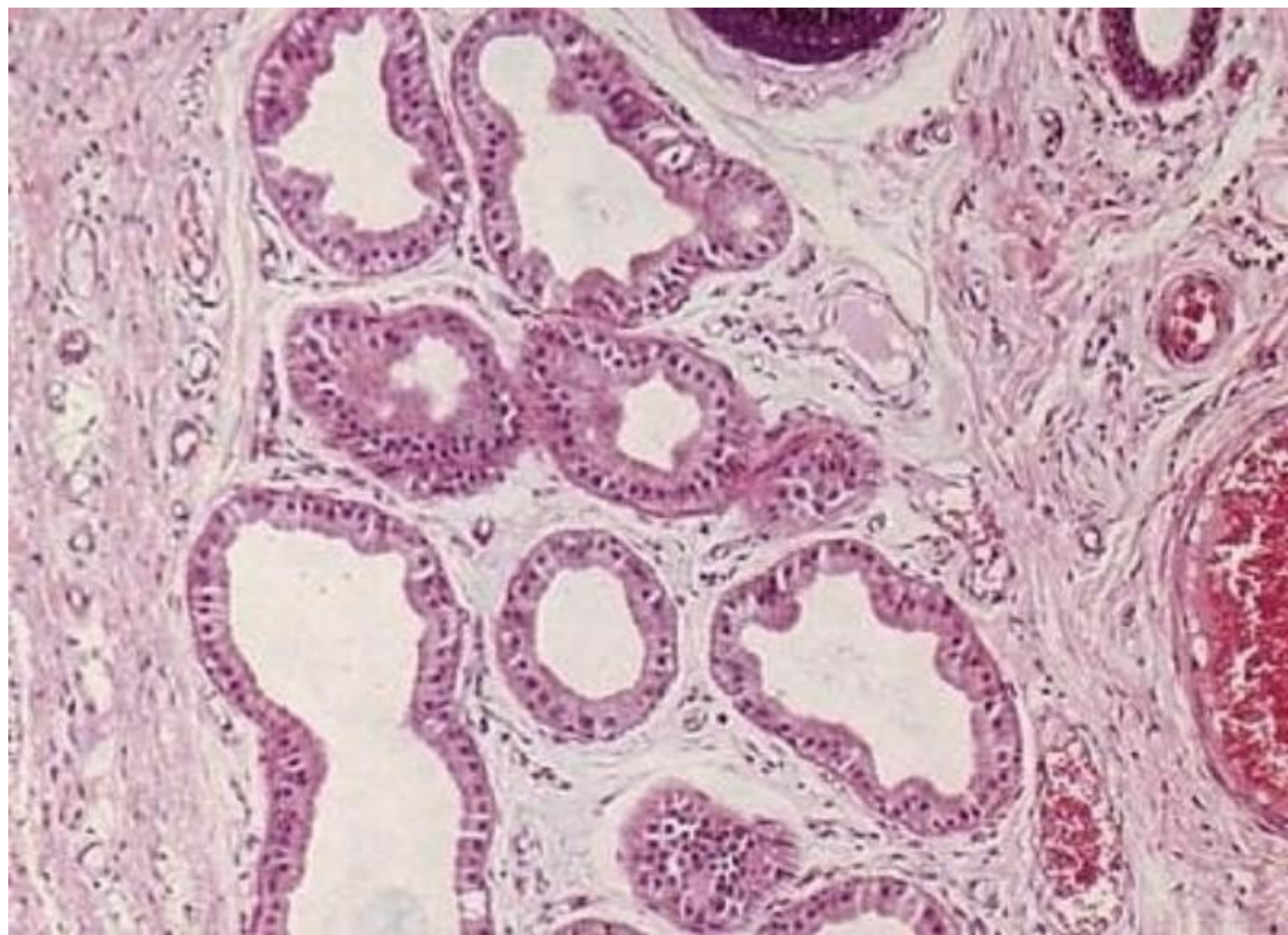
Tarsal glands and Glands of Moll in the eyelid



Ceruminous Glands



- **Modified apocrine sweat glands found in the inner lining of the external ear canal.**
- **Secrete a sticky waxy substance called cerumen (earwax).**
- **It combines with sebum and dead epidermal cells to form a bitter compound that offers a barrier to entry of the ear.**
- **It traps particles and deter insects.**



Skin as a general sense organ

- With its large surface coverage and external location, the skin functions as an extensive receiver for certain stimuli from the environment.
- A variety of sensory receptors are present in skin, including both simple nerve endings with no glial or collagenous covering and more complex, compound structures with sensory fibers enclosed by glia and delicate connective tissue capsules

Non-capsulated/simple receptors include the following:

1. **Tactile/Merkel discs** associated with the bases of epidermal tactile/Merkel cells, which function as receptors for light touch.
2. **Free nerve endings** in the papillary dermis and extending into lower epidermal layers, which respond primarily to high and low temperatures, pain, & itching, but also function as tactile receptors.
3. **Hair Root plexuses**, a web of sensory fibers surrounding the bases of hair follicles in the reticular dermis that detects movements of the hair & light touch.

The following encapsulated receptors are tactile mechanoreceptors:

1. Tactile corpuscles (also called Meissner corpuscles)

- are elliptical or oval-shaped structures (about 30–75 μm by 150 μm), located in the dermal papillae immediately beneath and perpendicular to the epidermis.
- Found in the papillary layer of the fingertips, soles of the feet, nipples, eyelids, lips and genitalia.
- Involved in the reception of light discriminatory touch, the degree of discrimination depending on the proximity of receptors to one another.

2. Lamellated (Vater Pacinian) corpuscle

- a pressure-sensitive mechano-receptor.
- *Named for Filippo Pacini (1812–1883), Professor of Anatomy and Physiology, successively at Pisa and then Florence, Italy*
- are large encapsulated oval-shaped sensory receptors responsive to pressure (sustained touch), coarse (crude) touch, high frequency vibrations and tension.
- Found in the deeper layers of the skin (reticular dermis or hypodermis), ligaments, joint capsules, in some serous membranes such as mesenteries, some viscera and some erogenous areas.
- Range from 1—4 mm in length, about 0.5 mm thick, 1 mm wide, and have an appearance of an onion in TS section.

Lamellated (Vater Pacinian) corpuscles

- found deep within an outer collagenous capsule and 15 to 50 thin, concentric lamellae of flat Schwann-type cells (teloglia cells) and delicate collagen fibers.
- Towards the center of the corpuscle the lamellae become closely packed and the core contains a single large unbranched non-myelinated nerve fiber with several club-like terminals which become myelinated as it leaves the corpuscle.
- Distortion of the capsule produces an amplified mechanical stimulus in the core which is transduced into the action potential in the sensory neuron.

3. Ruffini corpuscles (*Angelo Ruffini (1887–1929), Professor of Histology, Bologna, Italy*)

- are also simple encapsulated, robust spindle-shaped structures found particularly in the reticular dermis of the sole of the feet, finger and toe tips.
- pressure-sensing mechanoreceptors in dermis, but are more poorly characterized structurally.
- they detect continuous pressure and tissue distortion

4. Krause end bulbs (**Krause, also known for its glands of:** accessory lacrimal glands in the subconjunctival tissue of the upper fornix. *Karl Friedrich Theodor Krause (1797–1868), Professor of Anatomy, Hanover, Germany*)

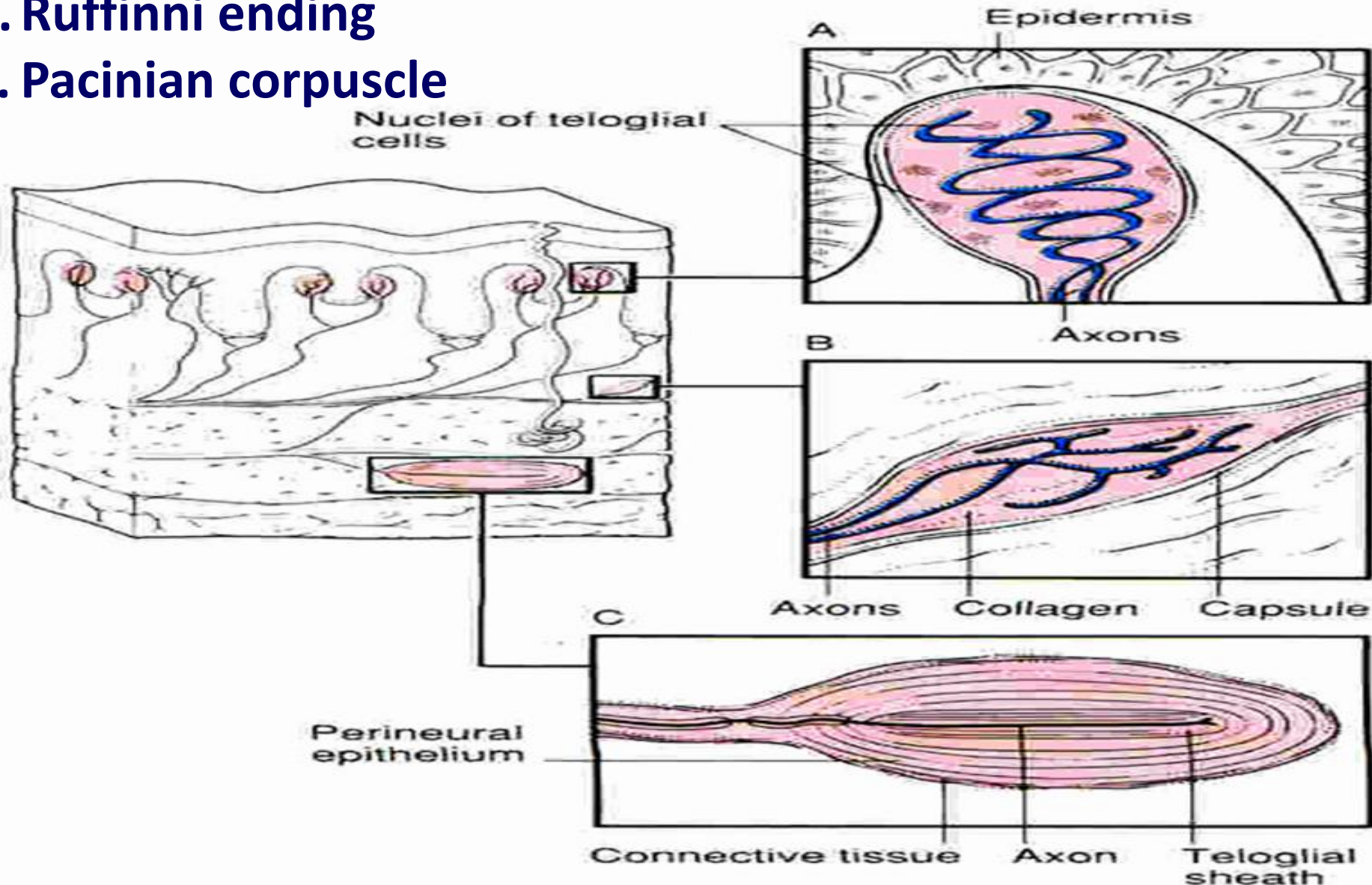
- are delicate receptors found in the linings of the oropharynx and the conjunctiva of the eye.
- but are more poorly characterized structurally
- they may detect pressure and low-frequency vibrations

Skin as a general sense organ

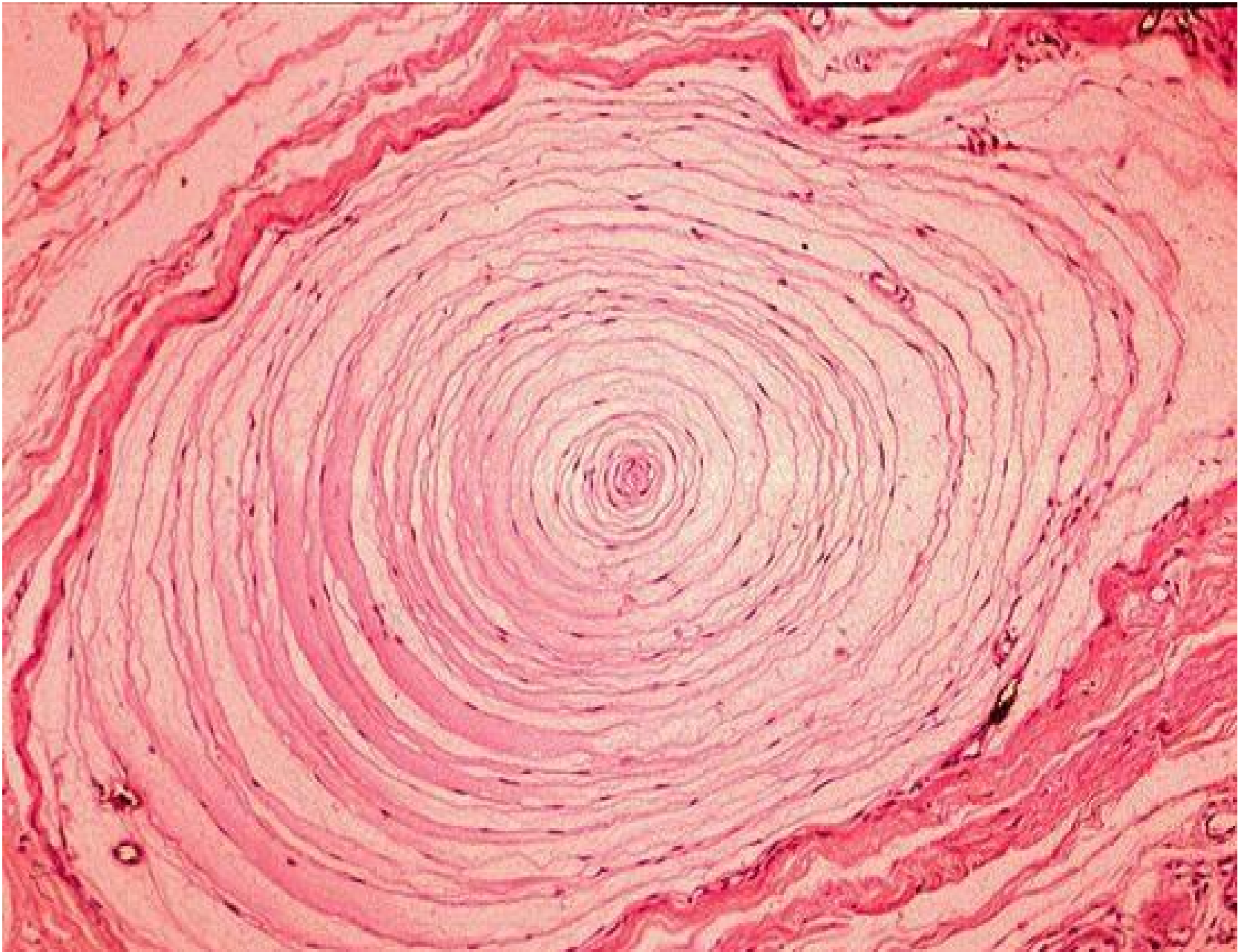
A. Meissner's corpuscle

B. Ruffini ending

C. Pacinian corpuscle



VATER PACIANIAN CORPUSCLE



Skin Cancer

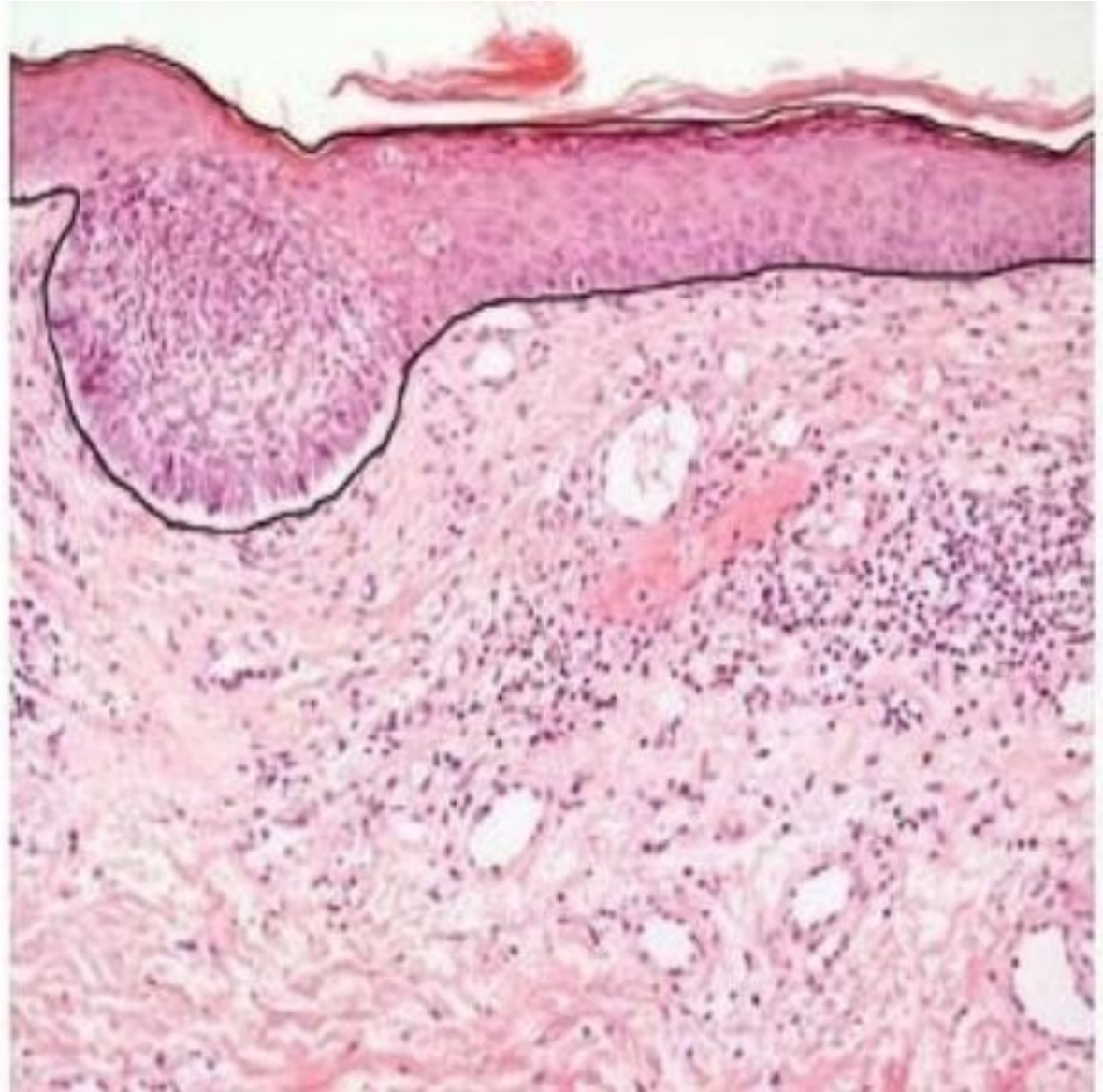
1. Basal cell carcinoma

- Most common (70%) type of skin cancers
- Least vicious
- Usually cured via surgical removal
- Consists of uncontrolled growth of cells of the stratum basale.
- They'll proliferate and invade the dermis and hypodermis.
- Often occurs on sun-exposed areas of face and neck



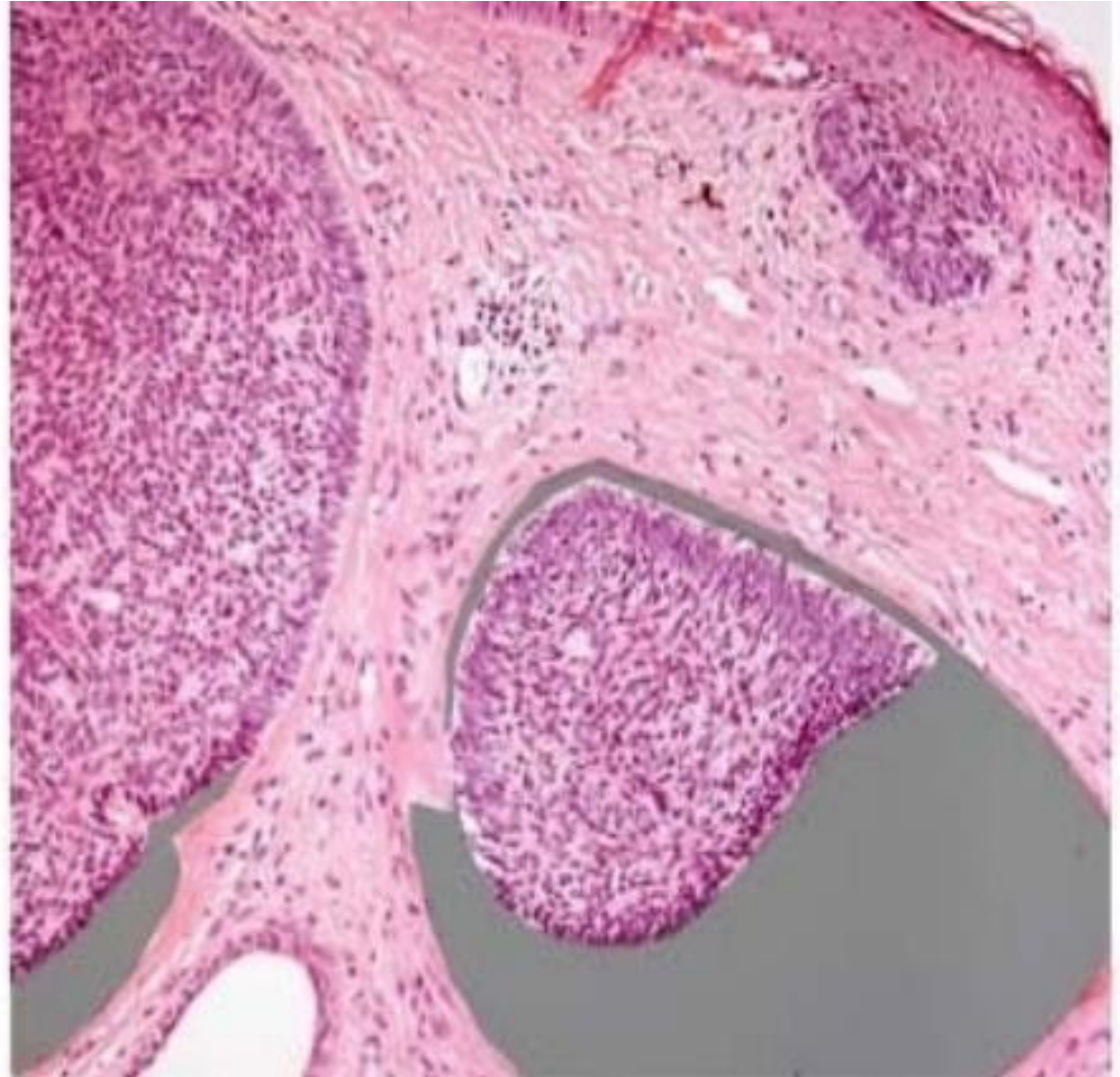
Skin, basal cell carcinoma, superficial type – Low power

- In a superficial basal cell carcinoma, the tumor nests are much smaller than in a nodular basal cell carcinoma, and all the nests are connected with the epidermis.



Skin, deep basal cell carcinoma - Low power

- Nests of tumor cells resembling the normal basal cell layer of the epidermis (basaloid cells) arise from the epidermis and occupy in the dermis or hypodermis.



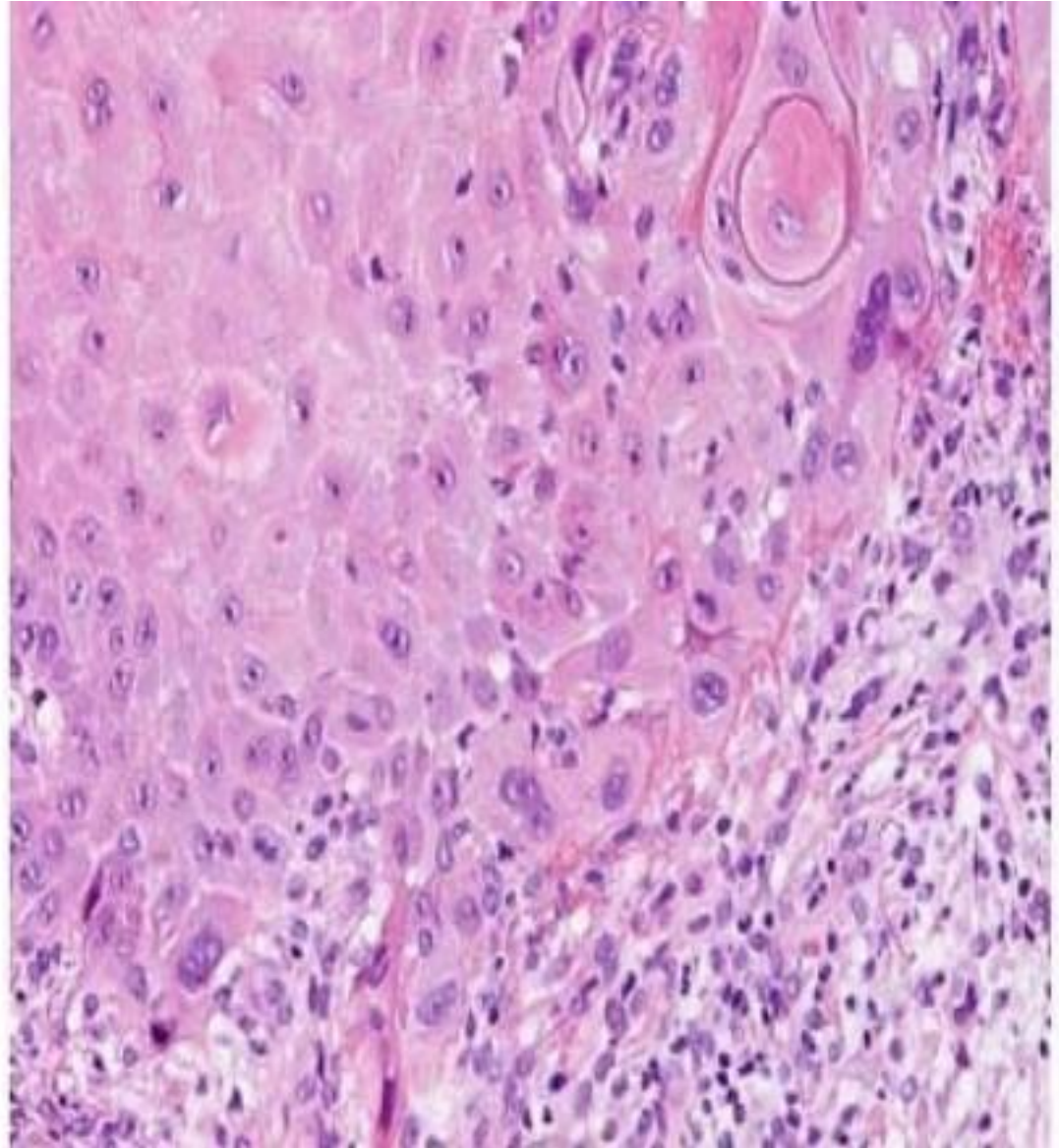
2. Squamous cell carcinoma

- Arises from keratinocytes of stratum spinosum.
- 25% of cases of skin cancer.
- Good prognosis if caught and treated early (surgical excision or radiation).
- Can be fatal if it metastasizes to the lymph nodes.



Skin, squamous cell carcinoma – Medium power

- An atypical squamous epithelial proliferation is present in the dermis, surrounded by a lymphocytic infiltrate.
- This image is part of a well-differentiated squamous cell carcinoma

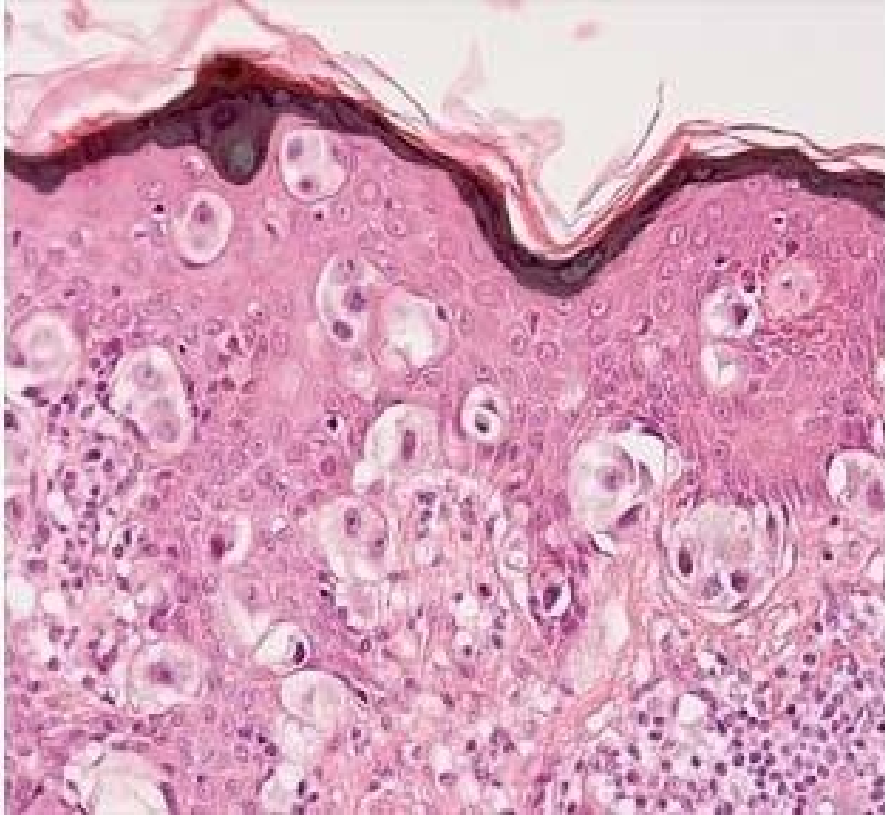


3. Malignant melanoma

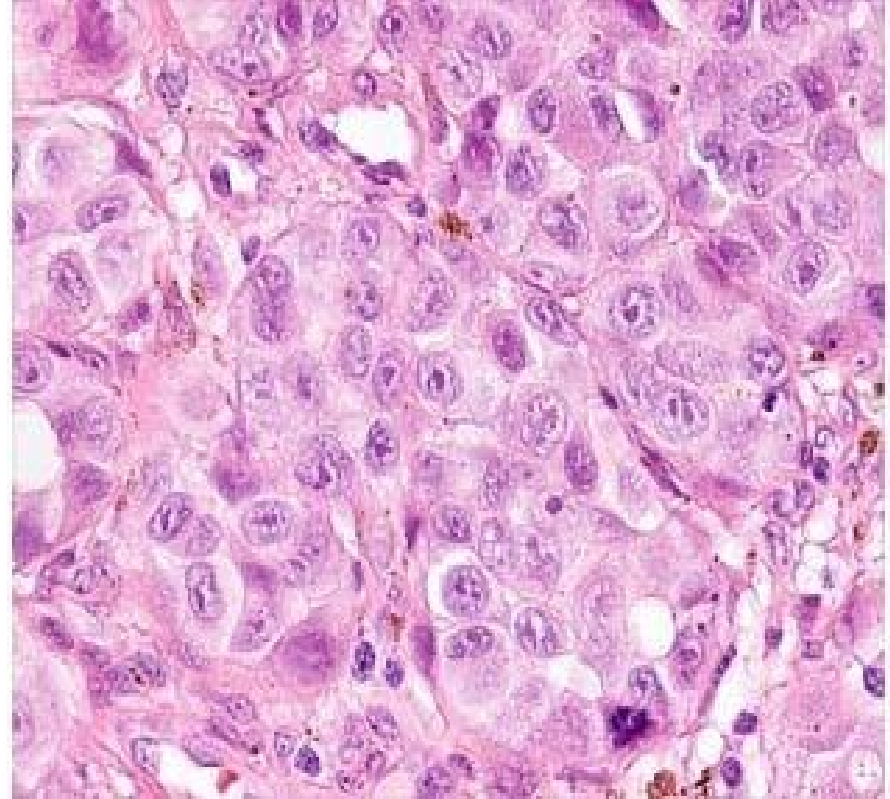
- Least common (<5%) and most dangerous.
- Cancer of melanocytes.
- Often arises from a pre-existing mole.
- Follow the ABCD rule for early detection:
 - Asymmetry (2 sides do not match)
 - Border irregularity in pigmentation and contour
 - Color (multiple)
 - Diameter (>6mm is bad!)



Skin, malignant melanoma in epidermis – Medium power



Skin, malignant melanoma – High power



The epidermis shows atypical melanocytic proliferation consisting of irregular nested and single melanocytes. Atypical melanocytes are visible in all layers of the epidermis from the basal layer to the granular layer. Normally, the epidermis is composed of different, interdependent cell types.

This high-power image of malignant melanoma cells shows striking nuclear pleomorphism and a mitotic figure.

Skin Burns

1. First-degree:

- only the epidermis is damaged
- Symptoms include localized redness, swelling & pain

2. Second-degree:

- the epidermis and upper portions of dermis are damaged
- symptoms mimic first degree burns, but blisters also appear

3. Third-degree:

- involve entire thickness of the skin, and may extend into the subcutaneous tissue, exposing the victim to fluid loss & infection
- Burned area appears gray-white, cherry red, or black, and there is neither initial edema nor pain (since nerve endings are destroyed)

Burns First-degree

(epidermis only; redness, swelling and pain)



(b)

Second-degree

(epidermis and dermis, in addition to symptoms of 1st degree burn, blisters also appear)



(c)

Third-degree

(full thickness, destroying epidermis, dermis, often part of hypodermis)



(d)

Critical Burns “Rule of 9” is used to estimate severity

- Over 10% of the body is prone to third-degree burns (burns of face, hands, or feet)
- 25 % of the body is subjected to second-degree burns
- When more than 20% of the body has been burned-out, the injury is often fatal.

